

Venzini

Software Requirements Specification

An AI-Assisted Yield Management Platform for Data-Driven Optimization in the Petroleum Industry

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Follows ISO/IEC/IEEE 29148:2018

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Document control

This SRS turns the validated Venzini Requirements Elicitation Document (final version) into a precise, testable specification for version 1, and records the author's decisions of 25 Oct 2026 (Appendix A).

Item	Value
Document	Venzini – Software Requirements Specification (SRS)
Version	Final version (FV), approved by the author on 25 Oct 2026
Author	Mohamed Zouari
Primary source	Venzini 1 – Requirement Elicitation FV
Other sources	Project Scope FV; AI-generated preliminary designs; decision log (Appendix A)
Precedence	Where Project Scope FV and the Elicitation document differ, the Elicitation document applies

Labels used in this document

- **Proposed** – a design decision suggested by this SRS. It is not a stakeholder requirement; all Proposed items were approved by the author (D-27, D-28).
- **Unresolved** – an open question still without an answer. No solution is invented for it.
- **Changed (D-xx)** – a requirement altered by a decision in Appendix A.
- **Inferred** – derived from stated goals; all inferred items were confirmed (D-24).

IDs and priority

- All IDs from the Elicitation FV are kept (UN, BG, FR, NFR, AI, BR, AS, CO, C, OQ, RK, UC, US). New items continue the sequence (FR-75 onward, BR-21 onward, NFR-19 onward).
- New ID families in this SRS: AIC (AI capability), DE (data entity), API, UI, ERR, OBS, TC (test case), AC (acceptance criterion), D (decision).
- Priority: **Must** = core of version 1; **Should** = rest of version 1. The **MVP** is a clickable frontend prototype with fake data that shows the Must screens and flows; the logic behind them (forecasting, routing, risk analysis, saved decisions) is built after the MVP (D-26).

1. Document Overview

1.1 Purpose

This document specifies what Venzini version 1 must do, how well it must do it, and how each requirement will be verified. It is precise enough for AI-assisted implementation and testing, and every item traces back to a requirement in the Elicitation FV or to a decision in Appendix A.

1.2 Scope of this document

- Covers version 1: the MVP core (Must) and the rest of version 1 (Should).
- Specifies requirements at the logical level. Concrete architecture, technology choices and detailed API contracts belong to the Software Design Document (SDD). Where this SRS gives them, they are marked **Proposed**.
- Does not cover future versions (station-manager interface, real data integrations), except where a v1 requirement prepares for them.

1.3 Intended audience

Reader	Uses this document to
Developer (Mohamed Zouari)	Build the MVP and the rest of v1
AI coding assistant	Generate code and tests from Section 20 and the referenced requirements
Academic supervisors and reviewers	Assess completeness, quality and traceability
Distributor management (business stakeholder)	Confirm goals, scope and rules
Tester	Derive and run test cases (Sections 16–18)

1.4 Definitions

Term	Meaning in Venzini
Station	A fuel station in the distributor's network, with one tank per fuel type
Warehouse (depot)	A storage site that holds fuel stock and is the home base of tankers
Tanker	A fuel truck with a capacity in litres and one home warehouse
Delivery day	The calendar day for which resources are uploaded and a plan is produced
Trip	One tanker journey, from the moment it leaves its

Term	Meaning in Venzini
	home warehouse until it returns. A trip may last more than one day (D-09)
Route	The ordered list of stops a tanker visits during one trip, starting and ending at its home warehouse
Stop	One station visit within a route, with the litres delivered per fuel type
Plan	All routes, quantities, costs and risks for one delivery day
Optimal plan	The plan recommended by the system before any manager edit
Edited plan	A plan changed by the manager (stop order, tanker assignment or quantities)
Recommended action	One unit of the plan the manager approves or rejects: a station delivery (station, fuel type, litres, tanker, arrival time)
Safety stock	The minimum level a station tank should never fall below
Free storage	Tank capacity minus current stock
Time to stockout	Hours until a tank's projected stock reaches zero at forecast demand
Station risk	High, medium or low, from time to stockout (BR-26)
Delivery priority	The rank of a station in the plan, combining station risk with resources, distance and demand (FR-25)
Risk (of an edit)	A warning raised when an edited plan is worse than the optimal plan on a defined dimension (FR-62)
Confidence	A 0–100% value showing how reliable a forecast is
Uncertain forecast	A forecast with confidence below 70% (BR-27)
Yield management	Getting the best overall result from limited resources (tankers, drivers, fuel, time)
Route fuel cost	Distance × tanker consumption rate × fuel price (BR-16)
MAPE	Mean Absolute Percentage Error: the average % gap

Term	Meaning in Venzini
	between forecast and actual demand
Template explanation	A pre-written sentence whose blanks are filled only with data values (D-11)
MVP	Minimum viable product: the Must requirements
Proposed / Unresolved / Changed / Inferred	See Document control

1.5 References

Ref	Document	Use in this SRS
R1	Venzini 1 – Requirement Elicitation FV, M. Zouari	Primary source of all requirements and IDs
R2	Venzini – Project Scope FV, M. Zouari	Scope, MVP split, out of scope; superseded by R1 where they differ
R3	Earlier Requirements Specification drafts (cited in R1 as "RS FRxx / RS NFRxx")	Historical source; superseded by R1, not used directly
R4	AI-generated preliminary designs “FuelFlow” and “Venzini” (2 images)	Visual inspiration only for Section 13; not requirements
R5	W3C, Web Content Accessibility Guidelines (WCAG) 2.1	Contrast level AA (NFR-11)
R6	ISO/IEC/IEEE 29148 – Requirements engineering	Structure and quality attributes of requirements (Proposed reference)

2. System Scope

Version 1 delivers forecasting, risk scoring, multi-tanker delivery planning, route editing with live risk analysis, approval and an immutable decision log, all running on sample data with simulated forecasts, routes and external data.

2.1 In scope

Module	What it covers	Must (v1 core)	Rest of v1 (Should)	Main FRs
Access and roles	Login, three roles, role-based access	Yes	–	FR-01–04
User and station administration	User accounts; station records	Users	Stations (MVP uses preloaded	FR-05–06

Module	What it covers	Must (v1 core)	Rest of v1 (Should)	Main FRs
			stations)	
Resource management	Daily tankers, drivers, warehouse stock; trip times	Yes	Other resource types (OQ-20)	FR-07–13, FR-78–79, FR-84
Network overview	Map, tank levels, stockout alerts	Yes	–	FR-14–16
Demand forecasting	7-day forecast, confidence, factors, uncertain flag	Yes (simulated)	–	FR-17–23, FR-83, FR-85
Urgency and priority	Station risk, delivery priority, reasons	Simple priority rule	Full yield-management allocation	FR-24–27
Delivery planning	Quantities, warehouse links, multi-tanker routes, shortage handling	Yes	–	FR-28–39
Route cost	Cost per route, in optimization, edit comparison	Yes	–	FR-40–44
Route map	Directed routes, numbering, urgency colors, details on click, tanker filter	Yes	–	FR-45–54
Route editing	Drag-and-drop reorder and move, live recalculation, reset	Yes	Quantity edits with limits, automatic re-plan	FR-55–60, FR-75–76
Risk analysis	Edit-vs-optimal comparison, severity, reasons, highlights, overload block	Yes	–	FR-61–67, FR-77
Recommendations and approval	Per action and whole plan, Adjust shortcut	Yes	–	FR-68–71, FR-80–82
Decision history	Immutable log, read-only page for all roles	Yes	–	FR-72–73
Data	Sample data set and simulations	Yes	–	FR-74

2.2 Out of scope

- Setting pump prices (BR-14, NFR-17).
- Controlling fuel systems or any equipment (BR-14, NFR-17).

- Platform access for tanker drivers, in any version (BR-15).
- Real distributor data and live integrations with weather, events, map or telematics services (CO-02, AS-02).
- Mobile application; v1 targets desktop browsers only (CO-03).
- Large language models (LLMs) and autonomous AI agents (D-11).
- Automatic execution or dispatch of any action without manager approval (BR-01, CO-05).
- Live tanker tracking; actual trip times are entered by the dispatcher (D-09).
- Station-manager interface (future version).
- Handing the approved plan to drivers and depot inside the platform (OQ-15, **Unresolved**; Proposed: none in v1).

2.3 System boundaries

- **Inside the system:** user accounts and roles; station, warehouse, tank and resource data; forecasting, risk, priority, routing and cost modules; plans, edits, risks and decisions; all screens.
- **At the boundary (through defined interfaces, NFR-15):** weather data, local events data, distance and travel-time data. In v1 each is a simulated provider behind an interface that can later point to a real source.
- **Outside the system:** drivers, depot staff and fleet manager (they act on the approved plan offline); regulators; real fuel equipment and pricing systems.

3. Product Vision and Goals

Vision: Venzini gives petroleum distributors a clear 7-day view of fuel demand and turns it into an explained, editable delivery plan that uses every available tanker, while the operations manager keeps the final say on every action.

It applies yield-management thinking: getting the best overall result from limited tankers, drivers, fuel and time.

ID	Goal	Success indicator	Target	Verified by
BG-01	Reduce stockouts at stations	Simulated stockout events per period vs. a baseline plan made without Venzini	**Unresolved** (OQ-11)	TC-AI-05 simulation comparison
BG-02	Reduce excess inventory	Average litres above target stock level vs. baseline	**Unresolved** (OQ-11)	TC-AI-05
BG-03	Reduce unnecessary trips and route fuel cost	Total km and route fuel cost per delivered litre vs. baseline	**Unresolved** (OQ-11)	TC-AI-05
BG-04	Keep managers in control with explainable recommendations	Share of executed actions with a logged manager approval; share of forecasts, priorities and risks with a reason	100% and 100%	AC-14, AC-07

ID	Goal	Success indicator	Target	Verified by
BG-05	Flexible planning that adapts to each day's resources	Manager can reorder stops and move stations between tankers and see the impact without leaving the map	Achieved for 100% of edits in UC-06 tests	AC-10

Baseline (Proposed): a simple rule-based plan that refills every station to capacity when it falls under safety stock, using the nearest free tanker, without forecasting. This gives BG-01 to BG-03 a comparison point in simulation.

4. Actors and Roles

Three roles have accounts in v1. Each role owns one part of the work: the dispatcher owns resources, the operations manager owns station orders and plan decisions, and the system admin owns accounts; stations are managed by both the system admin and the operations manager (D-05, D-10, D-15).

4.1 Primary actors (platform users)

Role	Responsibility	Can change	Cannot change
Operations manager	Monitors the network, reviews forecasts, edits routes and quantities, approves or rejects actions	Stop order, tanker assignment of a station, delivery quantity per station and fuel type, decisions, station records (D-15)	Tankers, drivers, warehouse stock, trip times, users (D-05)
Dispatcher	Uploads each delivery day's resources and records actual trip departure and return times	Tankers, drivers, warehouse stock for a day; trip actual times	Plans, quantities, decisions; cannot see station forecasts, map or plans (D-10); can read per-warehouse forecast totals (D-17)
System admin	Manages user accounts, roles and station records	Users, roles, stations	Resources, plans, decisions

4.2 Secondary and external actors (no platform access)

Actor	Relation to Venzini
Tanker drivers	Execute approved routes; no access in any version (BR-15)
Warehouse / depot team	Load tankers with approved quantities

Actor	Relation to Venzini
Fleet manager	Manages tanker availability and consumption
Distributor management / owners	Sponsor; benefit from the goals; no account in v1
Finance team	Follows route costs; no account in v1
Regulators	Set storage and transport rules (NFR-18, OQ-13)
Data providers	Weather and events data; simulated in v1. A dedicated data-provider role is planned for a later version (D-14)
Station managers	Future dedicated interface; not in v1

4.3 Permission matrix

The matrix is enforced on every page and every data request (NFR-05). “R” = read only, “RW” = read and change, “–” = no access (page hidden and request refused).

Function / page	Operations manager	Dispatcher	System admin	Source
Login and own profile	RW	RW	RW	FR-01
Network overview (map, alerts, tank levels)	R	–	–	FR-14–16, D-10
Station forecast	R	–	–	FR-17–23, D-10
Forecast totals (network and per warehouse)	R	R (per warehouse only)	–	FR-85, D-17
Resources page (tankers, drivers, warehouse stock)	R	RW	–	FR-04, FR-07–13, D-05
Trip actual departure and return times	R	RW	–	FR-79, D-09
Delivery planner (generate, view)	RW	–	–	FR-28–54
Route editing and quantity editing	RW	–	–	FR-55–60, FR-75–76
Recommendations page: approve, reject, adjust	RW	–	–	FR-03, FR-68–71, FR-80–82
Decision history	R	R	R	FR-73, D-10
User administration	–	–	RW	FR-05

Function / page	Operations manager	Dispatcher	System admin	Source
Station administration	RW	–	RW	FR-06, D-15
Edit or delete a decision log entry	–	–	–	NFR-14, BR-13

Changed (D-05, D-10): FR-02 now points to this matrix. The manager can no longer "edit tankers", the dispatcher has no access to planning pages, and the decision history is read-only for all three roles.
Changed (D-15, D-17): the manager also manages station records, and the dispatcher can read per-warehouse forecast totals.

5. System Context

Venzini is a single web application used by three roles; it reads simulated external data through defined interfaces and uses no external AI service in v1.

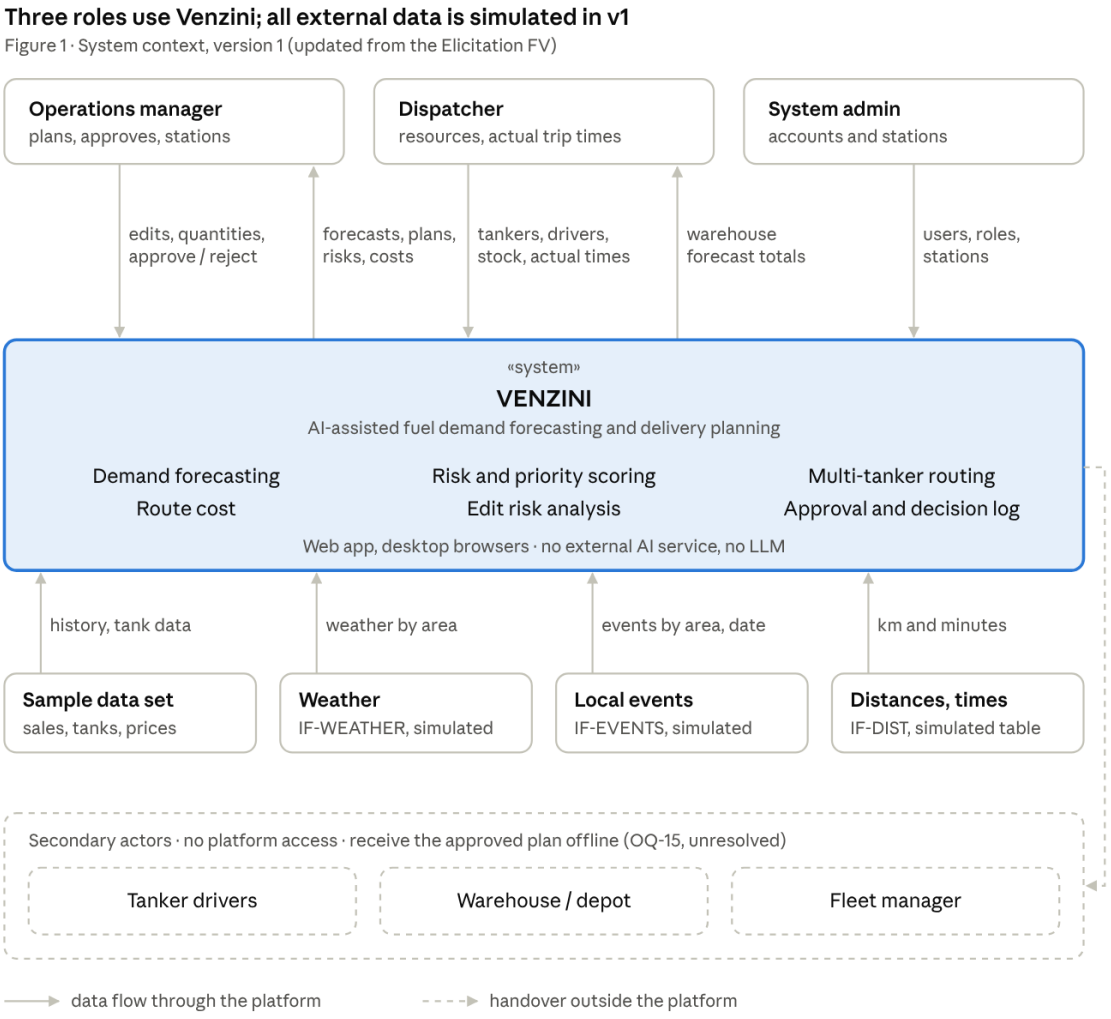


Figure 1 · System context, version 1

The dispatcher sends resources and actual trip times and reads warehouse forecast totals; the approved plan reaches drivers and the depot outside the platform (OQ-15).

5.1 Users

- **Operations manager:** monitors, plans, edits, approves or rejects; manages station records.
- **Dispatcher:** enters daily resources and actual trip times; reads per-warehouse forecast totals.
- **System admin:** manages users and stations.
- Everyone else (drivers, depot, fleet, finance, management) receives the approved plan outside the platform (OQ-15, **Unresolved**).

5.2 System

One web application (Proposed: browser client + server API + relational database) containing these logical components:

Component	Responsibility	Replaceable (NFR-12)
CMP-AUTH	Login, sessions, role checks	–
CMP-ADMIN	Users, roles, stations	–
CMP-RES	Resources per delivery day, trip times, locking	–
CMP-FC	Demand forecasting (AIC-01)	Yes
CMP-RISK	Station risk and delivery priority (AIC-02)	Yes
CMP-ROUTE	Warehouse links, quantities, tanker assignment, routes (AIC-03)	Yes
CMP-COST	Route fuel cost	Yes
CMP-EDIT	Edits, recalculation, edit-vs-optimal risk analysis (AIC-04)	Yes
CMP-EXPL	Template explanations (AIC-04)	Yes
CMP-DEC	Approvals, rejections, decision log	–
CMP-UI	All screens	–
CMP-DATA	Sample data set, simulated providers	Yes

5.3 External systems

External system	Data	v1 status	Interface
Weather source	Daily temperature and conditions per area	Simulated	IF-WEATHER
Local events	Events with date, area and expected	Simulated	IF-EVENTS

External system	Data	v1 status	Interface
source	impact		
Map and distance service	Distances and travel times between all points; map tiles for display	Simulated distance table (AS-08); map tiles from a map library (Proposed)	IF-DIST, map tile provider

5.4 APIs

- **Internal API:** the browser calls the Venzini server through a REST/JSON API (Section 9, Proposed).
- **Provider interfaces:** IF-WEATHER, IF-EVENTS and IF-DIST are internal interfaces with one simulated implementation each in v1. A real implementation can replace each one without changing the UI (NFR-12, NFR-15).
- **No public API** is offered to third parties in v1.

5.5 Data sources

- Sample data set (FR-74): stations, tanks (capacity, safety stock, current level, delivery window), warehouses, sales history, tanker consumption rates, fuel price.
- Dispatcher input: tankers, drivers, warehouse stock per day; trip actual times.
- Admin input: users, stations.
- Simulated providers: weather, events, distances.

5.6 AI services

No external AI service, LLM or agent is used in v1 (D-11). The four AI capabilities (forecasting, risk and priority scoring, route optimization, risk analysis and explanation) run inside Venzini as replaceable modules. In v1 the forecast is simulated and explanations are template-based.

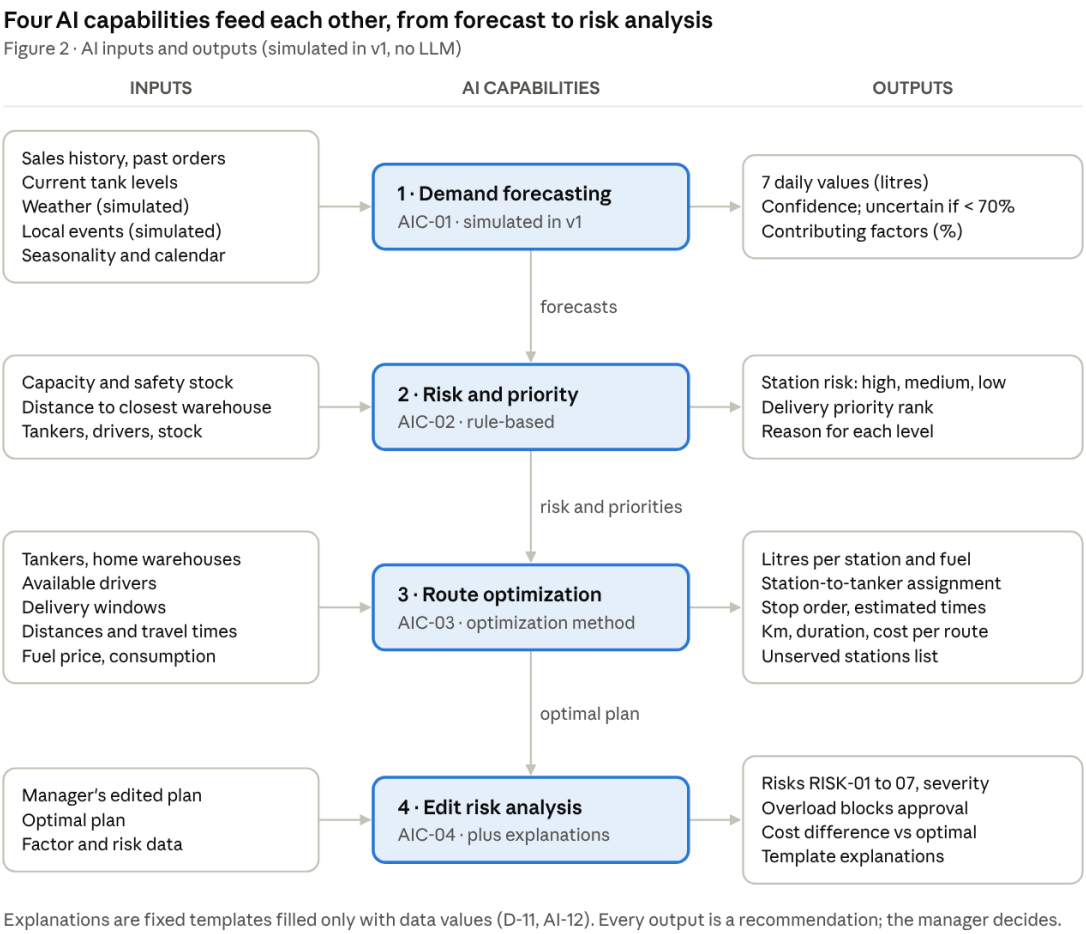


Figure 2 · AI inputs and outputs

Each capability feeds the next; the overload rule is an output of risk analysis, and every explanation is template-based.

6. Functional Specification

Version 1 has 85 functional requirements, all listed in the Elicitation document (Requirement Elicitation FV). “Changed” and “New” labels point to the decisions behind them (Appendix A).

Each requirement uses the same 12 fields. “None” means the field was checked and nothing applies. Traceability lists the user need (UN), use case (UC), user story (US), business rules (BR), and the test case (TC) and acceptance criterion (AC) that verify it.

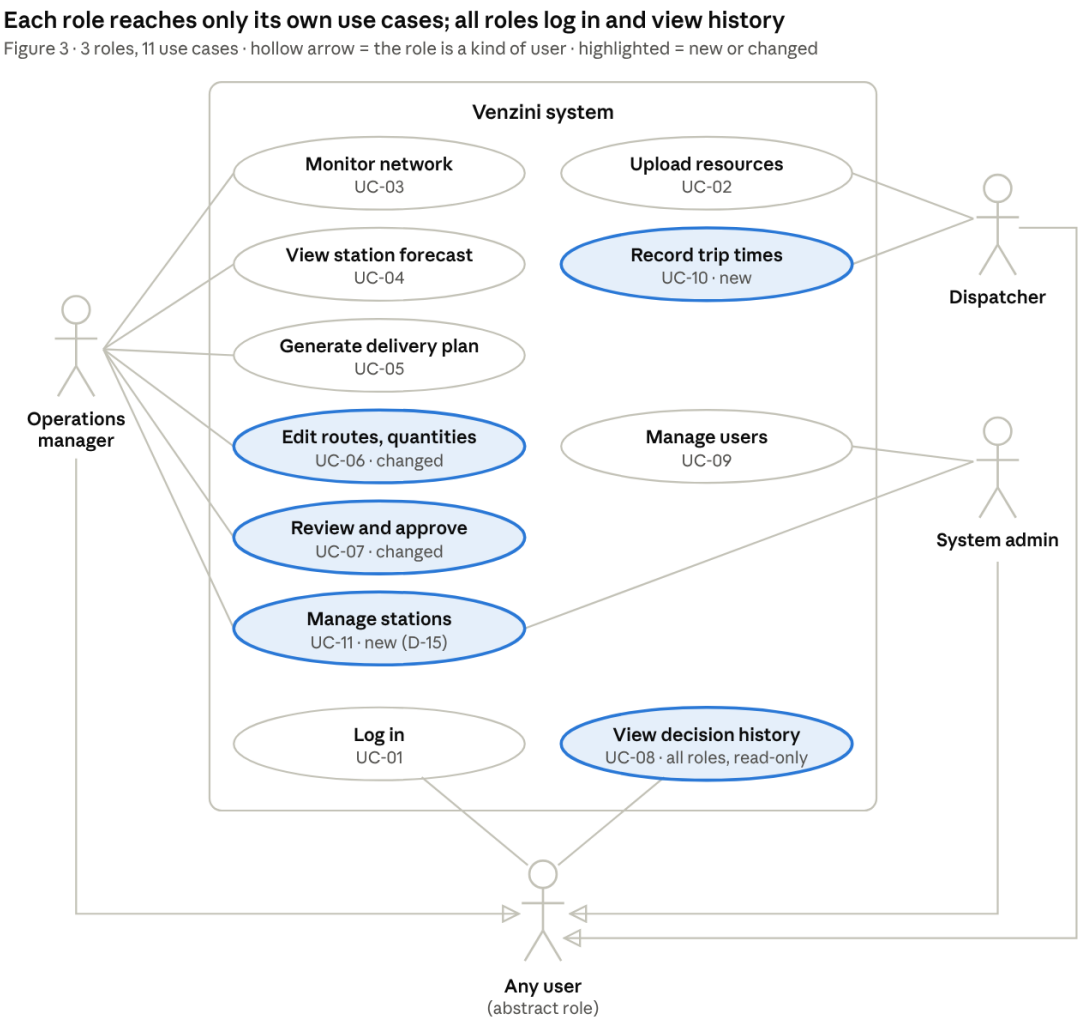


Figure 3 · Use case diagram, version 1

Each role reaches only its own use cases; all three roles log in and view the decision history read-only.

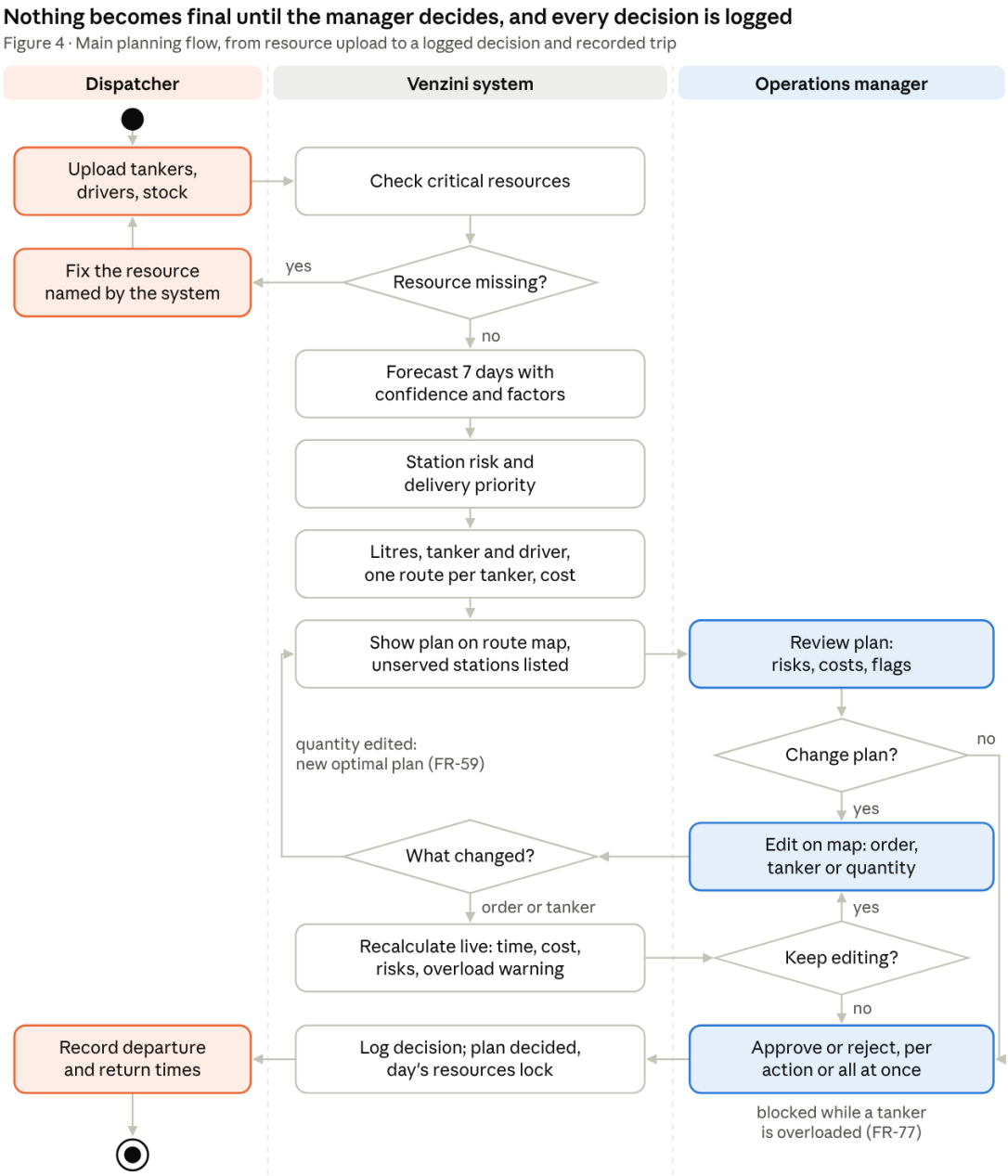


Figure 4 · Main planning flow (activity diagram)

The missing-resource check comes first (FR-39). Approval can be per action or for the whole plan and is blocked while a tanker is overloaded; after the decision, the dispatcher records actual trip times.

6.1 Business rules register

The functional requirements below refer to these 31 rules. All of them are in the Elicitation document; BR-21 to BR-31 come from the 25 Oct 2026 decisions.

ID	Rule	Source	Status

ID	Rule	Source	Status
BR-01	No recommendation becomes final without an operations manager's approval	RS FR53	Valid
BR-02	Only operations managers can approve or reject	RS NFR7	Valid
BR-03	Only dispatchers can upload or update operational resources	RS NFR7; stakeholder	Valid
BR-04	A recommended quantity cannot exceed the station's free storage (capacity minus current stock)	RS FR21	Valid
BR-05	Recommended plans keep stations above safety stock when resources allow	RS FR21	Valid
BR-06	A recommended tanker load cannot exceed the tanker's capacity, even when resources are short	RS FR22	Valid
BR-07	Recommended deliveries arrive within the station's delivery window	RS FR22	Valid
BR-08	Recommended totals per fuel type cannot exceed the serving warehouse's stock	RS FR12, FR15	Valid (confirmed, D-24)
BR-09	Urgency colors: red = high, orange = medium, green = low	RS FR32	Valid
BR-10	Tanker route lines never use red, orange or green	RS FR37	Valid
BR-11	Risks are warnings and do not block approval, **except** “capacity exceeded” (BR-21)	RS FR49; D-02	**Changed (D-02)**
BR-12	Reordering stops or moving stations triggers recalculation; editing quantities triggers a new optimal plan	RS FR41, FR42; D-05	**Changed (D-05)**: “stations or tankers” edits removed
BR-13	Decision log entries cannot be edited or deleted	RS NFR8	Valid
BR-14	The platform never sets pump prices or controls fuel systems	RS NFR13	Valid
BR-15	Tanker drivers have no access to the platform in any version	Stakeholder	Valid
BR-16	Route cost covers only the tanker's own fuel consumption	Stakeholder	Valid
BR-17	Each station is served from its closest warehouse;	Stakeholder	Valid

ID	Rule	Source	Status
	there can be several warehouses		
BR-18	Each tanker is tied to one home warehouse	Stakeholder	Valid
BR-19	A tanker may load at another warehouse of the same company; process and approver undefined	Stakeholder	**Unresolved** (OQ-07); not modelled in v1 (AS-06)
BR-20	The system always produces a plan unless a critical resource is missing (no tankers, no drivers or no warehouse stock)	Stakeholder	Valid
BR-21	A plan in which any tanker load exceeds its capacity cannot be approved, in whole or in part, until the overload is removed	D-02	New
BR-22	The operations manager cannot change resources; the manager may change only stop order, a station's tanker assignment and delivery quantities	D-05	New
BR-23	A trip lasts from the tanker's departure from its home warehouse until its return, and may span more than one day; a tanker starts at most one trip per delivery day	D-09	New
BR-24	Each station receives its delivery from exactly one tanker per plan (no split delivery)	D-09	New
BR-25	Every route ends at the tanker's home warehouse; the return leg counts in distance, duration and cost	D-09	New
BR-26	Station risk per fuel type: **high** if time to stockout < 24 h; **medium** if 24 h to 72 h; **low** if > 72 h	D-07	New
BR-27	A forecast is uncertain when its confidence is below 70%	D-08	New
BR-28	The maximum quantity the manager can enter for a stop is the smallest of: the free space in the tanker's compartments for that fuel type, the station's free storage, and the remaining stock of that fuel at the serving warehouse	D-06	New
BR-29	The system calculates estimated trip times; only the dispatcher records actual departure and return times	D-09	New

ID	Rule	Source	Status
BR-30	The manager can decide each action or the whole plan. A plan is decided when every action is approved or rejected; the day's resources lock at that moment	D-03	New
BR-31	A tanker has compartments: it can carry several fuel types on one trip; each compartment holds one fuel type and never more than its size	D-19	New

6.2 Access and roles

FR-01 – Login required (Must)

- **Requirement:** The system shall require users to log in before accessing any page.
- **Description:** Every page and API request except the login page needs an authenticated session.
- **Actors:** All users.
- **Preconditions:** The user has an active account (FR-05).
- **Inputs:** Login (unique), password.
- **Processing:** Verify the login exists and is active; compare the password with the stored hash (NFR-06); create a session; open the home page of the user's role (manager: Network overview; dispatcher: Resources; admin: Administration).
- **Outputs:** Authenticated session; role home page.
- **Business rules:** None specific.
- **Alternative flows:** A user opening a protected URL without a session is redirected to login, then returned to that URL if their role allows it.
- **Exception handling:** Wrong credentials → ERR-AUTH-01, stay on login. Deactivated account → ERR-AUTH-02. 5 failed attempts in 15 min → ERR-AUTH-03, account locked 15 min (Proposed, NFR-21).
- **Acceptance criteria:** Opening any page without a session redirects to login; a valid login opens the role home page; a wrong password shows ERR-AUTH-01.
- **Traceability:** UN-13; UC-01; RS NFR7 → TC-FR-01; AC-01.

FR-02 – Three roles with defined permissions (Must) – Changed (D-05, D-10)

- **Requirement:** The system shall support three roles – operations manager, dispatcher and system admin – with the permissions in the matrix of Section 4.3.
- **Description:** Each account has exactly one role. Menus show only allowed pages; the server refuses everything else.
- **Actors:** All users; system admin assigns roles.
- **Preconditions:** Logged in.
- **Inputs:** Session role; requested page or action.
- **Processing:** Check the role against the permission matrix on every request (NFR-05).
- **Outputs:** Allowed page or ERR-AUTH-04 (403).
- **Business rules:** BR-02, BR-03, BR-22.
- **Alternative flows:** None.

- **Exception handling:** Forbidden request → ERR-AUTH-04, logged as a security event (OBS-L3).
- **Acceptance criteria:** For each role and each matrix row, a test account sees and can use exactly what the matrix allows; direct API calls to forbidden functions return 403.
- **Traceability:** UN-13; UC-01, UC-09; US-15; RS FR19 → TC-FR-02; AC-01.

FR-03 – Only managers approve or reject (Must)

- **Requirement:** The system shall allow only operations managers to approve or reject recommendations.
- **Description:** Approve, reject, approve-all and reject-all controls exist only for the manager role.
- **Actors:** Operations manager.
- **Preconditions:** Logged in as manager.
- **Inputs:** Decision request.
- **Processing:** Role check before any decision is stored.
- **Outputs:** Decision stored, or ERR-AUTH-04.
- **Business rules:** BR-01, BR-02.
- **Alternative flows:** None.
- **Exception handling:** Non-manager decision request → ERR-AUTH-04.
- **Acceptance criteria:** Dispatcher and admin accounts cannot see or call approve/reject; a direct request returns 403 and stores nothing.
- **Traceability:** UN-11; UC-07; US-13 → TC-FR-03; AC-01, AC-14.

FR-04 – Only dispatchers change resources (Must)

- **Requirement:** The system shall allow only dispatchers to upload or update resources and trip actual times.
- **Description:** The manager can read resources but not change them (D-05); the admin has no access to them.
- **Actors:** Dispatcher.
- **Preconditions:** Logged in as dispatcher.
- **Inputs:** Resource save request.
- **Processing:** Role check before saving.
- **Outputs:** Saved resources, or ERR-AUTH-04.
- **Business rules:** BR-03, BR-22, BR-29.
- **Alternative flows:** None.
- **Exception handling:** Non-dispatcher save → ERR-AUTH-04.
- **Acceptance criteria:** Manager and admin accounts cannot save resource data or trip times; direct requests return 403.
- **Traceability:** UN-05; UC-02; US-05 → TC-FR-04; AC-01.

6.3 User and station administration

FR-05 – Manage user accounts (Must)

- **Requirement:** The system shall allow the system admin to create, edit and deactivate user accounts and assign their role.

- **Description:** Accounts are never deleted, only deactivated, so decision log entries keep a valid user reference (BR-13).
- **Actors:** System admin.
- **Preconditions:** Logged in as admin.
- **Inputs:** Full name, login, initial password, role (one of three), status (active / inactive).
- **Processing:** Validate fields (Section 8.4); check login uniqueness; hash the password; save; log the change (OBS-L4).
- **Outputs:** Created or updated account; confirmation message.
- **Business rules:** BR-15 (no driver accounts: role list has only the three roles).
- **Alternative flows:** Edit an existing account; reactivate an inactive account.
- **Exception handling:** Duplicate login → ERR-VAL-03. Admin deactivating their own account → ERR-VAL-07 (Proposed: refused, so at least one active admin remains).
- **Acceptance criteria:** A new account can log in with its role; a deactivated account cannot log in; a duplicate login is rejected.
- **Traceability:** UN-13; UC-09; US-15; RS FR19 → TC-FR-05; AC-01.

FR-06 – Manage station records (Should) – Changed (D-15)

- **Requirement:** The system shall allow the system admin and the operations manager to create, edit and deactivate station records.
- **Description:** In the MVP, stations come preloaded from the sample data set. Station data includes location, tanks per fuel type (capacity, safety stock, current level) and delivery window.
- **Actors:** System admin, operations manager (D-15).
- **Preconditions:** Logged in as admin or operations manager.
- **Inputs:** Station ID (unique), name, latitude, longitude, region, delivery window (opening and closing time), and per fuel type: tank capacity (L), safety stock (L), current level (L).
- **Processing:** Validate (Section 8.4); save; recompute the station's closest warehouse (BR-17); log the change.
- **Outputs:** Station visible on the map and usable in plans; a deactivated station is excluded from new plans.
- **Business rules:** BR-17.
- **Alternative flows:** Deactivate or reactivate a station.
- **Exception handling:** Duplicate station ID → ERR-VAL-03; safety stock above capacity, or current level above capacity → ERR-VAL-02 naming the field.
- **Acceptance criteria:** A new station appears on the map; a deactivated station no longer appears in newly generated plans; invalid values are refused with the field named.
- **Traceability:** UN-14; UC-11; US-16; RS FR20; D-15 → TC-FR-06; AC-02. Capacity, safety stock and delivery windows are maintained by the system admin or the operations manager (D-15).

6.4 Resource management

FR-07 – Upload available tankers (Must)

- **Requirement:** The system shall allow the dispatcher to upload the tankers available for a delivery day, with each tanker's capacity in litres and its home warehouse.

- **Description:** The tanker fleet (ID, capacity, consumption rate, home warehouse) exists as master data; the dispatcher marks which tankers are available for the day. Entry through a form in v1.
- **Actors:** Dispatcher.
- **Preconditions:** Logged in as dispatcher; the day's plan is not decided (BR-30).
- **Inputs:** Delivery day; per tanker: tanker ID, compartment sizes (L), total capacity (L, sum of compartments), home warehouse, available (yes/no).
- **Processing:** Validate; warn if a tanker has an open trip (FR-79) (Proposed: warning, not a block); store with user and timestamp (FR-13).
- **Outputs:** Available tanker list for the day, shown on the Resources page and the planner.
- **Business rules:** BR-03, BR-18, BR-23.
- **Alternative flows:** Resources already exist for the day → edit and save (UC-02).
- **Exception handling:** Capacity ≤ 0 or missing → ERR-VAL-02 naming the field; day already decided → ERR-VAL-04; tanker with an open trip → WARN-RES-01.
- **Acceptance criteria:** Saved tankers, capacities and home warehouses are shown on the planner for that day.
- **Traceability:** UN-05; UC-02; US-05; RS FR15 → TC-FR-07; AC-02.

FR-08 – Upload available drivers (Must) – confirmed (D-01)

- **Requirement:** The system shall allow the dispatcher to upload the drivers available for a delivery day.
- **Description:** Drivers are data only; they never log in (BR-15).
- **Actors:** Dispatcher.
- **Preconditions:** As FR-07.
- **Inputs:** Delivery day; per driver: driver ID, name, home warehouse, available (yes/no).
- **Processing:** Validate; store with user and timestamp.
- **Outputs:** Available driver list for the day.
- **Business rules:** BR-03, BR-15.
- **Alternative flows:** Edit existing list.
- **Exception handling:** Missing name or ID → ERR-VAL-01; day decided → ERR-VAL-04.
- **Acceptance criteria:** Saved drivers are shown on the planner and used when tankers are assigned (FR-35).
- **Traceability:** UN-05; UC-02; US-05; D-01 → TC-FR-08; AC-02.

FR-09 – Upload warehouse fuel stock (Must)

- **Requirement:** The system shall allow the dispatcher to upload the fuel stock of each warehouse, per fuel type, in litres.
- **Description:** This stock caps what can be delivered from each warehouse (BR-08).
- **Actors:** Dispatcher.
- **Preconditions:** As FR-07.
- **Inputs:** Delivery day; per warehouse and fuel type: stock (L).
- **Processing:** Validate (≥ 0); store.
- **Outputs:** Stock table for the day.
- **Business rules:** BR-03, BR-08.
- **Alternative flows:** Edit existing values.

- **Exception handling:** Negative value → ERR-VAL-02; day decided → ERR-VAL-04.
- **Acceptance criteria:** Saved stock values per warehouse and fuel type are shown on the planner.
- **Traceability:** UN-05; UC-02; US-05; RS FR15 → TC-FR-09; AC-02.

FR-10 – Upload other resource data (Should)

- **Requirement:** The system shall allow the dispatcher to upload other operational resource data once it is defined.
- **Description:** Placeholder for resource types not yet identified.
- **Actors:** Dispatcher.
- **Preconditions:** The resource type is defined.
- **Inputs: Unresolved** (OQ-20).
- **Processing: Unresolved** (OQ-20).
- **Outputs:** New resource visible on the planner.
- **Business rules:** BR-03.
- **Alternative flows:** None.
- **Exception handling:** As FR-07.
- **Acceptance criteria:** A newly defined resource type can be uploaded and is visible on the planner.
- **Traceability:** UN-05; OQ-20 → TC-FR-10; AC-02. Not implementable until OQ-20 is answered.

FR-11 – Update resources until the plan is decided (Must) – Changed (D-03)

- **Requirement:** The system shall allow the dispatcher to update the operational resources of a delivery day until that day's plan is decided.
- **Description:** “Decided” = every action approved or rejected (BR-30). From that moment the day's resources are read-only.
- **Actors:** Dispatcher.
- **Preconditions:** Resources exist for the day.
- **Inputs:** Changed resource values.
- **Processing:** Check the day's plan status; save if not decided.
- **Outputs:** Updated resources, or refusal.
- **Business rules:** BR-30; AS-05.
- **Alternative flows:** If a draft plan exists, the manager sees a “resources changed” notice and can regenerate (FR-12).
- **Exception handling:** Day decided → ERR-VAL-04.
- **Acceptance criteria:** Updates are accepted before the plan is decided and refused after.
- **Traceability:** UN-05; UC-02; RS FR16; AS-05 → TC-FR-11; AC-02.

FR-12 – Plan uses the latest resources (Must)

- **Requirement:** The system shall use the most recently saved resources when generating a delivery plan.
- **Description:** Each plan records the resource version it was built from (AI-18).
- **Actors:** System; operations manager.
- **Preconditions:** Resources saved for the day.
- **Inputs:** Latest resource version for the day.

- **Processing:** Load the latest version at generation time; store its version ID on the plan.
- **Outputs:** Plan linked to its resource version.
- **Business rules:** None specific.
- **Alternative flows:** Resources changed after the draft → notice to the manager with a "Regenerate" action.
- **Exception handling:** None.
- **Acceptance criteria:** After an update, a newly generated plan uses the new tanker count and stock.
- **Traceability:** UN-05; RS FR17; AI-18 → TC-FR-12; AC-02.

FR-13 – Show last resource update (Must)

- **Requirement:** The system shall display the date, time and user of the last resource update.
- **Description:** Shown on the Resources page and the planner.
- **Actors:** Dispatcher, operations manager.
- **Preconditions:** Resources saved at least once for the day.
- **Inputs:** Stored update metadata.
- **Processing:** None beyond display.
- **Outputs:** "Last updated by <name>, <date> <time>".
- **Business rules:** None.
- **Alternative flows:** No resources yet → "No resources uploaded for this day".
- **Exception handling:** None.
- **Acceptance criteria:** After an update, the planner shows the correct user and timestamp.
- **Traceability:** UN-05; RS FR18 → TC-FR-13; AC-02.

FR-78 – Estimated trip times (Must) – New (D-09)

- **Requirement:** The system shall calculate, for each route in a plan, the estimated departure time from the home warehouse, the estimated arrival time at each stop and the estimated return time.
- **Description:** A trip may end on a later day (BR-23). Estimates come from distances and travel times (IF-DIST) and an unloading time per stop that depends on the delivered quantity; v1 uses sample values, and real unloading data will later be uploaded by data managers to improve accuracy (D-25).
- **Actors:** System.
- **Preconditions:** A plan is generated or recalculated.
- **Inputs:** Route stop order, travel times, delivery windows, unloading time.
- **Processing:** Start at the earliest departure that lets every stop meet its window (BR-07); add travel and unloading times; include the return leg (BR-25).
- **Outputs:** Estimated times per route and stop.
- **Business rules:** BR-07, BR-23, BR-25, BR-29.
- **Alternative flows:** Recalculated after every edit (FR-57).
- **Exception handling:** No feasible departure → "missed delivery window" risk (FR-62).
- **Acceptance criteria:** For a test route, estimated return = departure + all travel times + all unloading times, to the minute.
- **Traceability:** D-09; UN-06 → TC-FR-78; AC-19.

FR-79 – Record actual trip times (Must) – New (D-09)

- **Requirement:** The system shall allow the dispatcher to record the actual departure time and the actual return time of each trip of an approved plan.
- **Description:** A trip is open from recorded departure until recorded return. These times are for traceability; drivers have no access (BR-15).
- **Actors:** Dispatcher.
- **Preconditions:** The trip belongs to a decided plan with at least one approved action.
- **Inputs:** Trip ID; actual departure date-time; actual return date-time.
- **Processing:** Validate (return after departure; departure not in the future); store with user and timestamp; set trip status (Section 8.6).
- **Outputs:** Trip status: planned → departed → returned.
- **Business rules:** BR-23, BR-29.
- **Alternative flows:** Correct a time before the return is recorded (Proposed).
- **Exception handling:** Return before departure → ERR-VAL-02; trip not approved → ERR-VAL-04.
- **Acceptance criteria:** A recorded departure sets the trip to "departed"; a recorded return sets it to "returned"; manager and admin cannot record times.
- **Traceability:** D-09; UN-05, UN-12 → TC-FR-79; AC-19.

FR-84 – Estimated vs. actual trip time (Should) – New (D-16)

- **Requirement:** The system shall store, for each trip with recorded actual times, the difference between the estimated and the actual trip duration, and show it to the operations manager.
- **Description:** Gap = actual duration (return − departure) − estimated duration, in minutes and %. It builds the data needed to improve travel-time estimates in a later version; that learning is not part of v1.
- **Actors:** System; operations manager (reads); dispatcher (records times, FR-79).
- **Preconditions:** Actual departure and return recorded (FR-79).
- **Inputs:** Estimated departure and return (FR-78); actual departure and return.
- **Processing:** Compute the gap when the return is recorded; store it with the trip (DE-16); show it in the trip list.
- **Outputs:** Gap per trip (e.g. "+45 min, +9.4%"); gap history per route and tanker.
- **Business rules:** BR-29.
- **Alternative flows:** Cancelled trip → no gap.
- **Exception handling:** A corrected actual time → gap recalculated.
- **Acceptance criteria:** For a trip estimated at 8 h 00 min and recorded at 8 h 45 min, the gap shows +45 min (+9.4%).
- **Traceability:** D-16; UN-12 → TC-FR-84; AC-19.

6.5 Network overview

FR-14 – Map of all active stations (Must)

- **Requirement:** The system shall display all active stations on a map.
- **Description:** One marker per active station, plus warehouse markers. Default label: station name and ID only (FR-49).
- **Actors:** Operations manager.

- **Preconditions:** Logged in as manager.
- **Inputs:** Active stations with coordinates; warehouses.
- **Processing:** Load and place markers; color by the station's highest fuel-type risk (FR-47).
- **Outputs:** Network overview map.
- **Business rules:** BR-09.
- **Alternative flows:** Selecting a station opens its details (UC-03 step 3).
- **Exception handling:** Missing station data (e.g. no coordinates or no tank level) → station listed as “Data unavailable” (ERR-DATA-01) instead of a colored marker.
- **Acceptance criteria:** The number of station markers equals the number of active stations in the data set.
- **Traceability:** UN-01; UC-03; US-01; RS FR1 → TC-FR-14; AC-03.

FR-15 – Stockout-risk alerts (Must)

- **Requirement:** The system shall display a stockout-risk alert for each station and fuel type with a high risk level.
- **Description:** Alerts list station, fuel type, time to stockout and the reason (FR-27). Medium-risk stations are shown on the map but raise no alert.
- **Actors:** Operations manager.
- **Preconditions:** Station risk calculated (FR-24).
- **Inputs:** Station risk per fuel type.
- **Processing:** Select all high-risk pairs; sort by time to stockout, shortest first.
- **Outputs:** Alert list and alert count.
- **Business rules:** BR-26.
- **Alternative flows:** No high-risk pair → “No stations at high risk” message, no alert.
- **Exception handling:** Risk not computable (no forecast) → station flagged for manual planning (FR-83).
- **Acceptance criteria:** A sample station set to high risk shows an alert; a low-risk station does not.
- **Traceability:** UN-01; UC-03; US-01; RS FR2 → TC-FR-15; AC-03, AC-06.

FR-16 – Current tank levels (Must)

- **Requirement:** The system shall display the current tank level per station and fuel type.
- **Description:** Shown in litres and as a percentage of capacity, with the safety-stock level.
- **Actors:** Operations manager.
- **Preconditions:** Tank data available.
- **Inputs:** Current level, capacity, safety stock per tank.
- **Processing:** Compute % = level / capacity × 100, rounded to the nearest whole number.
- **Outputs:** Tank levels in the station details.
- **Business rules:** None specific.
- **Alternative flows:** None.
- **Exception handling:** Missing level → “Data unavailable”.
- **Acceptance criteria:** Displayed levels match the data set exactly.
- **Traceability:** UN-01; UC-03; RS FR3 → TC-FR-16; AC-03.

6.6 Demand forecasting

FR-17 – Select station and fuel type (Must)

- **Requirement:** The system shall allow the manager to select a station and a fuel type.
- **Description:** Two selectors on the Station forecast page; fuel types listed are those the station has tanks for.
- **Actors:** Operations manager.
- **Preconditions:** Logged in as manager.
- **Inputs:** Station ID, fuel type.
- **Processing:** Load or compute the matching forecast (FR-18).
- **Outputs:** Forecast view for the pair.
- **Business rules:** None.
- **Alternative flows:** Opening the page from a map marker or alert pre-selects that station.
- **Exception handling:** Pair with no tank → fuel type not offered.
- **Acceptance criteria:** Selecting a station and fuel type loads the matching forecast.
- **Traceability:** UN-02; UC-04; US-02; RS FR4 → TC-FR-17; AC-04.

FR-18 – 7-day daily forecast (Must)

- **Requirement:** The system shall produce a daily demand forecast in litres for the next 7 days for the selected station and fuel type.
- **Description:** Day 1 = the day after the current date (or after the selected delivery day). In v1 the forecast is simulated (AS-02, AIC-01).
- **Actors:** System (CMP-FC).
- **Preconditions:** Historical data exists for the pair.
- **Inputs:** Station ID, fuel type, reference date, forecast inputs (FR-19).
- **Processing:** Run the forecasting module; store the result with its model version and input data version (AI-18).
- **Outputs:** 7 values in litres (whole numbers, ≥ 0), one per day.
- **Business rules:** None specific.
- **Alternative flows:** None.
- **Exception handling:** Forecast cannot be produced → FR-83.
- **Acceptance criteria:** Exactly 7 daily values are shown, starting from the next day.
- **Traceability:** UN-02; UC-04; US-02; RS FR5; AI-01 → TC-FR-18; AC-04.

FR-19 – Forecast inputs (Must)

- **Requirement:** The system shall use historical sales, previous orders, current tank level, weather, local events and seasonality as forecast inputs.
- **Description:** Weather, events and seasonality come through IF-WEATHER, IF-EVENTS and a calendar; all simulated in v1.
- **Actors:** System.
- **Preconditions:** Inputs available.
- **Inputs:** Sales history per day; previous orders (past deliveries per station and fuel type, D-28); current tank level; weather per day; events per day and area; calendar (weekday, month, holidays).

- **Processing:** Combine inputs in the forecasting module (method: Proposed in Section 10).
- **Outputs:** Forecast and factor breakdown.
- **Business rules:** None specific.
- **Alternative flows:** One input missing → forecast still produced, flagged low confidence with the missing input named (AI-20).
- **Exception handling:** Sales history missing → FR-83.
- **Acceptance criteria:** Changing one input in the sample data changes the forecast or its factor breakdown.
- **Traceability:** UN-02; RS FR6; AS-02 → TC-FR-19; AC-04. History: at least 365 days.

FR-20 – Forecast chart (Must)

- **Requirement:** The system shall display the forecast as a chart.
- **Description:** Line of 7 forecast values with its confidence band, next to the last 7 days of actual sales for context (Proposed, inspired by R4).
- **Actors:** Operations manager.
- **Preconditions:** Forecast produced.
- **Inputs:** Forecast values, confidence band, recent actuals.
- **Processing:** Render chart; litres on the y-axis, dates on the x-axis.
- **Outputs:** Chart with labeled axes and legend.
- **Business rules:** NFR-10 (not color-only: forecast and actuals differ by line style).
- **Alternative flows:** None.
- **Exception handling:** No forecast → message instead of chart (FR-83).
- **Acceptance criteria:** The station page shows a chart with the 7 forecast values.
- **Traceability:** UN-02; UC-04; RS FR7 → TC-FR-20; AC-04.

FR-21 – Confidence level (Must)

- **Requirement:** The system shall display a confidence level for each forecast.
- **Description:** A 0–100% value per daily value, and an overall value for the 7 days (Proposed: the lowest daily value).
- **Actors:** Operations manager.
- **Preconditions:** Forecast produced.
- **Inputs:** Confidence values from CMP-FC.
- **Processing:** Display as a whole percentage.
- **Outputs:** Confidence per day and overall.
- **Business rules:** BR-27.
- **Alternative flows:** None.
- **Exception handling:** None.
- **Acceptance criteria:** Every forecast value has a visible confidence level.
- **Traceability:** UN-03; UC-04; US-03; RS FR8; AI-09 → TC-FR-21; AC-05.

FR-22 – Contributing factors (Must)

- **Requirement:** The system shall display the factors that influenced each forecast and their relative contribution.

- **Description:** Each factor (recent sales, previous orders, current tank level, seasonality, weather, local events) shows a signed % impact and a template explanation (AIC-04).
- **Actors:** Operations manager.
- **Preconditions:** Forecast produced.
- **Inputs:** Factor breakdown from CMP-FC.
- **Processing:** Sort factors by absolute impact; fill explanation templates with data values only (AI-12).
- **Outputs:** Factor list with % impact and one-line explanation.
- **Business rules:** None specific.
- **Alternative flows:** None.
- **Exception handling:** Factor data missing → that factor shown as "not available" and named in the confidence note (AI-20).
- **Acceptance criteria:** Each forecast shows at least its main contributing factors with their share; every number shown exists in the factor data.
- **Traceability:** UN-03; UC-04; US-03; RS FR9; AI-14 → TC-FR-22; AC-07.

FR-23 – Uncertain forecast flag (Must)

- **Requirement:** The system shall flag a forecast as uncertain when its confidence is below 70%.
- **Description:** The flag shows on the station page, in the plan and on the recommendations page (AI-11).
- **Actors:** System; operations manager.
- **Preconditions:** Confidence computed.
- **Inputs:** Overall confidence.
- **Processing:** If confidence < 70% → set flag; propagate to every recommendation based on that forecast.
- **Outputs:** "Uncertain forecast – review manually" label with icon and reason.
- **Business rules:** BR-27.
- **Alternative flows:** None.
- **Exception handling:** None.
- **Acceptance criteria:** A forecast at 69% shows the flag on the station page, in the plan and on the recommendations page; a forecast at 70% does not.
- **Traceability:** UN-03; UC-04, UC-05; RS FR10; AI-10, AI-11; D-08 → TC-FR-23; AC-05.

FR-83 – Manual planning flag (Must) – New (from AI-21)

- **Requirement:** The system shall flag a station and fuel type for manual planning when no forecast can be produced, and shall not recommend a quantity for it.
- **Description:** Makes AI-21 a testable functional requirement.
- **Actors:** System; operations manager.
- **Preconditions:** Forecasting failed for the pair.
- **Inputs:** Failure reason (e.g. no sales history).
- **Processing:** Mark the pair "manual planning"; exclude it from recommended quantities; list it on the planner with its reason.
- **Outputs:** "Manual planning needed" label and reason (ERR-AI-01).
- **Business rules:** BR-01.

- **Alternative flows:** None.
- **Exception handling:** None.
- **Acceptance criteria:** A station with no sales history shows the manual-planning label and receives no recommended quantity.
- **Traceability:** AI-21; UC-04 exception → TC-FR-83; AC-05.

FR-85 – Network and warehouse forecast totals (Must) – New (D-17)

- **Requirement:** The system shall display, for each of the next 7 days and each fuel type, the total forecast demand of all active stations, and the total for the stations served by each warehouse.
- **Description:** Saves the warehouse from adding up station forecasts by hand. Totals are sums of station forecasts (FR-18); per-warehouse totals use the closest-warehouse link (BR-17).
- **Actors:** Operations manager (network and per-warehouse totals); dispatcher (per-warehouse totals only, read-only, D-17).
- **Preconditions:** Station forecasts available.
- **Inputs:** Station forecasts; station–warehouse links.
- **Processing:** Sum daily litres per fuel type; total band = sum of station band lows and highs (Proposed); uncertain share = litres from uncertain forecasts ÷ total litres × 100.
- **Outputs:** Table and chart per day and fuel type, with the uncertain share and the number of stations left out.
- **Business rules:** BR-17, BR-27.
- **Alternative flows:** Stations in manual planning (FR-83) are left out and listed, e.g. “3 stations not included: manual planning”.
- **Exception handling:** No forecasts available → message instead of totals.
- **Acceptance criteria:** For each day and fuel type, the network total equals the sum of the included station forecasts, and the per-warehouse totals add up to the network total; the dispatcher sees per-warehouse totals but no station-level forecast.
- **Traceability:** UN-02; D-17 → TC-FR-85; AC-04.

6.7 Urgency and priority

FR-24 – Station risk level (Must)

- **Requirement:** The system shall calculate a station risk level (high, medium or low) for each station and fuel type.
- **Description:** Station risk uses stock and forecast only (D-07).
- **Actors:** System (CMP-RISK).
- **Preconditions:** Current level and forecast available.
- **Inputs:** Current level (L), 7-day forecast (L/day).
- **Processing:** Time to stockout = hours until the projected level reaches 0, with each day's forecast spread evenly over 24 h (Proposed). Apply BR-26: < 24 h high; 24–72 h medium; > 72 h low. If the level stays above 0 for all 7 days → low.
- **Outputs:** Risk level and time to stockout per pair.
- **Business rules:** BR-26.
- **Alternative flows:** None.
- **Exception handling:** No forecast → FR-83.

- **Acceptance criteria:** Every station and fuel type has one of the three levels; test pairs with 23 h, 24 h, 72 h and 73 h to stockout get high, medium, medium and low.
- **Traceability:** UN-04; UC-05; US-04; RS FR11; AI-02; D-07 → TC-FR-24; AC-06.

FR-25 – Delivery priority (Must)

- **Requirement:** The system shall calculate a delivery priority for each station that needs fuel, using station risk, quantity to deliver, number of stations in need, distance from the warehouse, available tankers and warehouse stock.
- **Description:** A station "needs fuel" when any tank is projected to fall below safety stock within the 7-day horizon. MVP uses a simple rule (AS-07).
- **Actors:** System (CMP-RISK).
- **Preconditions:** Station risk calculated; resources uploaded.
- **Inputs:** As in the requirement.
- **Processing (Proposed MVP rule):** Rank by (1) risk level, high first; (2) time to stockout, shortest first; (3) quantity needed, largest first; (4) distance from the serving warehouse, shortest first. Number of stations in need, tankers and warehouse stock decide how far down the ranking the plan can serve (FR-38).
- **Outputs:** Ranked list with a priority number per station.
- **Business rules:** BR-26; AI-16 (remoteness or low volume alone never lowers a high-risk station).
- **Alternative flows:** None.
- **Exception handling:** None.
- **Acceptance criteria:** Stations in the plan are ordered by priority; changing one input (e.g. time to stockout) changes the ranking as the rule predicts.
- **Traceability:** UN-04; UC-05; RS FR12; AS-07 → TC-FR-25; AC-06.

FR-26 – Yield-management allocation (Should)

- **Requirement:** The system shall allocate limited tankers, fuel and time across stations using yield-management principles.
- **Description:** Replaces the MVP rule after the MVP.
- **Actors:** System.
- **Preconditions:** Resources below total demand.
- **Inputs:** As FR-25.
- **Processing: Unresolved** – the documented rule is not yet defined. Proposed objective: minimize total projected stockout hours across the network, then total route cost.
- **Outputs:** Allocation of tankers, litres and stations.
- **Business rules:** BR-04 to BR-08, BR-24.
- **Alternative flows:** None.
- **Exception handling:** As FR-37 to FR-39.
- **Acceptance criteria:** With resources below total demand, allocation follows the documented rule (once defined).
- **Traceability:** UN-04; RS FR13; AS-07 → TC-FR-26; AC-09.

FR-27 – Reasons for urgency (Must)

- **Requirement:** The system shall display the reasons for each station's urgency level.

- **Description:** Template explanation, e.g. “ST027 is high risk: current Diesel stock covers 14 hours of forecast demand.”
- **Actors:** Operations manager.
- **Preconditions:** Risk and priority calculated.
- **Inputs:** Risk inputs and calculated values.
- **Processing:** Fill the matching template (AIC-04) with data values only.
- **Outputs:** At least one reason per station in the plan.
- **Business rules:** AI-12.
- **Alternative flows:** None.
- **Exception handling:** Template data missing → generic reason with the risk level and time to stockout only.
- **Acceptance criteria:** Every station in the plan shows at least one reason for its level, and every number in it matches the data.
- **Traceability:** UN-04; US-04; RS FR14; AI-15 → TC-FR-27; AC-07.

6.8 Delivery planning

FR-28 – Recommended litres per station and fuel type (Must)

- **Requirement:** The system shall recommend the litres to deliver per station and fuel type.
- **Description:** Applies to every station that needs fuel and is served in the plan.
- **Actors:** System (CMP-ROUTE).
- **Preconditions:** Forecast, tank data and resources available.
- **Inputs:** Current level, capacity, safety stock, 7-day forecast.
- **Processing (Proposed rule):** Quantity = (safety stock + forecast demand from now until the end of the 7-day horizon) − current level, capped by free storage (BR-04), rounded down to the nearest 100 L; then reduced if needed by BR-06 and BR-08.
- **Outputs:** Recommended litres per station and fuel type, with reason.
- **Business rules:** BR-04, BR-05, BR-06, BR-08.
- **Alternative flows:** Uncertain forecast → quantity carries the uncertain flag (FR-23).
- **Exception handling:** Manual-planning pair → no quantity (FR-83).
- **Acceptance criteria:** Each station in the plan shows a recommended quantity per fuel type.
- **Traceability:** UN-06; UC-05; US-06; RS FR21 → TC-FR-28; AC-08.

FR-29 – Never above free storage (Must)

- **Requirement:** The system shall not recommend a quantity above the station's storage capacity minus its current stock.
- **Description:** Hard constraint on every recommended plan.
- **Actors:** System.
- **Preconditions:** Tank data available.
- **Inputs:** Capacity, current level, quantity.
- **Processing:** Cap quantity at free storage; validate the whole plan before display.
- **Outputs:** Compliant quantities.
- **Business rules:** BR-04.
- **Alternative flows:** None.

- **Exception handling:** Violation detected at validation → internal error ERR-SYS-02, plan not shown.
- **Acceptance criteria:** No recommendation in a test set exceeds free storage.
- **Traceability:** UN-06; RS FR21; AI-25 → TC-FR-29; AC-08.

FR-30 – Keep stations above safety stock (Must)

- **Requirement:** The system shall recommend quantities that keep each station above its safety-stock level over the planning period, when resources allow.
- **Description:** "Resources allow" = enough tankers, drivers, capacity and warehouse stock.
- **Actors:** System.
- **Preconditions:** As FR-28.
- **Inputs:** Forecast, levels, safety stock, resources.
- **Processing:** Simulate stock day by day with planned deliveries; check level ≥ safety stock.
- **Outputs:** Plan meeting safety stock, or shortage list (FR-38).
- **Business rules:** BR-05.
- **Alternative flows:** Resources short → FR-38.
- **Exception handling:** None.
- **Acceptance criteria:** With enough resources, no station falls below safety stock in the simulated period.
- **Traceability:** UN-06; RS FR21 → TC-FR-30; AC-08.

FR-31 – Never above tanker capacity (Must) – Changed (D-19)

- **Requirement:** The system shall not assign a tanker more litres than its capacity in a recommended plan.
- **Description:** A tanker has compartments and can carry several fuel types on one trip (BR-31, D-19); its total load across all fuel types and stops ≤ its capacity. Each compartment holds one fuel type per trip and never more than its size (BR-31); the load of a fuel type ≤ the total size of the compartments assigned to it.
- **Actors:** System.
- **Preconditions:** Resources uploaded.
- **Inputs:** Tanker capacity, assigned quantities.
- **Processing:** Constraint in route building; validation before display.
- **Outputs:** Compliant routes.
- **Business rules:** BR-06.
- **Alternative flows:** None.
- **Exception handling:** As FR-29.
- **Acceptance criteria:** Each tanker's total load in any recommended plan is at or below its capacity, and no compartment is above its size or holds two fuel types.
- **Traceability:** UN-06; RS FR22; AI-25 → TC-FR-31; AC-08.

FR-32 – Within delivery windows (Must)

- **Requirement:** The system shall schedule recommended deliveries within each station's delivery window.
- **Description:** Estimated arrival (FR-78) must fall between window open and close.

- **Actors:** System.
- **Preconditions:** Windows defined per station.
- **Inputs:** Windows, travel times, unloading time.
- **Processing:** Time-window constraint in route building.
- **Outputs:** Arrival times inside windows.
- **Business rules:** BR-07.
- **Alternative flows:** No feasible arrival for a station → station listed as unserved with reason (FR-38).
- **Exception handling:** None.
- **Acceptance criteria:** Every estimated arrival in a recommended plan falls inside the station's window.
- **Traceability:** UN-06; RS FR22 → TC-FR-32; AC-08.

FR-33 – Within warehouse stock (Must) – Confirmed (D-24)

- **Requirement:** The system shall not recommend delivering more of a fuel type than the stock of that fuel type at the warehouse serving the stations.
- **Description:** Sum per warehouse and fuel type ≤ uploaded stock.
- **Actors:** System.
- **Preconditions:** Warehouse stock uploaded.
- **Inputs:** Stock per warehouse and fuel type; quantities.
- **Processing:** Allocate stock by priority (FR-25); validate.
- **Outputs:** Compliant totals.
- **Business rules:** BR-08.
- **Alternative flows:** Stock short → lower-priority stations unserved or partly served, with reason (FR-38).
- **Exception handling:** As FR-29.
- **Acceptance criteria:** For each warehouse, the plan total per fuel type is at or below its uploaded stock.
- **Traceability:** UN-06; RS FR12, FR15 → TC-FR-33; AC-08.

FR-34 – Closest warehouse (Must)

- **Requirement:** The system shall serve each station from its closest warehouse.
- **Description:** Closest = shortest road distance from IF-DIST.
- **Actors:** System.
- **Preconditions:** Distances available.
- **Inputs:** Station and warehouse locations; distance table.
- **Processing:** For each station pick the warehouse with minimum distance; ties → lower warehouse ID (Proposed, for determinism, NFR-13).
- **Outputs:** Station → serving warehouse link.
- **Business rules:** BR-17.
- **Alternative flows:** None (loading elsewhere not modelled, AS-06).
- **Exception handling:** Distance missing → ERR-DATA-02, station excluded and listed.
- **Acceptance criteria:** In a test plan, every station is linked to the warehouse with the shortest distance.

- **Traceability:** UN-06; BR-17 → TC-FR-34; AC-08.

FR-35 – One route per tanker, each with a driver (Must) – Changed (D-09, D-19)

- **Requirement:** The system shall assign the stations that need fuel to the available tankers, each with an available driver, and produce one optimal multi-stop route (one trip) per tanker for the delivery day.
- **Description:** A tanker can only serve stations linked to its home warehouse (FR-34, FR-36). Each station is on exactly one route (BR-24); a route may carry several fuel types (BR-31). "Optimal" = the best plan found by the routing module for the objective in AIC-03.
- **Actors:** System (CMP-ROUTE).
- **Preconditions:** Tankers, drivers and stock available.
- **Inputs:** Ranked stations, quantities, tankers, drivers, distances, windows.
- **Processing:** Pair tankers with drivers (Proposed: driver from the same home warehouse, in ID order); assign stations by priority; build and improve routes (method Proposed in AIC-03).
- **Outputs:** Up to N routes, N = number of tanker–driver pairs.
- **Business rules:** BR-06, BR-07, BR-18, BR-23, BR-24, BR-25.
- **Alternative flows:** More tankers than drivers → only paired tankers get routes; unpaired tankers listed as "no driver".
- **Exception handling:** No pair at all → FR-39.
- **Acceptance criteria:** With N tanker–driver pairs, the plan contains at most N routes, each tanker starts at most one trip, every served station is on exactly one route.
- **Traceability:** UN-06; UC-05; US-06; RS FR23, FR24 → TC-FR-35; AC-09.

FR-36 – Routes start and end at the home warehouse (Must) – Changed (D-09)

- **Requirement:** The system shall start each tanker's route at the tanker's home warehouse and end it there.
- **Description:** The return leg is part of the route (BR-25).
- **Actors:** System.
- **Preconditions:** Tanker has a home warehouse.
- **Inputs:** Home warehouse.
- **Processing:** First and last point = home warehouse.
- **Outputs:** Closed route.
- **Business rules:** BR-18, BR-25; AS-06.
- **Alternative flows:** None.
- **Exception handling:** None.
- **Acceptance criteria:** The first and last point of every route is its tanker's home warehouse; totals include the return leg.
- **Traceability:** UN-06; BR-18; AS-06; D-09 → TC-FR-36; AC-09.

FR-37 – Always produce a plan (Must)

- **Requirement:** The system shall always produce a delivery plan when no critical resource is missing, even if resources cannot cover every station.
- **Description:** Critical resources: at least one tanker, one driver, and stock at the serving warehouse.

- **Actors:** System.
- **Preconditions:** No critical resource missing.
- **Inputs:** All planning inputs.
- **Processing:** Plan with available resources; apply FR-38 for shortages.
- **Outputs:** Plan (possibly partial).
- **Business rules:** BR-20.
- **Alternative flows:** Short resources → FR-38.
- **Exception handling:** Critical resource missing → FR-39.
- **Acceptance criteria:** With fewer tankers than needed, a plan is still produced.
- **Traceability:** UN-06; AI-22 → TC-FR-37; AC-09.

FR-38 – Highest priority first; list unserved (Must)

- **Requirement:** The system shall serve the highest-priority stations first when resources are short, and list the stations left unserved with their risks.
- **Description:** The unserved list shows station, fuel type, risk level, time to stockout and the reason (e.g. “no tanker capacity left”).
- **Actors:** System; operations manager.
- **Preconditions:** Resources below demand.
- **Inputs:** Priority ranking; resources.
- **Processing:** Fill tankers in priority order; anything left goes to the unserved list.
- **Outputs:** Plan + unserved list.
- **Business rules:** BR-20; AI-16.
- **Alternative flows:** None.
- **Exception handling:** None.
- **Acceptance criteria:** With resources short, unserved stations are the lowest-priority ones and each shows its risk and reason.
- **Traceability:** UN-06, UN-10; UC-05; AI-22 → TC-FR-38; AC-09.

FR-39 – Stop when a critical resource is missing (Must)

- **Requirement:** The system shall not produce a plan when a critical resource is missing (no tankers, no drivers, or no stock at the serving warehouse) and shall name the missing resource.
- **Description:** “No stock at the serving warehouse” blocks only the stations linked to that warehouse (Proposed); the plan stops entirely only when no station can be served.
- **Actors:** System; operations manager.
- **Preconditions:** Planning requested.
- **Inputs:** Resources for the day.
- **Processing:** Check critical resources before planning.
- **Outputs:** ERR-PLAN-01 naming the missing resource.
- **Business rules:** BR-20.
- **Alternative flows:** No resources uploaded at all → ERR-PLAN-02 “Resources for this day are not uploaded yet.”
- **Exception handling:** As outputs.
- **Acceptance criteria:** Removing all tankers, all drivers or a warehouse's stock stops planning (for the affected stations) with a message naming the missing resource.

- **Traceability:** UN-05, UN-06; UC-05; AI-22 → TC-FR-39; AC-09.

6.9 Route cost

All costs use one currency: **US dollar (USD, $)**, with 3 decimals stored and 2 shown (D-20).

FR-40 – Route fuel cost (Must)

- **Requirement:** The system shall calculate the fuel cost of each route from its distance, the tanker's fuel consumption rate and the fuel price.
- **Description:** Covers only the tanker's own fuel (BR-16), including the return leg (BR-25).
- **Actors:** System (CMP-COST).
- **Preconditions:** Route, consumption rate and fuel price available.
- **Inputs:** Route distance (km); consumption rate (L/km); fuel price (USD/L, public price, D-20).
- **Processing:** Cost = distance × consumption rate × fuel price.
- **Outputs:** Cost per route.
- **Business rules:** BR-16, BR-25.
- **Alternative flows:** None.
- **Exception handling:** Consumption rate or fuel price missing → ERR-DATA-03, cost shown as “not available” and the plan flagged.
- **Acceptance criteria:** For a test route, displayed cost = distance × consumption rate × fuel price, to $0.01.
- **Traceability:** UN-07; US-07; RS FR25 → TC-FR-40; AC-16. **Unresolved:** who provides tanker consumption rates (OQ-04; the fuel price is public, D-20).

FR-41 – Cost in route optimization (Must)

- **Requirement:** The system shall include route cost in route optimization together with urgency, delivery windows and stockout risk.
- **Description:** Hard constraints first (BR-04 to BR-08, BR-24), then urgency and stockout risk, then cost.
- **Actors:** System.
- **Preconditions:** Costs computable.
- **Inputs:** Candidate plans.
- **Processing:** Compare feasible plans lexicographically: (1) fewer projected stockouts, (2) higher-priority stations served, (3) lower total cost (Proposed order).
- **Outputs:** Recommended plan.
- **Business rules:** BR-04 to BR-08.
- **Alternative flows:** None.
- **Exception handling:** None.
- **Acceptance criteria:** Between two feasible plans with equal risk, the recommended one has the lower cost.
- **Traceability:** UN-07; RS FR26 → TC-FR-41; AC-16.

FR-42 – Show route and total cost (Must)

- **Requirement:** The system shall display the cost of each route and the total cost of all routes.
- **Description:** On the planner's route summary.

- **Actors:** Operations manager.
- **Preconditions:** Plan produced.
- **Inputs:** Route costs.
- **Processing:** Total = sum of route costs.
- **Outputs:** Cost per route and plan total.
- **Business rules:** None specific.
- **Alternative flows:** None.
- **Exception handling:** As FR-40.
- **Acceptance criteria:** The planner shows each route cost and a total equal to their sum.
- **Traceability:** UN-07; RS FR27 → TC-FR-42; AC-16.

FR-43 – Optimal vs. edited cost (Must)

- **Requirement:** The system shall display, for an edited plan, the optimal cost, the new cost and the difference in amount and percentage.
- **Description:** Difference = new − optimal; % = difference ÷ optimal × 100, one decimal; positive = more expensive.
- **Actors:** Operations manager.
- **Preconditions:** Plan edited.
- **Inputs:** Optimal and edited totals.
- **Processing:** As description.
- **Outputs:** Three values in the comparison panel.
- **Business rules:** None specific.
- **Alternative flows:** No edit → panel shows “Optimal plan, no changes”.
- **Exception handling:** None.
- **Acceptance criteria:** After an edit, the three values appear and the difference is correct.
- **Traceability:** UN-07, UN-10; RS FR28 → TC-FR-43; AC-16.

FR-44 – Saving and risks shown together (Must)

- **Requirement:** The system shall show the cost change and the risks together when an edit lowers cost but creates risks.
- **Description:** Prevents a saving from hiding a risk.
- **Actors:** Operations manager.
- **Preconditions:** Edited plan cheaper than optimal with ≥ 1 risk.
- **Inputs:** Cost difference; risk list.
- **Processing:** Show both in the same panel.
- **Outputs:** Saving and risk list side by side.
- **Business rules:** None specific.
- **Alternative flows:** None.
- **Exception handling:** None.
- **Acceptance criteria:** A cheaper but risky edit shows both the saving and the risk list.
- **Traceability:** UN-10; RS FR29 → TC-FR-44; AC-12.

6.10 Route map

All route-map requirements share these fields: **Actors:** operations manager. **Preconditions:** a plan (optimal or edited) is open in the planner. **Alternative flows:** none unless stated. **Exception handling:** map tiles unavailable → routes still drawn on a plain background with ERR-SYS-03 notice (Proposed).

FR-45 – Directed route paths (Must)

- **Requirement:** The system shall display each route on the map as a directed path with arrows.
- **Description:** Arrows show the driving direction from stop to stop and back to the warehouse.
- **Inputs:** Route stop order and coordinates.
- **Processing:** Draw a polyline per route with arrowheads on each segment.
- **Outputs:** Directed paths.
- **Business rules:** BR-10, BR-25.
- **Acceptance criteria:** Arrows on every route follow the stop order, including the return leg.
- **Traceability:** UN-08; US-08; RS FR30 → TC-FR-45; AC-10.

FR-46 – Stop numbers (Must)

- **Requirement:** The system shall number each station by its stop order in its route.
- **Description:** 1 = first stop after leaving the warehouse.
- **Inputs:** Stop order.
- **Processing:** Label each station marker with its number; renumber after every edit (FR-55).
- **Outputs:** Numbered markers.
- **Business rules:** None specific.
- **Acceptance criteria:** Numbers match the stop order in the plan.
- **Traceability:** UN-08; RS FR31 → TC-FR-46; AC-10.

FR-47 – Urgency colors (Must)

- **Requirement:** The system shall color each station marker by urgency: red for high, orange for medium, green for low.
- **Description:** A station with several fuel types takes its highest risk level.
- **Inputs:** Station risk levels.
- **Processing:** Map level to color (BR-09).
- **Outputs:** Colored markers.
- **Business rules:** BR-09.
- **Acceptance criteria:** Marker colors match the urgency levels in the data.
- **Traceability:** UN-08; RS FR32 → TC-FR-47; AC-18.

FR-48 – Urgency not by color alone (Must)

- **Requirement:** The system shall show urgency with a label or icon in addition to color.
- **Description:** Proposed icons: triangle-with-exclamation (high), circle-with-exclamation (medium), check (low), plus the text “High / Medium / Low” in details and lists.
- **Inputs:** Risk level.
- **Processing:** Add icon to marker; add label in lists.
- **Outputs:** Icon + label.
- **Business rules:** NFR-10.

- **Acceptance criteria:** Urgency can be identified on a grayscale screenshot.
- **Traceability:** UN-08; RS NFR9 → TC-FR-48; AC-18.

FR-49 – Clean map by default (Must)

- **Requirement:** The system shall show only the station name and ID on the map by default.
- **Description:** No quantities, times or costs on the map until a station is selected.
- **Inputs:** Station name and ID.
- **Processing:** Default marker label only.
- **Outputs:** Uncluttered map.
- **Business rules:** None specific.
- **Acceptance criteria:** No quantities are visible until a station is selected.
- **Traceability:** UN-08; US-08; RS FR33 → TC-FR-49; AC-10.

FR-50 – Station details box (Must)

- **Requirement:** The system shall show a details box next to a station when the user selects it, with its stop order and litres per fuel type.
- **Description:** Also shows tanker, estimated arrival, risk level and the uncertain flag if any. Selection = click.
- **Inputs:** Selected station.
- **Processing:** Open box next to the marker; close on click elsewhere or on Esc.
- **Outputs:** Details box.
- **Business rules:** None specific.
- **Alternative flows:** Selecting another station moves the box to it.
- **Acceptance criteria:** Selecting a station opens the box with correct values; selecting elsewhere closes it.
- **Traceability:** UN-08; RS FR34; AS-12 → TC-FR-50; AC-10.

FR-51 – Segment labels (Must)

- **Requirement:** The system shall label each route segment with its distance and travel time.
- **Description:** Format “24 km · 32 min”. Proposed: shown when the route is selected or zoomed in, to keep FR-49's clean default.
- **Inputs:** Segment distance and time (IF-DIST).
- **Processing:** Place a label at each segment's midpoint.
- **Outputs:** Segment labels.
- **Business rules:** None specific.
- **Acceptance criteria:** Every segment has a distance and time label.
- **Traceability:** UN-08; RS FR35 → TC-FR-51; AC-10.

FR-52 – Route and plan totals (Must)

- **Requirement:** The system shall display the total distance, duration, litres and cost of each route and of the whole plan.
- **Description:** Duration includes unloading and the return leg.
- **Inputs:** Segments, stops, costs.
- **Processing:** Sum per route; sum routes for the plan.

- **Outputs:** Route summary and plan totals.
- **Business rules:** BR-25.
- **Acceptance criteria:** Totals equal the sum of the segments and stops.
- **Traceability:** UN-08; RS FR36 → TC-FR-52; AC-16.

FR-53 – One line color per tanker (Must)

- **Requirement:** The system shall draw each tanker's route in its own line color, never red, orange or green.
- **Description:** Proposed palette: blue, purple, teal-blue, brown, gray, dark blue, magenta, black (8 colors; beyond 8, dashed patterns are added).
- **Inputs:** Tanker list.
- **Processing:** Assign colors in tanker ID order.
- **Outputs:** Distinct route colors plus a legend with tanker IDs.
- **Business rules:** BR-10.
- **Acceptance criteria:** Each route has a distinct color and none uses the urgency colors.
- **Traceability:** UN-08; RS FR37 → TC-FR-53; AC-10.

FR-54 – Tanker filter (Must)

- **Requirement:** The system shall allow the user to show all routes or only one tanker's route.
- **Description:** A filter control with “All routes” and one entry per tanker.
- **Inputs:** Filter choice.
- **Processing:** Hide other routes and their stops.
- **Outputs:** Filtered map.
- **Business rules:** None specific.
- **Acceptance criteria:** Selecting a tanker hides the other routes; “All routes” shows them again.
- **Traceability:** UN-08; RS FR37 → TC-FR-54; AC-10.

6.11 Route and quantity editing

The manager can change three things in a plan: stop order, a station's tanker, and delivery quantities. Resources stay as the dispatcher uploaded them (BR-22).

FR-55 – Reorder stops by drag and drop (Must)

- **Requirement:** The system shall allow the manager to reorder stops within a route by drag and drop.
- **Description:** Works on the map and in the route stop list. Keyboard alternative: move up / move down buttons (NFR-27).
- **Actors:** Operations manager.
- **Preconditions:** Draft plan open; plan not decided.
- **Inputs:** Route, stop, new position.
- **Processing:** Update stop order; renumber (FR-46); trigger FR-57.
- **Outputs:** Edited plan with new numbering.
- **Business rules:** BR-12, BR-22.
- **Alternative flows:** Drop on the original position → no change.
- **Exception handling:** Plan decided → editing disabled (ERR-VAL-04).

- **Acceptance criteria:** Dragging a stop to a new position changes the stop numbers accordingly.
- **Traceability:** UN-09; UC-06; US-09; RS FR38 → TC-FR-55; AC-10.

FR-56 – Move a station to another tanker (Must) – Changed (D-02)

- **Requirement:** The system shall allow the manager to move a station from one tanker's route to another by drag and drop.
- **Description:** The move is allowed even if it overloads the receiving tanker; the overload then raises a high-severity "capacity exceeded" risk and blocks approval (FR-77). Only routes of tankers from the station's serving warehouse accept it (BR-17, AS-06).
- **Actors:** Operations manager.
- **Preconditions:** As FR-55; at least two routes from the same warehouse.
- **Inputs:** Station, source route, target route, position.
- **Processing:** Remove from source; insert into target; trigger FR-57 and FR-61.
- **Outputs:** Edited plan.
- **Business rules:** BR-12, BR-17, BR-21, BR-24.
- **Alternative flows:** Target route belongs to another warehouse → drop refused with WARN-EDIT-01 "This tanker loads at another warehouse."
- **Exception handling:** As FR-55.
- **Acceptance criteria:** The station disappears from the first route and appears in the second; an overloading move is accepted and raises "capacity exceeded".
- **Traceability:** UN-09; UC-06; US-09; RS FR39; D-02 → TC-FR-56; AC-10, AC-13.

FR-57 – Live recalculation (Must)

- **Requirement:** The system shall recalculate the duration, distance, arrival time per station and cost of every affected route after a reorder or a move.
- **Description:** Recalculation only – no re-optimization (BR-12).
- **Actors:** System (CMP-EDIT).
- **Preconditions:** Edit made.
- **Inputs:** Edited routes; distances; windows.
- **Processing:** Recompute FR-78 times, FR-52 totals and FR-40 costs for affected routes.
- **Outputs:** Updated values on map, summary and comparison panel.
- **Business rules:** BR-12, BR-25.
- **Alternative flows:** None.
- **Exception handling:** Recalculation fails → ERR-SYS-01, edit undone, message shown.
- **Acceptance criteria:** Updated values appear within NFR-01 (≤ 1 s for 95% of edits) and are correct for a test case.
- **Traceability:** UN-09, UN-10; UC-06; US-10; RS FR39, FR41 → TC-FR-57; AC-10.

FR-58 – Edit delivery quantities (Should) – Changed (D-05)

- **Requirement:** The system shall allow the manager to edit the delivery quantity per station and fuel type in a plan.
- **Description:** The manager is responsible for station orders. Editing "stations included" and "tankers used" is removed (D-05). Proposed: a quantity of 0 removes that fuel type from the stop; 0 for all fuel types removes the station from the route.

- **Actors:** Operations manager.
- **Preconditions:** Draft plan; plan not decided.
- **Inputs:** Station, fuel type, new quantity (L, whole number ≥ 0).
- **Processing:** Validate against BR-28 (FR-75, FR-76); save in the draft; trigger FR-59.
- **Outputs:** Draft with edited quantity.
- **Business rules:** BR-12, BR-22, BR-28.
- **Alternative flows:** Cancel → previous value kept.
- **Exception handling:** Above the limit → FR-76.
- **Acceptance criteria:** An edited value within the limit is saved in the draft plan.
- **Traceability:** UN-09; RS FR40; D-05 → TC-FR-58; AC-11.

FR-59 – New optimal plan after quantity edit (Should) – Changed (D-05)

- **Requirement:** The system shall generate a new optimal plan when quantities are edited.
- **Description:** Edited quantities are kept fixed; routes are rebuilt. Proposed: if the manager has reordered or moved stops, a confirmation asks before those route edits are replaced.
- **Actors:** System.
- **Preconditions:** FR-58 saved.
- **Inputs:** Edited quantities + all planning inputs.
- **Processing:** Re-run planning (FR-28 to FR-41) with fixed quantities.
- **Outputs:** New optimal plan; the comparison baseline becomes this plan.
- **Business rules:** BR-04 to BR-08, BR-12.
- **Alternative flows:** Manager declines the confirmation → quantity edit undone.
- **Exception handling:** No feasible plan with the fixed quantities → ERR-PLAN-03 naming the blocking constraint.
- **Acceptance criteria:** After a quantity edit, a new plan is produced that respects BR-04 to BR-08.
- **Traceability:** UN-09; RS FR42 → TC-FR-59; AC-11.

FR-60 – Reset to optimal (Must)

- **Requirement:** The system shall provide a “Reset to optimal” action that restores the recommended plan.
- **Description:** Removes all route edits since the last optimal plan.
- **Actors:** Operations manager.
- **Preconditions:** Plan edited; not decided.
- **Inputs:** Reset request.
- **Processing:** Replace the draft with the stored optimal plan; clear risks.
- **Outputs:** Optimal plan shown.
- **Business rules:** None specific.
- **Alternative flows:** None.
- **Exception handling:** None.
- **Acceptance criteria:** After reset, the plan matches the original recommendation exactly.
- **Traceability:** UN-09; US-12; RS FR43 → TC-FR-60; AC-10.

FR-75 – Show the quantity limit while editing (Should) – New (D-06)

- **Requirement:** The system shall display, under the quantity input, the maximum quantity allowed for that stop and which limit sets it.
- **Description:** Maximum = smallest of the free space in the tanker's compartments for that fuel type, the station's free storage and the remaining warehouse stock of that fuel (BR-28).
- **Actors:** Operations manager.
- **Preconditions:** Quantity input open (FR-58).
- **Inputs:** Tanker capacity and load of other stops; tank capacity and level; warehouse stock and quantities already planned.
- **Processing:** Compute the three limits; show the smallest and its source; update as the manager types.
- **Outputs:** Hint, e.g. “Max for this stop: 8,000 L. Tanker T-01 has 8,000 L left (36,000 L capacity, 28,000 L already assigned).”
- **Business rules:** BR-28.
- **Alternative flows:** None.
- **Exception handling:** A limit not computable → hint names the missing data and save is disabled.
- **Acceptance criteria:** For each of three test cases (tanker, tank, warehouse as binding limit), the hint shows the correct maximum and names the correct limit.
- **Traceability:** D-06; UN-09, UN-10 → TC-FR-75; AC-11.

FR-76 – Block save above the limit (Should) – New (D-06)

- **Requirement:** The system shall disable the Save button and show an error message directly under the quantity input when the entered quantity exceeds the maximum allowed.
- **Description:** The message names the binding limit and the maximum.
- **Actors:** Operations manager.
- **Preconditions:** Quantity input open.
- **Inputs:** Entered quantity; maximum (FR-75).
- **Processing:** If quantity > maximum → disable Save; show the matching message (ERR-VAL-05a/b/c). Server re-checks on save.
- **Outputs:** One of: “This quantity exceeds the space left on tanker {T-ID}. Enter {max} L or less.” / “This quantity exceeds the free space in the station's tank. Enter {max} L or less.” / “This quantity exceeds the {fuel} stock at {warehouse}. Enter {max} L or less.”
- **Business rules:** BR-28.
- **Alternative flows:** Value corrected to ≤ maximum → message disappears, Save enabled.
- **Exception handling:** A forced request above the limit → server refuses with the same code.
- **Acceptance criteria:** Entering maximum + 1 L disables Save and shows the correct message; entering the maximum enables Save; a direct API request above the limit is refused.
- **Traceability:** D-06; UN-09 → TC-FR-76; AC-11.

6.12 Risk analysis

Risk types and severity (adopted for v1, D-18):

Risk type	Raised when (edited vs. optimal plan)	Low	Medium	High	Blocks approval
RISK-01 Capacity exceeded	A tanker's load > its capacity, or a compartment over its size	–	–	Always	**Yes** (BR-21)
RISK-02 Stockout before arrival	Projected level reaches 0 before the stop's arrival time	–	–	Always	No
RISK-03 Missed delivery window	Arrival outside the station's window	–	–	Always	No
RISK-04 Late arrival	Arrival later than in the optimal plan	≤ 60 min	61–180 min	> 180 min	No
RISK-05 Extra distance	Plan distance above optimal	< 10%	10–25%	> 25%	No
RISK-06 Extra time	Plan duration above optimal	< 10%	10–25%	> 25%	No
RISK-07 Extra cost	Plan cost above optimal	< 10%	10–25%	> 25%	No

FR-61 – Compare edited vs. optimal (Must)

- **Requirement:** The system shall compare the edited plan with the optimal plan after each edit.
- **Description:** Comparison covers distance, duration, cost, arrival per station and all risk types above.
- **Actors:** System (CMP-EDIT); operations manager.
- **Preconditions:** Edit made.
- **Inputs:** Optimal plan; edited plan.
- **Processing:** Compute differences; run risk checks (FR-62).
- **Outputs:** Comparison panel.
- **Business rules:** None specific.
- **Alternative flows:** No difference → FR-66.
- **Exception handling:** As FR-57.
- **Acceptance criteria:** The comparison panel updates after every edit, within NFR-01.
- **Traceability:** UN-10; UC-06; US-11; RS FR44 → TC-FR-61; AC-12.

FR-62 – Detect risks (Must)

- **Requirement:** The system shall detect these risks: late arrival compared with the optimal plan, stockout before arrival, missed delivery window, tanker capacity exceeded, and extra distance, time or cost.
- **Description:** RISK-01 to RISK-07.

- **Actors:** System.
- **Preconditions:** Comparison computed.
- **Inputs:** Edited plan, optimal plan, forecasts, windows, capacities.
- **Processing:** Apply each risk rule per stop, route and plan.
- **Outputs:** Risk list.
- **Business rules:** BR-06, BR-07, BR-21.
- **Alternative flows:** None.
- **Exception handling:** None.
- **Acceptance criteria:** Each risk type is raised in a test scenario built to trigger it, and not raised otherwise.
- **Traceability:** UN-10; UC-06; RS FR45 → TC-FR-62; AC-12.

FR-63 – Severity per risk (Must)

- **Requirement:** The system shall give each risk a severity level: low, medium or high.
- **Description:** Thresholds as in the table above (D-18).
- **Actors:** System.
- **Preconditions:** Risk detected.
- **Inputs:** Risk value.
- **Processing:** Apply thresholds.
- **Outputs:** Severity with label and icon (NFR-10).
- **Business rules:** Severity table above (D-18).
- **Alternative flows:** None.
- **Exception handling:** None.
- **Acceptance criteria:** Every risk shown has exactly one severity level; boundary values (60/61 min, 10%/25%) map as the table states.
- **Traceability:** UN-10; RS FR46; D-18 → TC-FR-63; AC-12.

FR-64 – Reason per risk (Must)

- **Requirement:** The system shall give each risk a reason, including why the edited version was not chosen as optimal.
- **Description:** Template explanation, e.g. “Moving ST032 to Tanker 02 adds 38 km (+12%) and makes ST014 arrive 95 min later than planned.”
- **Actors:** System (CMP-EXPL).
- **Preconditions:** Risk detected.
- **Inputs:** Risk data.
- **Processing:** Fill risk template with data values only (AI-12).
- **Outputs:** Reason text naming the affected station, route or value.
- **Business rules:** AI-12, AI-15.
- **Alternative flows:** None.
- **Exception handling:** None.
- **Acceptance criteria:** Every risk has a reason that names the affected station, route or value, and every number matches the data.
- **Traceability:** UN-10; US-11; RS FR46, FR47 → TC-FR-64; AC-07.

FR-65 – Highlight affected elements (Must)

- **Requirement:** The system shall highlight the stations and segments affected by each risk on the map.
- **Description:** Highlight = thicker outline and pulsing halo (Proposed), not color only.
- **Actors:** Operations manager.
- **Preconditions:** Risk list shown.
- **Inputs:** Selected risk.
- **Processing:** Highlight its elements; dim others.
- **Outputs:** Highlighted map elements.
- **Business rules:** NFR-10.
- **Alternative flows:** Deselect → normal map.
- **Exception handling:** None.
- **Acceptance criteria:** Selecting a risk highlights the matching elements on the map.
- **Traceability:** UN-10; UC-06; RS FR48 → TC-FR-65; AC-12.

FR-66 – "No risk" message (Must)

- **Requirement:** The system shall state clearly when an edit creates no risk.
- **Description:** Message "No risk: this change keeps all deliveries on time and within limits."
- **Actors:** Operations manager.
- **Preconditions:** Edit made; risk list empty.
- **Inputs:** Empty risk list.
- **Processing:** Show message.
- **Outputs:** Explicit no-risk message.
- **Business rules:** None specific.
- **Alternative flows:** None.
- **Exception handling:** None.
- **Acceptance criteria:** A no-risk edit shows an explicit "no risk" message.
- **Traceability:** UN-10; RS FR50 → TC-FR-66; AC-12.

FR-67 – Approval allowed with warnings (Must) – Changed (D-02)

- **Requirement:** The system shall allow approval of a plan that has risk warnings, except a plan with a "capacity exceeded" risk (FR-77).
- **Description:** RISK-02 to RISK-07 are warnings; the manager can approve knowingly.
- **Actors:** Operations manager.
- **Preconditions:** Plan with ≥ 1 warning, no RISK-01.
- **Inputs:** Approval request.
- **Processing:** Store approval with the risks shown (FR-72).
- **Outputs:** Approved actions.
- **Business rules:** BR-11 (Changed), BR-21.
- **Alternative flows:** None.
- **Exception handling:** RISK-01 present → FR-77.
- **Acceptance criteria:** A plan with medium- or high-severity warnings (other than RISK-01) can be approved and its risks are logged.
- **Traceability:** UN-09; RS FR49; D-02 → TC-FR-67; AC-14.

FR-77 – Block approval on capacity exceeded (Must) – New (D-02)

- **Requirement:** The system shall block approval of the whole plan and of any action on an overloaded tanker while any tanker load exceeds its capacity, and shall show a warning naming the tanker, its load and its capacity.
- **Description:** The overload can be removed by moving a station back, moving a station to another tanker, lowering a quantity, or Reset to optimal.
- **Actors:** System; operations manager.
- **Preconditions:** RISK-01 present.
- **Inputs:** Tanker loads and capacities.
- **Processing:** Disable Approve and Approve all for affected actions and the plan; server refuses approval requests while RISK-01 exists.
- **Outputs:** Warning ERR-PLAN-04: "Tanker {T-ID} is overloaded: {load} L on a {capacity} L tanker. Remove {excess} L before approving."
- **Business rules:** BR-21.
- **Alternative flows:** Overload removed → approval enabled again.
- **Exception handling:** Forced approval request → refused with ERR-PLAN-04.
- **Acceptance criteria:** With a test plan overloaded by 1 L, Approve and Approve all are disabled and a direct API approval is refused; after removing the overload, approval works.
- **Traceability:** D-02; UN-11 → TC-FR-77; AC-13.

6.13 Recommendations and approval

Approval works at two levels (D-03): each recommended action, and the whole plan in one step. A plan is **decided** when every action is approved or rejected; the day's resources then lock (BR-30).

FR-68 – List recommended actions (Must)

- **Requirement:** The system shall list all recommended actions for the delivery day.
- **Description:** One row per action: station, fuel types and litres, tanker, stop order, estimated arrival, priority reason, forecast confidence and uncertain flag, risk level, risks from edits, status.
- **Actors:** Operations manager.
- **Preconditions:** A draft plan exists (UC-05).
- **Inputs:** Current plan.
- **Processing:** Build list in priority order.
- **Outputs:** Recommendations page list, with route totals and plan cost.
- **Business rules:** AI-11.
- **Alternative flows:** No plan yet → link to the planner.
- **Exception handling:** None.
- **Acceptance criteria:** The list contains every station delivery and route of the plan.
- **Traceability:** UN-11; UC-07; US-13; RS FR51 → TC-FR-68; AC-14.

FR-69 – Approve or reject each action (Must)

- **Requirement:** The system shall allow the manager to approve or reject each recommended action.

- **Description:** Status per action: pending → approved or rejected (Section 8.6). Proposed: an optional note on rejection.
- **Actors:** Operations manager.
- **Preconditions:** Action pending; no RISK-01 on its tanker (FR-77).
- **Inputs:** Action ID; decision; optional note.
- **Processing:** Store decision (FR-72); apply FR-82 for rejections; check whether the plan is now decided (BR-30).
- **Outputs:** Updated status.
- **Business rules:** BR-01, BR-02, BR-21, BR-30.
- **Alternative flows:** Adjust before deciding → FR-81.
- **Exception handling:** RISK-01 → ERR-PLAN-04; already decided → ERR-VAL-04.
- **Acceptance criteria:** Each action can be approved or rejected and shows its status.
- **Traceability:** UN-11; UC-07; US-13; RS FR52; D-03 → TC-FR-69; AC-14.

FR-70 – Approve or reject an edited plan (Must)

- **Requirement:** The system shall allow the manager to approve or reject an edited plan.
- **Description:** The final edited version is stored together with the original recommendation (FR-72).
- **Actors:** Operations manager.
- **Preconditions:** Edited plan; no RISK-01.
- **Inputs:** Plan decision.
- **Processing:** As FR-80.
- **Outputs:** Decided plan; stored final version.
- **Business rules:** BR-01, BR-21, BR-30.
- **Alternative flows:** None.
- **Exception handling:** As FR-69.
- **Acceptance criteria:** An edited plan can be approved and its final version is stored.
- **Traceability:** UN-11; UC-07; RS FR52 → TC-FR-70; AC-14.

FR-71 – Nothing final without approval (Must)

- **Requirement:** The system shall not mark any action as final unless a manager has approved it.
- **Description:** No automatic approval, no timeout approval, no system user approval.
- **Actors:** System.
- **Preconditions:** None.
- **Inputs:** Any status change.
- **Processing:** “Approved” is only set by a decision request from a manager session, with a log entry in the same transaction.
- **Outputs:** None.
- **Business rules:** BR-01; AI-04, AI-05, AI-23.
- **Alternative flows:** None.
- **Exception handling:** Log write fails → decision not stored (ERR-SYS-04).
- **Acceptance criteria:** No action reaches “approved” status without a manager approval record.
- **Traceability:** UN-11; RS FR53 → TC-FR-71; AC-14.

FR-80 – Approve or reject the whole plan (Must) – New (D-03)

- **Requirement:** The system shall allow the manager to approve all pending actions, or reject all pending actions, in one step.
- **Description:** A confirmation shows the number of actions, total litres, total cost and any warnings before the step is applied.
- **Actors:** Operations manager.
- **Preconditions:** ≥ 1 pending action; no RISK-01 for “approve all”.
- **Inputs:** Plan ID; decision (approve all / reject all).
- **Processing:** Apply the decision to each pending action; one log entry per action plus one plan-level entry (FR-72); plan becomes decided.
- **Outputs:** All actions decided; resources locked.
- **Business rules:** BR-01, BR-21, BR-30.
- **Alternative flows:** Cancel in the confirmation → nothing changes.
- **Exception handling:** RISK-01 → ERR-PLAN-04.
- **Acceptance criteria:** “Approve all” on a 5-action plan sets all 5 to approved, creates 6 log entries and locks the day's resources.
- **Traceability:** D-03; UN-11 → TC-FR-80; AC-14.

FR-81 – Adjust opens the planner (Must) – New (D-04)

- **Requirement:** The system shall make the “Adjust” control of an action open the delivery planner with that station selected.
- **Description:** One editing system only.
- **Actors:** Operations manager.
- **Preconditions:** Action pending.
- **Inputs:** Action ID.
- **Processing:** Navigate to the planner; select the station, open its details box and its route.
- **Outputs:** Planner focused on the station.
- **Business rules:** BR-22.
- **Alternative flows:** Returning to the recommendations page shows the recalculated values and risks.
- **Exception handling:** None.
- **Acceptance criteria:** Clicking Adjust on an action opens the planner with that station selected and its route visible.
- **Traceability:** D-04; UC-07 → TC-FR-81; AC-14.

FR-82 – Effect of a rejected action (Must) – New (D-23)

- **Requirement:** When the manager rejects a station delivery, the system shall generate a new optimal plan for the delivery day without that station, and show the station as “not dispatched” with its risk.
- **Description:** The new plan can give the freed tanker capacity to other stations, which matters most when a high-priority station was left unserved (D-23). Before the rejection applies, a confirmation shows the station's risk, e.g. “ST014 is high risk: rejecting it leaves it unserved today.”
- **Actors:** Operations manager; system.

- **Preconditions:** Action pending; plan not decided.
- **Inputs:** Rejected action; current plan; all planning inputs.
- **Processing:** Exclude the station for this delivery day; re-run planning (FR-28 to FR-41); compare the new plan with the previous one. Actions whose station, quantities, tanker and arrival time are unchanged keep their approval (D-23); all others return to pending.
- **Outputs:** New optimal plan; “not dispatched” entry; list of actions returned to pending.
- **Business rules:** BR-01, BR-12, BR-30.
- **Alternative flows:** Manager cancels the confirmation → nothing changes. All actions rejected → plan decided with no trips.
- **Exception handling:** No feasible new plan → ERR-PLAN-03; the previous plan is kept and the rejection is not applied.
- **Acceptance criteria:** Rejecting one station produces a new plan without it within NFR-19; freed capacity serves a previously unserved station when one exists; unchanged approved actions stay approved and changed ones return to pending.
- **Traceability:** UN-11; FR-69; D-23 → TC-FR-82; AC-14.

6.14 Decision history and data

FR-72 – Record every decision (Must)

- **Requirement:** The system shall record each approval and rejection with the user, date and time, the original recommendation, whether the approved plan is the suggested plan or an edited version, the edits performed, the final version and the risks shown.
- **Description:** Also records the plan's input versions (data, resources, model versions; AI-18), the cost difference vs. optimal, and the optional note. Every manager override is logged with the recommendation it replaced (AI-07).
- **Actors:** System (CMP-DEC).
- **Preconditions:** A decision is submitted.
- **Inputs:** Decision; plan state.
- **Processing:** Write one append-only entry per action decision, plus one per whole-plan decision (FR-80), in the same transaction as the status change (FR-71).
- **Outputs:** Log entry (fields in Section 8, DE-17).
- **Business rules:** BR-13; AS-11.
- **Alternative flows:** None.
- **Exception handling:** Write fails → decision not applied (ERR-SYS-04).
- **Acceptance criteria:** Each decision creates one complete log entry, including whether the plan was edited and which edits were made.
- **Traceability:** UN-12; UC-07; US-14; RS FR54; AI-07 → TC-FR-72; AC-15.

FR-73 – View decision history (Must) – Changed (D-10)

- **Requirement:** The system shall allow users of all three roles to view the decision history in chronological order, newest first, read-only.
- **Description:** List with filters (date range, action type, station, manager) and a detail view showing original recommendation, final version, edits and risks (Proposed, inspired by R4).
- **Actors:** Operations manager, dispatcher, system admin.

- **Preconditions:** Logged in.
- **Inputs:** Filters (optional).
- **Processing:** Query log; no edit or delete operation exists (NFR-14).
- **Outputs:** Read-only list and detail view.
- **Business rules:** BR-13.
- **Alternative flows:** No decisions yet → "No decisions recorded yet."
- **Exception handling:** None.
- **Acceptance criteria:** Past decisions are listed newest first with all recorded fields; all three roles can open the page; no role sees edit or delete controls, and such requests are refused.
- **Traceability:** UN-12; UC-08; US-14; RS FR55; D-10 → TC-FR-73; AC-15.

FR-74 – Run on sample data (Must)

- **Requirement:** The system shall run on a sample data set with simulated forecasts, routes and costs.
- **Description:** Sample set size: 20 stations, 3 warehouses, 6 tankers, 6 drivers, 2 fuel types (Petrol 95, Diesel), 365 days of daily sales history per station and fuel type. It covers several seasons, station sizes and locations (AI-17).
- **Actors:** Developer; system.
- **Preconditions:** None.
- **Inputs:** Sample data files; simulated providers (weather, events, distances).
- **Processing:** Load sample data at setup; simulated modules are deterministic for the same inputs (NFR-13).
- **Outputs:** A complete demo environment.
- **Business rules:** CO-02.
- **Alternative flows:** Reset demo data to its initial state (Proposed, admin-only command outside the UI).
- **Exception handling:** Invalid sample file → setup stops with the file and row named.
- **Acceptance criteria:** All Must requirements can be demonstrated without real distributor data.
- **Traceability:** UN-16; RS FR56; AS-02, AS-03 → TC-FR-74; AC-20.

7. Non-Functional Specification

Every NFR below has a measurable target and a verification method. All 31 are in the Elicitation document; NFR-19 to NFR-31 cover the categories added during review.

Reference data set: 20 stations, 3 warehouses, 6 tankers, 6 drivers, 2 fuel types, 365 days of history.
Reference environment (Proposed): a laptop with 4 CPU cores and 8 GB RAM running server and database, latest Chrome. All performance targets are measured on these.

ID	Category	Requirement	Measurable target	Verification	Status
NFR-01	Performance	Show recalculated route values after a reorder	95% of edits update in ≤ 1 s; 100% in ≤ 2 s	TC-PERF-01: 100 scripted edits, timed from drop to	Valid (data size Proposed)

ID	Category	Requirement	Measurable target	Verification	Status
		or move		render	
NFR-02	Performance	Load the station forecast page and the delivery plan	95% of loads ≤ 3 s	TC-PERF-02: 50 loads each	Valid
NFR-19	Performance	Generate a full optimal plan	≤ 10 s on the reference data set	TC-PERF-03: 20 runs	Approved (D-24)
NFR-20	Performance	Load the network overview and the decision history	95% of loads ≤ 3 s	TC-PERF-02	Proposed
NFR-03	Availability	Available during planned test and demo sessions	0 unplanned downtime during scheduled sessions; production target **Unresolved** (OQ-17)	Session log	Valid
NFR-04	Scalability	Add stations, warehouses, tankers and drivers through data, without code changes	100% of additions usable in planning with no code change	TC-SCAL-01	Valid
NFR-24	Scalability	Handle a larger network	100 stations, 10 warehouses, 30 tankers: plan generation ≤ 30 s; edits ≤ 2 s (95%)	TC-PERF-04	Proposed
NFR-05	Security	Role-based access on every function and data request	100% of forbidden requests refused (403)	TC-SEC-01: matrix test	Valid
NFR-06	Security	Passwords not stored in readable form	Salted hash (Proposed: bcrypt, cost ≥ 10); 0 plain-text passwords	TC-SEC-02: database inspection	Valid

ID	Category	Requirement	Measurable target	Verification	Status
NFR-21	Security	Limit login guessing	Lock after 5 failed attempts in 15 min, for 15 min	TC-SEC-03	Proposed
NFR-22	Security	Session expiry	Expire after 30 min without activity; logout ends session at once	TC-SEC-04	Approved (D-24)
NFR-23	Security	Encrypted transport	All traffic over HTTPS (TLS 1.2+) on any non-local deployment	TC-SEC-05	Proposed
NFR-07	Privacy	Personal data only for user accounts (name, login, role) and driver name and ID (Changed, D-22)	0 personal fields outside DE-01 (User) and DE-09 (Driver name, operational data only)	Data model review	Valid, see note
NFR-25	Privacy	No sensitive data in logs	0 passwords, tokens or session IDs in logs; users referenced by ID	TC-SEC-06: log scan	Proposed
NFR-13	Reliability	Same results for the same inputs	2 runs on identical inputs give identical forecasts, priorities, routes and costs (byte-equal output)	TC-REL-01	Valid
NFR-14	Reliability / Auditability	Decision log entries cannot be edited or deleted	0 update/delete paths for any role; database blocks UPDATE/DELETE on the log table	TC-SEC-07	Valid
NFR-26	Reliability	Atomic saves	A failed save leaves the previous data unchanged in 100% of fault-injection tests	TC-REL-02	Proposed
NFR-12	Maintainability	Forecasting, priority, routing and cost modules replaceable	A second implementation of each module passes the same interface tests with 0 UI	TC-MAINT-01	Valid

ID	Category	Requirement	Measurable target	Verification	Status
		without UI changes	changes		
NFR-28	Maintainability	Tested core logic	≥ 80% line coverage on CMP-FC, CMP-RISK, CMP-ROUTE, CMP-COST, CMP-EDIT; 0 linter errors	CI report	Proposed
NFR-08	Usability	Non-technical manager generates, edits and approves a plan without training	≥ 4 of 5 test users complete UC-05 to UC-07 unaided in ≤ 10 min	TC-USE-01	Valid (target Proposed)
NFR-09	Usability	Plain-language explanations, reasons and risks	0 unexplained technical terms in a review of all templates; sentences ≤ 25 words	TC-USE-02: template review	Valid
NFR-10	Accessibility	Color never the only way to convey information	100% of urgency and risk information readable in grayscale	TC-ACC-01	Valid
NFR-11	Accessibility	WCAG 2.1 AA contrast for text and essential map elements	0 AA contrast failures on the main screens	TC-ACC-02: contrast checker	Valid
NFR-27	Accessibility	Keyboard use	Every function, including each drag-and-drop action, has a keyboard alternative; visible focus on all controls	TC-ACC-03	Proposed
NFR-29	Observability	Structured logging of key events	100% of requests, plan generations, decisions, edits and errors logged with timestamp, user ID and correlation ID (Section 15)	TC-OBS-01	Proposed

ID	Category	Requirement	Measurable target	Verification	Status
NFR-30	Portability	Run anywhere for development and demo	Starts with one command (Proposed: container setup) on Windows, macOS and Linux	TC-PORT-01	Proposed
NFR-15	Interoperability	External data through defined interfaces	Each source (weather, events, distances) accessed through exactly one documented interface	Code review	Valid
NFR-16	Interoperability	Current major desktop browsers	UC-01 to UC-11 pass on the latest Chrome, Firefox and Edge, at 1366×768 and above	TC-COMP-01	Valid
NFR-17	Compliance	No pump prices, no fuel-system control	0 functions that change prices or send commands to equipment	Code review; TC-SEC-08	Valid
NFR-18	Compliance	Plans respect storage and tanker capacity	0 violations in recommended plans; other regulatory rules **Unresolved** (OQ-13)	TC-AI-03	Valid
NFR-31	Usability	Consistent units and formats	Litres as whole numbers with thousands separators; times 24 h; dates DD MMM YYYY; currency USD ($) (D-20)	UI review	Proposed

Note on NFR-07 (D-01): drivers are now stored with a name and an ID. A driver name is personal data, so NFR-07's wording (“only user accounts”) needs a small update: personal data is limited to user accounts and driver name and ID, used only for planning. This change is final (D-22).

8. Data Specification

Venzini stores 23 data entities. All data is operational except user accounts and driver names (Section 8.8). The model is logical; table design belongs to the SDD.

8.1 Data entities and attributes

Types: **ID** = short text key; **int** = whole number; **dec** = decimal; **dt** = date-time (UTC stored, local Africa/Tunis shown, Proposed); **enum** = fixed list.

ID	Entity	Key attributes (type)	Validation rules
DE-01	User	user_id (ID), full_name (text 80), login (text 40), password_hash (text), role (enum: manager, dispatcher, admin), status (enum: active, inactive), created_at (dt)	login unique, 3–40 chars; password ≥ 10 chars at creation (Proposed); exactly one role
DE-02	Station	station_id (ID, e.g. ST014), name (text 80), latitude (dec), longitude (dec), region (text), window_open (time), window_close (time), serving_warehouse_id (ID, derived), status (enum: active, inactive)	ID unique; lat −90..90, lon −180..180; window_open < window_close
DE-03	Tank	station_id, fuel_type_id, capacity_l (int), safety_stock_l (int), current_level_l (int), level_updated_at (dt)	one tank per station and fuel type; 0 ≤ safety_stock < capacity; 0 ≤ current_level ≤ capacity
DE-04	Fuel type	fuel_type_id (ID), name (text, e.g. Petrol 95, Diesel)	name unique; list is data, not code (NFR-04)
DE-05	Warehouse	warehouse_id (ID), name, latitude, longitude	ID unique; coordinates valid
DE-06	Warehouse stock	delivery_day (date), warehouse_id, fuel_type_id, stock_l (int), resource_version_id	stock ≥ 0; one row per day, warehouse and fuel type
DE-07	Tanker	tanker_id (ID), compartments (list of size_l, int), capacity_l (int, sum of compartments), consumption_l_per_km (dec), home_warehouse_id	capacity > 0; consumption > 0; one home warehouse (BR-18)
DE-08	Tanker availability	delivery_day, tanker_id, available (bool), resource_version_id	one row per day and tanker
DE-09	Driver	driver_id (ID), full_name (text 80), home_warehouse_id	ID unique; no login, no contact data (BR-15, NFR-07)
DE-10	Driver availability	delivery_day, driver_id, available (bool), resource_version_id	one row per day and driver
DE-11	Resource version	resource_version_id, delivery_day, saved_by (user_id), saved_at (dt), locked (bool)	new version on every save; locked when the day's plan is decided

ID	Entity	Key attributes (type)	Validation rules
			(BR-30)
DE-12	Sales history	station_id, fuel_type_id, date, litres_sold (int)	litres ≥ 0; one row per day; gaps allowed but counted (data quality)
DE-13	External factor	date, area, kind (enum: weather, event, calendar), value (structured), source (enum: simulated, real)	one row per date, area and kind
DE-14	Forecast	forecast_id, station_id, fuel_type_id, reference_date, model_version, input_version, daily_values (7 × {date, litres, confidence, band_low, band_high}), factors (list of {factor, impact_pct, explanation}), overall_confidence (int %), uncertain (bool), status (enum: ok, manual_planning)	exactly 7 days; litres ≥ 0; confidence 0–100; uncertain = confidence < 70
DE-15	Plan	plan_id, delivery_day, kind (enum: optimal, edited), parent_plan_id, resource_version_id, data_version, model_versions, status (enum: draft, edited, partly decided, decided), created_by, created_at, total_km, total_minutes, total_litres, total_cost_usd	one current draft per day; decided plans read-only
DE-16	Route (trip)	route_id, plan_id, tanker_id, driver_id, compartment_plan (list of {compartment, fuel type, litres}), color, est_departure (dt), est_return (dt), actual_departure (dt), actual_return (dt), km, minutes, litres, cost_usd, time_gap_min (derived, FR-84), trip_status (enum: planned, departed, returned, cancelled)	one route per tanker per plan (BR-23); load ≤ capacity in optimal plans (BR-06); actual_return > actual_departure
DE-17	Decision log entry	entry_id, timestamp (dt), user_id, scope (enum: action, plan), plan_id, stop_id, decision (enum: approved, rejected), original_recommendation (snapshot), final_version (snapshot), was_edited (bool), edits (list), risks_shown (list), cost_optimal, cost_final, cost_diff_pct, note (text 500), input_versions	append-only (BR-13); all fields required except note and stop_id (plan scope)
DE-18	Stop (recommended	stop_id, route_id, station_id, stop_order (int), est_arrival (dt), priority_rank (int), risk_level	stop_order 1..n without gaps; one stop

ID	Entity	Key attributes (type)	Validation rules
	action)	(enum), reason (text), uncertain (bool), decision_status (enum: pending, approved, rejected)	per station per plan (BR-24)
DE-19	Stop quantity	stop_id, fuel_type_id, recommended_l (int), final_l (int)	0 ≤ litres ≤ BR-28 maximum; multiples of 100 for recommendations (Proposed)
DE-20	Risk	risk_id, plan_id, type (RISK-01..07), severity (enum: low, medium, high), affected (list of stop/segment IDs), reason (text), blocks_approval (bool)	blocks_approval true only for RISK-01
DE-21	Distance entry	from_point_id, to_point_id, km (dec), minutes (int), source (enum: simulated, real)	km > 0, minutes > 0; one entry per ordered pair
DE-22	Parameter	key, value, unit, valid_from (dt), changed_by	e.g. fuel_price_usd_per_l (public price, D-20), unloading_minutes; history kept
DE-23	Plan edit	edit_id, plan_id, user_id, timestamp, kind (enum: reorder, move, quantity, reset), before (snapshot), after (snapshot)	append-only within a plan

8.2 Relationships

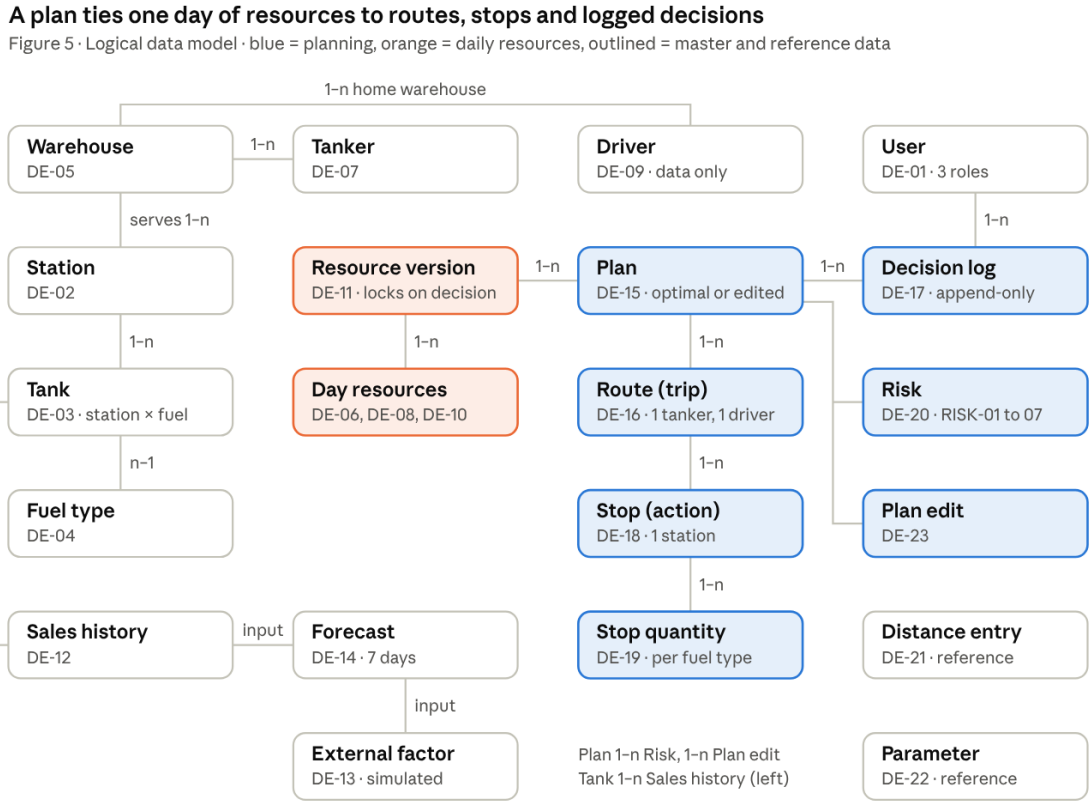


Figure 5 · Logical data model (ERD)

To keep the picture readable, three links are listed here rather than drawn: a route uses one tanker and one driver, a stop refers to one station, and a resource version records the user who saved it. DE-22 (Parameter) and DE-21 (Distance entry) are reference data used by the cost and routing modules.

- A **warehouse** has many tankers, drivers and stock rows; each **station** is served by one warehouse (derived, BR-17).
- A **station** has one **tank** per fuel type, many sales-history rows and many forecasts.
- A **delivery day** has many resource versions; each **plan** uses exactly one resource version.
- A **plan** has many **routes**; each route uses one tanker and one driver; a route has many **stops**; a stop has one quantity per fuel type.
- A **plan** has many **risks** and many **edits**; an edited plan points to its optimal parent.
- A **decision log entry** refers to one plan and, for action scope, one stop; it stores snapshots so it stays valid if other data changes.

8.3 Data ownership

Data	Owner (creates/changes)	Readers
Users	System admin	System (auth)

Data	Owner (creates/changes)	Readers
Stations, tanks (capacity, safety stock, window)	System admin or operations manager (D-15); sample data in MVP	Manager, system
Tank current levels, sales history	Sample data (v1)	System, manager
Tankers, drivers, warehouse stock per day; trip actual times	Dispatcher	Manager (read), system
Fuel price, consumption rates	Fuel price: public source (D-20); consumption rates: **Unresolved** (OQ-04); sample values in v1	System
Forecasts, plans, routes, risks	System	Manager
Edits, decisions	Operations manager	All roles (history, read-only)

8.4 Validation rules (cross-entity)

- Recommended quantities: BR-04, BR-06, BR-08 checked on every generated plan before display.
- Manager quantity edits: BR-28 checked in the browser and again on the server.
- Resource changes refused when the day's resource version is locked (BR-30).
- Every plan, route and stop refers to existing, active stations, tankers and drivers at creation time.
- Error responses name the field that failed (UC-02 exception).

8.5 Data lifecycle

1. **Master data** (stations, tanks, warehouses, tankers, drivers, fuel types): created by admin or sample data; deactivated, never deleted when referenced.
2. **Daily resources:** created by the dispatcher per delivery day; versioned on each save; locked when the plan is decided.
3. **Forecasts:** generated on demand or at plan generation; kept with their versions.
4. **Plans:** draft → edited → partly decided → decided (Figure 6). Decided plans are read-only.
5. **Trips:** planned → departed → returned, or cancelled (Figure 6).
6. **Decision log:** written once, never changed.

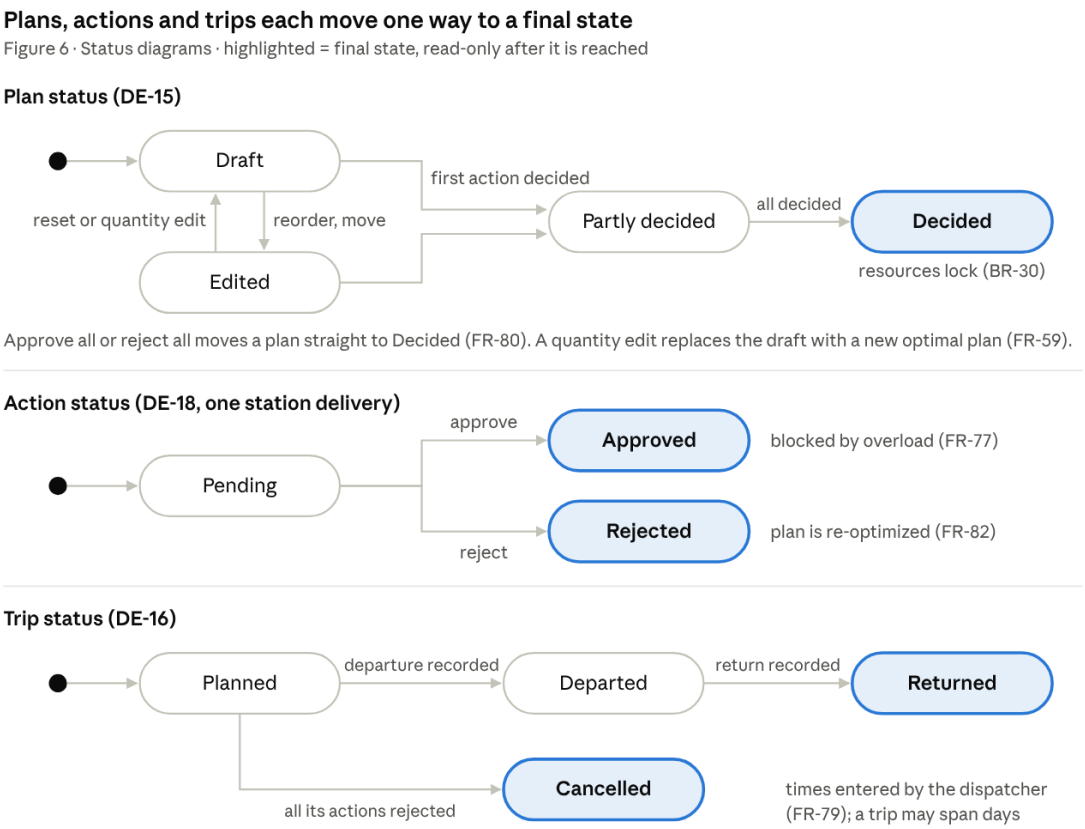


Figure 6 · Status diagrams for plan, action and trip

No state goes backwards once a decision is made: decided plans, decided actions and returned or cancelled trips are read-only.

8.6 Retention

Data	Retention
Decision log entries, decided plans and their snapshots	Kept for the life of the v1 system, never deleted (BR-13). Production period **Unresolved** (OQ-16)
Draft plans not decided	Proposed: kept 30 days after the delivery day, then removed
Forecasts	Proposed: kept with any plan that used them; others 30 days
Application logs	Proposed: 90 days
Inactive user accounts	Kept (referenced by the log); login disabled

8.7 Data quality requirements

- Sales history: ≥ 365 days per station and fuel type in the sample set; missing days ≤ 5% per series, otherwise the forecast is flagged low confidence (AI-20).
- Tank data: every active station has capacity, safety stock, current level and window; otherwise "Data unavailable" (FR-14).
- Distances: complete matrix between all warehouses and stations; a missing pair excludes the station with ERR-DATA-02.
- Sample data covers at least 4 seasons, 3 station size classes and 3 regions (AI-17).
- Every data row records its source (simulated or real) so v1 results are never mistaken for real-world performance.

8.8 Privacy classification

Class	Data	Handling
Personal – restricted	User password hash	Never shown, never logged
Personal – internal	User full name, login, role; driver full name and ID	Visible to roles that need it; never sent to external services
Operational – internal	Stations, tanks, resources, forecasts, plans, costs	Role-based access (Section 4.3)
Audit – internal, immutable	Decision log	Read-only for all roles
Public / simulated	Weather, events, distances (v1)	No restriction

9. API Specification

The whole API is **Proposed**: 19 logical REST/JSON endpoints between the browser and the Venzini server. Exact schemas, versioning and framework choices belong to the SDD. Each endpoint traces to the FRs it serves.

9.1 Common conventions (apply to every endpoint)

- **Base path:** `/api/v1`. JSON in and out, UTF-8. Litres as integers, money as decimals in USD, date-times ISO 8601.
- **Authentication:** session cookie set by API-01 (HttpOnly, Secure, SameSite=Strict); expires per NFR-22. Every endpoint except API-01 requires it → otherwise `401 ERR-AUTH-05`.
- **Authorization:** role check against Section 4.3 on every call → otherwise `403 ERR-AUTH-04`.
- **Validation:** server-side on every write, even when the browser already checked; errors name the field.
- **Error format:** `{"error": {"code": "ERR-VAL-02", "message": "Capacity must be greater than 0.", "field": "tankers[2].capacity_l"}}`.
- **Common errors:** 400 validation (ERR-VAL-xx), 401, 403, 404 not found (ERR-API-01), 409 conflict / locked (ERR-VAL-04), 429 rate limit (ERR-API-02), 500 internal (ERR-SYS-xx).

- **Rate limits:** login 10 requests/min per IP; plan generation 6/min per user; all others 120/min per user.
- **Writes** that change a plan return the full updated plan summary, so the UI never computes business values itself.

9.2 Endpoints

ID	Method and endpoint	Purpose	Roles	Request	Response	Specific errors	FRs
API-01	POST /auth/login	Log in	Public	login, password	user, role, home page	401 ERR-AUTH-01/02/03	FR-01
API-02	POST /auth/logout	Log out	All	–	204	–	FR-01
API-03	GET /me	Current user and permissions	All	–	user, role, allowed pages	–	FR-02
API-04	GET, POST /users; PATCH /users/{id}	List, create, edit, (de)activate users	Admin	full_name, login, password (create), role, status	user	400 ERR-VAL-03 duplicate login; ERR-VAL-07	FR-05
API-05	GET, POST /stations; PATCH /stations/{id}	List, create, edit, (de)activate stations and tanks	Manager, admin (D-15)	station fields + tanks[]	station	ERR-VAL-02, ERR-VAL-03	FR-06, FR-14
API-06	GET /resources/{day}	Day's tankers, drivers, warehouse stock, last update	Dispatcher, manager	–	resources + version + updated_by/at	404 if none	FR-07–09, FR-13
API-07	PUT /resources/{day}	Save the day's resources (new version)	Dispatcher	tankers[], drivers[], stock[]	version, warnings[]	ERR-VAL-01/02, 409 ERR-VAL-04 locked	FR-07–12
API-08	PATCH	Record actual	Dispatche	actual_de	trip with	ERR-VAL-	FR-

ID	Method and endpoint	Purpose	Roles	Request	Response	Specific errors	FRs
	/trips/{routeId}	departure or return	r	parture and/or actual_return	status	02, ERR-VAL-04	79
API-09	GET /network	Stations, levels, risks, alerts	Manager	optional date	stations[], alerts[]	–	FR-14–16, FR-24
API-10	GET /forecasts?station={id}&fuel={id}	7-day forecast with confidence and factors	Manager	station, fuel	forecast (DE-14)	ERR-AI-01 manual planning	FR-17–23, FR-83
API-11	POST /plans	Generate the optimal plan for a day	Manager	delivery_day	plan, routes, stops, unserved[], risks	ERR-PLAN-01/02	FR-25–42, FR-78
API-12	GET /plans/{id}	Read a plan (optimal or edited) with comparison	Manager	–	plan + comparison vs optimal	404	FR-42–54, FR-61
API-13	POST /plans/{id}/edits	Apply an edit: reorder, move, quantity, reset	Manager	kind + parameters	updated plan, comparison, risks	ERR-VAL-04, ERR-VAL-05a/b/c, ERR-PLAN-03, WARN-EDIT-01	FR-55–61, FR-75–77
API-14	GET /plans/{id}/limits?stop={id}&fuel={id}	Maximum quantity and binding limit	Manager	stop, fuel	max_l, binding_limit, details	–	FR-75
API-15	POST	Approve or	Manager	decision,	stop	ERR-	FR-

ID	Method and endpoint	Purpose	Roles	Request	Response	Specific errors	FRs
	/plans/{id}/stops/{stopId}/decision	reject one action		note?	status, plan status	PLAN-04, ERR-VAL-04	69, FR-71, FR-72, FR-82
API-16	POST /plans/{id}/decision	Approve all or reject all pending actions	Manager	decision	plan status, log entry IDs	ERR-PLAN-04	FR-70, FR-80
API-17	GET /decisions	Decision history, newest first, filterable	All	from, to, type, station, user, page	entries[]	–	FR-73
API-18	GET /decisions/{id}	One log entry in full	All	–	entry (DE-17)	404	FR-73
API-19	GET /forecasts/totals?level={network|warehouse}&fuel={id}	7-day forecast totals per day and fuel type	Manager (both levels); dispatcher (warehouse level only)	level, fuel, warehouse?	totals[], band, uncertain_share, excluded[]	403 ERR-AUTH-04 if a dispatcher asks for the network level	FR-85

No endpoint exists to update or delete a decision log entry (NFR-14), to set prices, or to control equipment (NFR-17).

9.3 Example payloads

API-07 – save resources

```
PUT /api/v1/resources/2026-11-03
{
  "tankers": [{"tanker_id": "T-01", "available": true}, {"tanker_id": "T-02",
"available": false}],
  "drivers": [{"driver_id": "D-04", "available": true}],
  "stock": [{"warehouse_id": "WH-RIV", "fuel_type_id": "DIESEL", "stock_l": 480000}]
}
→ 200 {"resource_version_id": "RV-0192", "warnings": []}
```

API-13 – move a station to another tanker

```
POST /api/v1/plans/PL-2211/edits
{"kind": "move", "stop_id": "S-07", "to_route_id": "R-02", "position": 2}
→ 200 {"plan": {"plan_id": "PL-2212", "status": "edited", "total_cost_usd": 412.35},
        "comparison": {"cost_optimal": 388.10, "cost_new": 412.35, "diff": 24.25,
"diff_pct": 6.2},
        "risks": [{"type": "RISK-01", "severity": "high", "blocks_approval": true,
                   "reason": "Tanker T-02 is overloaded: 34,000 L on a 32,000 L
tanker."}]}
```

API-13 – quantity above the limit

```
POST /api/v1/plans/PL-2212/edits
{"kind": "quantity", "stop_id": "S-03", "fuel_type_id": "P95", "litres": 9000}
→ 400 {"error": {"code": "ERR-VAL-05a", "field": "litres",
        "message": "This quantity exceeds the space left on tanker T-01. Enter 8,000
L or less."}}
```

API-16 – approve all

```
POST /api/v1/plans/PL-2212/decision
{"decision": "approve_all"}
→ 409 {"error": {"code": "ERR-PLAN-04",
        "message": "Tanker T-02 is overloaded: 34,000 L on a 32,000 L tanker. Remove
2,000 L before approving."}}
```

10. AI System Specification

Venzini has four AI capabilities. None uses an LLM or acts on its own (D-11): they compute recommendations and explanations, and the operations manager decides. In v1 the forecast is simulated (AS-02), so accuracy figures prove the pipeline works, not real-world performance.

AIC-01 – Demand forecasting

Field	Specification
Objective	Predict daily litres per station and fuel type for the next 7 days
User need	UN-02, UN-03
Input	Sales history, previous orders, current tank level, weather, local events, calendar/seasonality (FR-19)
Output	7 daily values (L), confidence per day and overall (%), confidence band, factor breakdown with signed % impact (DE-14)
Model / task	Time-series regression. **v1 (Proposed):** deterministic simulation = baseline (mean of the same weekday over the last 4 weeks) × factor multipliers (seasonality, weather, events) + seeded noise. **Later:** a trained model behind the same interface (NFR-12)
Context	One station and fuel type, one reference date

Field	Specification
Data sources	DE-12 sales history, past delivered quantities (DE-18, DE-19 of decided plans), DE-03 tank, DE-13 external factors
Prompt / context requirements	Not applicable (no LLM, D-11)
Tools available	None; pure function called by CMP-RISK, CMP-ROUTE and the UI
Retrieval	Reads only the station's own history and its area's factors for the horizon
Expected behavior	Same inputs → same outputs (NFR-13); every factor shown is one actually used
Forbidden behavior	Inventing factors or events not in DE-13; negative litres; output for a fuel type the station has no tank for
Confidence / uncertainty	Confidence (0–100%) falls with historical variability and missing inputs; < 70% → uncertain flag (BR-27), shown everywhere the forecast is used (AI-11)
Human approval	Forecasts are information; quantities derived from them need approval (BR-01)
Failure behavior	Missing input → low confidence, missing input named (AI-20); no history → manual planning (FR-83, AI-21)
Safety constraints	No personal data used (AI-26)
Evaluation metrics	MAPE on held-back sample data; bias (mean signed error); band coverage
Acceptance thresholds	v1 pipeline check: MAPE ≤ 15%,
Monitoring	Log model version, input version, confidence per forecast (OBS-M3); with real data later, forecast vs. actual error over time (AI-19)
Traceability	FR-18–23, FR-83; AI-01, AI-08–AI-11, AI-14, AI-17, AI-20, AI-21 → TC-AI-01; AC-04, AC-05

AIC-02 – Risk and priority scoring

Field	Specification
Objective	Assign each station and fuel type a risk level, and each station in need a delivery priority, with reasons

Field	Specification
User need	UN-01, UN-04
Input	Current level, forecast, safety stock, distance, resources
Output	Risk level + time to stockout (FR-24); priority rank (FR-25); reason text (FR-27)
Model / task	Rule-based scoring (BR-26; Proposed MVP ranking in FR-25); yield-management allocation after MVP (FR-26, rule **Unresolved**)
Context	All active stations for one delivery day
Data sources	DE-03, DE-14, DE-21, day's resources
Prompt / context requirements	Not applicable
Tools available	Calls AIC-01 for forecasts
Retrieval	Not applicable
Expected behavior	Thresholds applied exactly; ranking reproducible
Forbidden behavior	Lowering a high-risk station only because it is remote or low-volume (AI-16)
Confidence / uncertainty	Inherits the forecast's uncertain flag
Human approval	Priority is advice; the plan needs approval
Failure behavior	No forecast → manual planning, excluded from ranking with reason
Safety constraints	None beyond BR-01
Evaluation metrics	Rule conformance on boundary cases
Acceptance thresholds	100% of boundary test cases (23/24/72/73 h) classified correctly
Monitoring	Count of high-risk stations per day; count of manual-planning flags (OBS-M4)
Traceability	FR-15, FR-24–27; AI-02, AI-15, AI-16 → TC-AI-02; AC-06

AIC-03 – Delivery and route optimization

Field	Specification
Objective	Produce quantities, tanker–driver pairs, station assignments and one route per tanker that respect all hard

Field	Specification
	constraints and are cheap
User need	UN-06, UN-07
Input	Priorities, quantities, tankers, drivers, warehouse stock, distances, windows, consumption rates, fuel price
Output	Plan (DE-15 to DE-19) + unserved list
Model / task	Optimization, not machine learning: capacitated vehicle routing with time windows. **Proposed method:** priority-ordered cheapest-insertion construction, then local search (2-opt, relocate) with a fixed seed and an 8 s time limit (NFR-19)
Context	One delivery day
Data sources	Outputs of AIC-01/02, resources, DE-21, DE-22
Prompt / context requirements	Not applicable
Tools available	Distance provider IF-DIST; cost module CMP-COST
Retrieval	Not applicable
Expected behavior	Hard constraints BR-04 to BR-08, BR-23 to BR-25 always met in recommended plans (AI-03, AI-25); objective order per FR-41
Forbidden behavior	Returning a plan that breaks any hard constraint; serving a station from a non-closest warehouse; loading at another warehouse (AS-06)
Confidence / uncertainty	Stops based on uncertain forecasts carry the flag
Human approval	Every action needs manager approval (BR-01)
Failure behavior	Short resources → partial plan + unserved list (FR-37, FR-38); critical resource missing → stop and name it (FR-39); validator finds a violation → plan withheld, ERR-SYS-02
Safety constraints	AI-23 to AI-25
Evaluation metrics	Constraint violations; cost gap vs. best known solution; runtime
Acceptance thresholds	0 violations over 100 generated test plans; cost within 5% of an exhaustive optimum on 10 small instances (≤ 8 stations) (Proposed); runtime per NFR-19

Field	Specification
Monitoring	Runtime, violations caught by the validator, unserved count per plan (OBS-M1, OBS-M5)
Traceability	FR-28–42, FR-78; AI-03, AI-22, AI-25 → TC-AI-03; AC-08, AC-09, AC-16

AIC-04 – Edit risk analysis and template explanations

Field	Specification
Objective	Compare an edited plan with the optimal one, detect and grade risks, and explain forecasts, priorities and risks in plain language
User need	UN-03, UN-04, UN-10
Input	Optimal plan, edited plan, forecasts, factor data
Output	Risks with severity, reason and affected elements (DE-20); explanation sentences
Model / task	Deterministic rules (RISK-01 to RISK-07) + fixed sentence templates filled only with data values (D-11)
Context	One plan and its optimal parent
Data sources	DE-14 to DE-20
Prompt / context requirements	Not applicable; templates are versioned text in the code base (Proposed)
Tools available	None
Retrieval	Not applicable
Expected behavior	Every number, name and event in a sentence comes from the data (AI-12); plain language (NFR-09)
Forbidden behavior	Free-text generation; any value not present in the data; blaming the user
Confidence / uncertainty	Explanations of uncertain forecasts say so and name the cause
Human approval	Not applicable (information)
Failure behavior	Missing template data → shorter generic sentence with only known values

Field	Specification
Safety constraints	AI-12, AI-13 (closed: no LLM)
Evaluation metrics	Groundedness (share of numbers in text that match data); coverage (share of forecasts, priorities and risks with an explanation); risk detection precision and recall on built scenarios
Acceptance thresholds	Groundedness 100%; coverage 100%; risk detection precision = recall = 100% on the 7 scenario sets
Monitoring	Template fill failures (OBS-M6)
Traceability	FR-22, FR-27, FR-61–66; AI-12–AI-15 → TC-AI-04; AC-07, AC-12

10.5 Human-in-the-loop summary

- AI recommends, the manager decides (AI-04, AI-05).
- The manager can override order, tanker assignment and quantities (AI-06, as narrowed by D-05).
- Every override is logged with the recommendation it replaced (AI-07, FR-72).
- The system never executes, dispatches, prices or controls anything (AI-23, AI-24).

11. AI Agent Specification

Not applicable in v1. Venzini v1 uses no AI agents and no LLM (D-11). No component plans its own steps, chooses tools or acts without a direct user request.

If an agent is added in a later version, it must meet these minimum rules before it is specified in full. They come from rules already valid in v1.

Aspect	Minimum rule for any future agent	Source
Agent role	Assistant to the operations manager only; never to dispatcher or admin tasks without a new requirement	BR-02
Goal	Prepare drafts and explanations; never final decisions	AI-04
Available tools	Read-only access to forecasts, risk, routing and cost modules	NFR-12
Tool permissions	No write access to resources, decisions, users, stations or the log	BR-03, BR-13, BR-22
Input / output contracts	Same structured contracts as AIC-01 to AIC-04; outputs validated against BR-04 to BR-08	AI-03

Aspect	Minimum rule for any future agent	Source
	before display	
Memory	No storage of personal data; no memory across users	AI-26, NFR-07
Planning	Bounded number of steps and time per request (to be set)	NFR-19
Decision boundaries	Can propose; cannot approve, reject, dispatch, price or control equipment	BR-01, BR-14
Human-in-the-loop	Every proposal passes through the existing approval flow	AI-05
Maximum autonomy	Level 0: suggest only	CO-05
Failure recovery	On failure, fall back to the v1 deterministic modules	AI-22
Escalation	Uncertain or failed output is flagged for manual review	AI-10, AI-21
Audit / logging	Every agent action and tool call logged with inputs, outputs and versions	AI-18

12. Security Specification

Security in v1 rests on three things: every request is authenticated and checked against the role matrix, the decision log cannot be changed, and no function can act outside the platform. LLM-specific threats do not apply because v1 has no LLM (D-11).

Area	Specification	Status	Refs
Authentication	Login + password; session cookie (HttpOnly, Secure, SameSite=Strict); expiry after 30 min idle; lockout after 5 failed attempts in 15 min; generic error message that does not reveal whether the login exists	Proposed (method), Valid (login required)	FR-01, NFR-21, NFR-22
Authorization	Server-side role check on every endpoint against Section 4.3; UI hiding is not a security control; default deny	Valid	FR-02–04, NFR-05
Identity management	Accounts created only by the system admin; no self-registration; one role per account; deactivate, never delete; at least one active admin always remains	Valid / Proposed	FR-05

Area	Specification	Status	Refs
Encryption	In transit: HTTPS (TLS 1.2+) outside localhost. At rest: passwords hashed with bcrypt (cost ≥ 10); database disk encryption left to the hosting environment	Proposed	NFR-06, NFR-23
Secrets	Database credentials and session signing key in environment variables, never in the code repository; sample data contains no real credentials	Proposed	–
Input validation	All inputs validated on the server: types, ranges, lengths, enums, referential integrity; quantities against BR-28; parameterized database queries only; output encoding in the UI against script injection	Valid / Proposed	Section 8.4, FR-76
API security	Rate limits (Section 9.1); CSRF protection for all state-changing requests; no CORS for other origins; errors without stack traces	Proposed	Section 9
Prompt injection	Not applicable in v1: no LLM. Free text fields (rejection note) are stored and shown as plain text only	N/A	D-11
Data leakage	No external AI service; no personal data sent outside the system; logs exclude passwords, tokens and session IDs; role-based filtering on every list and export	Proposed	NFR-25, AI-26
Model abuse	Not applicable in v1 for LLMs. For optimization: rate limit on plan generation (6/min/user) and an 8 s time limit stop resource exhaustion	Proposed	NFR-19
Logging	Security events logged: logins (success and failure), lockouts, 403 refusals, role changes, account (de)activation	Proposed	OBS-L3, OBS-L4
Auditability	Decision log append-only at application and database level (no UPDATE/DELETE grants); each entry stores user, time, snapshots and input versions	Valid	FR-72, NFR-14, BR-13
Out-of-platform actions	No function sends commands to equipment, sets prices or dispatches tankers	Valid	NFR-17, BR-14

13. UI/UX Specification

Venzini v1 has 8 screens for desktop browsers. Each role sees only its own screens in a left navigation menu (Section 4.3). Screen layouts are not final; Figures 7 and 8 are AI-generated preliminary designs used only as inspiration (R4).

Navigation per role: Manager – Network, Forecasts (stations and totals), Resources (read-only), Delivery plan, Approvals, Stations, Decision history. Dispatcher – Resources, Trips, Warehouse forecast (read-only totals), Decision history. Admin – Administration, Decision history.

Rules for all screens: delivery-day selector in the header where relevant; user name and role in the header; WCAG 2.1 AA contrast (NFR-11); no color-only meaning (NFR-10); full keyboard use with visible focus (NFR-27); minimum width 1366 px; loading state for waits over 300 ms; plain-language messages (NFR-09).

UI-01 – Login

Field	Specification
User goal	Get into the platform with the right permissions
Components	Login field, password field (show/hide), Log in button, error area
User actions	Enter credentials; submit with Enter or the button
System responses	Open the role's home page (FR-01)
Validation	Both fields required
Error states	ERR-AUTH-01 wrong credentials; ERR-AUTH-02 deactivated; ERR-AUTH-03 locked with minutes remaining
Accessibility	Labels tied to inputs; errors announced to screen readers; focus on first field

UI-02 – Network overview (manager)

Field	Specification
User goal	See at a glance which stations need action (UN-01)
Components	KPI cards (stations at high risk, tankers available, warehouse stock status, plan status); map with station and warehouse markers, risk legend; “Stations at highest risk” table (station, fuel type, risk, time to stockout, reason)
User actions	Click a marker or row → station details; open the station forecast; open the delivery planner
System responses	Markers colored and iconed by risk (FR-47, FR-48); alerts for high risk only (FR-15)

Field	Specification
Validation	None (read-only)
Error states	“Data unavailable” stations listed (ERR-DATA-01); “No stations at high risk” message; map tiles unavailable → plain background (ERR-SYS-03)
Accessibility	Risk table is the accessible alternative to the map; legend with icons and words

UI-03 – Station forecast (manager)

Field	Specification
User goal	Know expected demand and how much to trust it (UN-02, UN-03)
Components	Station and fuel-type selectors; confidence badge; 7-day chart with confidence band and recent actuals; factor list with % impact and one-line explanation; uncertain-forecast panel; Totals view with network and per-warehouse daily totals per fuel type (FR-85)
User actions	Change station or fuel type; hover a day for values
System responses	Load forecast ≤ 3 s (NFR-02); uncertain panel when confidence < 70% (FR-23)
Validation	Only fuel types the station has
Error states	Manual planning needed (ERR-AI-01) instead of the chart; missing input named under the confidence badge (AI-20)
Accessibility	Chart values also available as a table; forecast vs. actuals differ by line style, not only color

UI-04 – Resources and trips (dispatcher; manager read-only)

Field	Specification
User goal	Record the day's tankers, drivers and warehouse stock, and actual trip times (UN-05)
Components	Delivery-day picker; tankers table (ID, capacity, home warehouse, available checkbox); drivers table (name, home warehouse, available); warehouse stock table (warehouse × fuel type, litres); “Last updated by” line; Save button; Trips tab (route, tanker, driver, estimated departure/return, actual departure/return inputs, status, estimated-vs-actual gap, FR-84); Warehouse forecast tab (read-only 7-day totals per fuel type for each warehouse, FR-85)

Field	Specification
User actions	Toggle availability; type stock; Save; record departure or return
System responses	New resource version saved with user and time (FR-13); warning for a tanker with an open trip (WARN-RES-01); locked banner after the plan is decided
Validation	Stock ≥ 0 integer; at least one field changed to enable Save; return after departure
Error states	Field-level errors (ERR-VAL-01/02); locked day (ERR-VAL-04) with fields disabled
Accessibility	Tables with header cells; checkboxes reachable by Tab and Space

UI-05 – Delivery planner and route map (manager)

Field	Specification
User goal	Get a plan that uses all tankers, adapt it, and see the impact (UN-06 to UN-10)
Components	Map with one colored directed route per tanker, numbered stops, warehouse markers, urgency-colored station markers; tanker filter (All / each tanker); station details box; route summary cards (stops, km, time, litres, cost, estimated departure/return); plan totals; comparison panel (optimal cost, new cost, difference L and %); risk list with severity; unserved stations list; Reset to optimal; quantity editor with limit hint
User actions	Drag a stop to reorder (FR-55); drag a station to another route (FR-56); click a station (FR-50); click a risk to highlight (FR-65); edit a quantity (FR-58); Reset (FR-60); go to Approvals
System responses	Recalculated values ≤ 1 s (NFR-01); risks or “No risk” message (FR-66); overload warning and approval block (FR-77); new optimal plan after a quantity edit (FR-59)
Validation	Quantity ≤ maximum (FR-75, FR-76) with message under the input and Save disabled; drop only on routes of the same warehouse (WARN-EDIT-01)
Error states	Critical resource missing (ERR-PLAN-01); no resources (ERR-PLAN-02); infeasible after edit (ERR-PLAN-03); recalculation failure (ERR-SYS-01)
Accessibility	Stop list beside the map with Move up / Move down / Move to tanker controls as keyboard alternative to drag and drop; risk severity as icon + word

UI-06 – Review and approve (manager)

Field	Specification
User goal	Decide every recommended action knowingly (UN-11)
Components	Info banner “Recommendations require manager approval”; actions table (station, litres per fuel type, tanker, arrival, priority reason, confidence, uncertain flag, risk, status, Approve / Reject / Adjust); risk-level legend; Approve all and Reject all buttons with confirmation; overload warning
User actions	Approve, reject (optional note), adjust (FR-81), approve all, reject all (FR-80)
System responses	Status updates; log entries created (FR-72); plan decided when all actions are decided (BR-30)
Validation	Approval disabled while RISK-01 exists (FR-77)
Error states	ERR-PLAN-04 overload; ERR-VAL-04 already decided
Accessibility	Buttons with text labels; confirmation dialog traps focus and returns it after closing

UI-07 – Decision history (all roles, read-only)

Field	Specification
User goal	Trace who decided what and when (UN-12)
Components	Search; filters (action, date range); table (date and time, manager, action, plan status: as recommended / modified / not dispatched, cost difference, risk warnings); detail panel (original recommendation, final decision, edits, risks shown, note)
User actions	Filter, search, open an entry
System responses	Newest first (FR-73); “View only” label on details
Validation	Date range: from ≤ to
Error states	“No decisions recorded yet.”
Accessibility	Table sortable by keyboard; detail panel reachable without a mouse

UI-08 – Administration (admin; Stations tab also for the manager, D-15)

Field	Specification
User goal	Control who can access what, and keep stations current (UN-13, UN-

Field	Specification
	14)
Components	Users tab (list, create, edit, activate/deactivate, role selector); Stations tab (list, create, edit, deactivate, tanks per fuel type, delivery window) – Stations tab is Should (FR-06)
User actions	Create, edit, (de)activate users and stations
System responses	Confirmation messages; new station appears on the map
Validation	Section 8.1 rules; unique login and station ID
Error states	ERR-VAL-01/02/03/07
Accessibility	Forms with labels and field-level errors

13.9 Preliminary designs (inspiration only)

The two images below were generated with AI before the requirements were final. They are **not** requirements and **not** final designs. Where they differ from this SRS, this SRS applies (for example: Figure 7 shows a single route and no drivers; Figure 8 shows Petrol 91, which is not a confirmed fuel type).

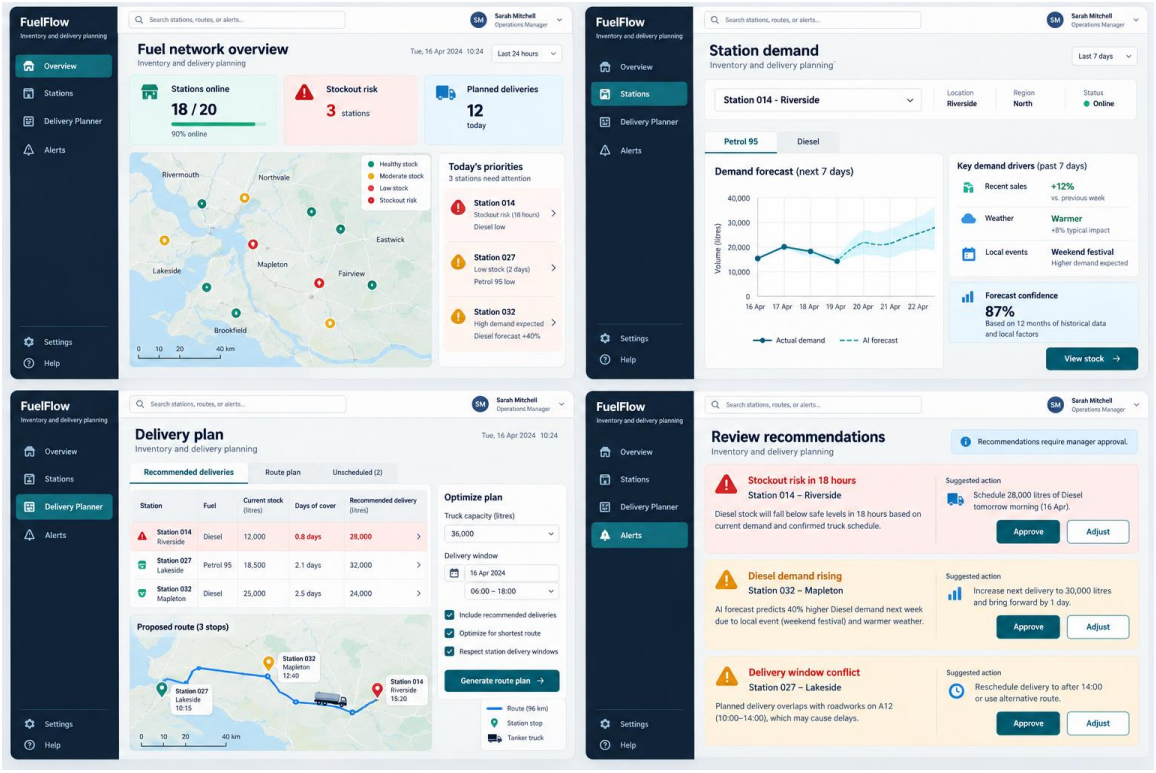


Figure 7 – AI-generated preliminary design "FuelFlow" (inspiration only, not a requirement): network overview, station demand, single-route delivery plan, review recommendations.

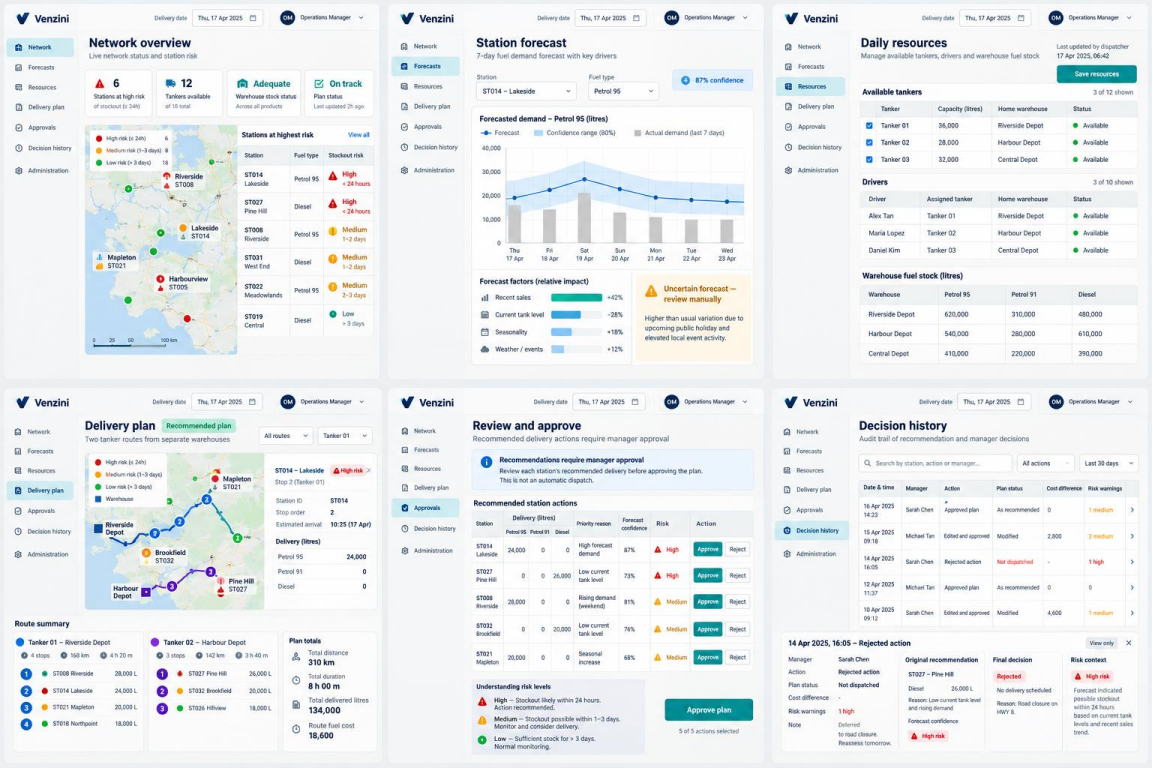


Figure 8 – AI-generated preliminary design "Venzini" (inspiration only, not a requirement): network overview, station forecast, daily resources, multi-tanker delivery plan, review and approve, decision history.

14. Error and Exception Specification

Every error has a code, a plain-language message that says what to do next, a recovery path and a log level. Messages never show stack traces or internal IDs (Section 12). Codes are Proposed; messages may be reworded but must keep their meaning.

14.1 Error categories

Category	Prefix	Meaning
Authentication and access	ERR-AUTH	Login, session and permission problems
Validation	ERR-VAL	Input breaks a rule; the user can fix it
Planning	ERR-PLAN	A plan cannot be produced, changed or approved
Data	ERR-DATA	Required master or reference data is missing
AI	ERR-AI	Forecasting or explanation cannot produce a normal result
API	ERR-API	Request-level problems (not found, too many requests)

Category	Prefix	Meaning
System	ERR-SYS	Unexpected internal failures
Warnings	WARN	The action is allowed but the user should know something

14.2 Error codes and messages

Code	HTTP	User-facing message	Recovery	Log level	Refs
ERR-AUTH-01	401	Login or password is incorrect.	Retry	Info	FR-01
ERR-AUTH-02	401	This account is deactivated. Contact your system admin.	Admin reactivates	Info	FR-05
ERR-AUTH-03	401	Too many attempts. Try again in {n} minutes.	Wait	Warning	NFR-21
ERR-AUTH-04	403	You don't have access to this page.	Return to home page	Warning (security)	FR-02
ERR-AUTH-05	401	Your session has expired. Please log in again.	Log in; return to page	Info	NFR-22
ERR-VAL-01	400	{Field} is required.	Fill the field	Info	Section 8.1
ERR-VAL-02	400	{Field}: {rule}, e.g. “Capacity must be greater than 0.”	Correct the value	Info	UC-02
ERR-VAL-03	400	This {login / station ID} already exists.	Choose another	Info	FR-05, FR-06
ERR-VAL-04	409	The plan for {day} is already decided. This data can no longer be changed.	None (read-only)	Info	BR-30
ERR-VAL-05a	400	This quantity exceeds the space left on tanker {T-ID}. Enter {max} L or less.	Lower the quantity	Info	FR-76
ERR-VAL-05b	400	This quantity exceeds the free space in the station's tank. Enter {max} L or less.	Lower the quantity	Info	FR-76
ERR-VAL-05c	400	This quantity exceeds the {fuel} stock at {warehouse}. Enter {max} L or less.	Lower the quantity	Info	FR-76
ERR-VAL-07	400	You can't deactivate your own admin account.	Ask another admin	Info	FR-05

Code	HTTP	User-facing message	Recovery	Log level	Refs
ERR-PLAN-01	422	No plan can be made: {no tankers / no drivers / no {fuel} stock at {warehouse}} for {day}.	Dispatcher uploads the resource	Warning	FR-39
ERR-PLAN-02	422	Resources for {day} are not uploaded yet.	Dispatcher uploads resources	Info	UC-05
ERR-PLAN-03	422	No feasible plan with these quantities: {constraint}.	Change the quantity or Reset	Warning	FR-59
ERR-PLAN-04	409	Tanker {T-ID} is overloaded: {load} L on a {capacity} L tanker. Remove {excess} L before approving.	Move a station, lower a quantity or Reset	Warning	FR-77
ERR-DATA-01	–	Data unavailable for {station}.	Admin completes station data	Warning	FR-14
ERR-DATA-02	–	No distance data for {station}; it is left out of the plan.	Complete distance table	Warning	FR-34
ERR-DATA-03	–	Route cost not available: missing {fuel price / consumption rate}.	Complete parameters	Warning	FR-40
ERR-AI-01	–	Manual planning needed for {station} {fuel}: {reason}.	Manager plans manually	Warning	FR-83, AI-21
ERR-AI-02	–	Low confidence: {input} is missing for this forecast.	Review manually	Info	AI-20
ERR-AI-03	–	Explanation shortened: some details are not available.	None	Info	AIC-04
ERR-API-01	404	We couldn't find this item. It may have been removed.	Go back	Info	Section 9
ERR-API-02	429	Too many requests. Please wait a moment.	Wait	Warning	Section 9
ERR-SYS-01	500	The change couldn't be calculated and was undone. Please try again.	Retry	Error	FR-57
ERR-SYS-	500	The plan failed a safety check and was not shown. Please try again or contact	Retry; developer	Critical	FR-29,

Code	HTTP	User-facing message	Recovery	Log level	Refs
02		support.	investigates		AIC-03
ERR-SYS-03	–	Map background unavailable; routes are still shown.	None	Warning	Section 6.10
ERR-SYS-04	500	Your decision couldn't be saved. Nothing was changed. Please try again.	Retry	Critical	FR-71, FR-72
WARN-RES-01	–	Tanker {T-ID} has not returned from trip {trip}. Check before marking it available.	Confirm or uncheck	Info	FR-07
WARN-EDIT-01	–	This tanker loads at another warehouse. Stations can only move between tankers of {warehouse}.	Drop on another route	Info	FR-56

14.3 Recovery mechanisms

- **Field errors:** shown under the field; the rest of the form keeps its values.
- **Edits:** a failed recalculation undoes only the last edit; Reset to optimal is always available.
- **Decisions:** status change and log entry are one transaction: both succeed or neither does.
- **Plans:** a plan that fails the constraint validator is never shown (fail safe).
- **Session expiry:** after login, the user returns to the page they were on; unsaved form data is lost (Proposed: warn 2 min before expiry).

14.4 Logging requirements

- Every error is logged with code, timestamp, user ID, correlation ID, endpoint and safe context (no passwords or tokens, NFR-25).
- Levels: Info (user-fixable), Warning (data or resource gaps, security refusals), Error (failed operations), Critical (safety check or decision failures; trigger an alert, Section 15).

14.5 AI-specific failures

Failure	Detection	Behavior	Code
No forecast possible	No sales history or module error	Manual planning flag; no quantity	ERR-AI-01
Degraded forecast	Missing input	Low confidence; uncertain flag if < 70%	ERR-AI-02
Unsafe plan	Constraint validator after optimization	Plan withheld; alert	ERR-SYS-02
Optimization timeout	8 s limit reached	Best feasible plan found so far is used, marked “time-limited”	–

Failure	Detection	Behavior	Code
		(Proposed)	
Incomplete explanation	Template field missing	Shorter generic sentence	ERR-AI-03

15. Observability Specification

Observability in v1 is sized for a solo developer running tests and demos: structured logs, a small set of metrics, and alerts for the two failures that matter most – an unsafe plan and a lost decision. All items are **Proposed** unless they restate an existing requirement.

15.1 Logs

ID	Log	Content	Retention
OBS-L1	Request log	Timestamp, correlation ID, user ID, role, method, endpoint, status, duration (ms)	90 days
OBS-L2	Error log	Section 14.4 fields	90 days
OBS-L3	Security log	Logins, failures, lockouts, 403 refusals, session expiry	90 days
OBS-L4	Admin change log	User and station create/edit/(de)activate, with before and after values	90 days
OBS-L5	Planning log	Plan ID, day, resource version, data version, model versions, runtime, routes, unserved count, validator result	90 days
–	Decision log	Business audit record (FR-72), separate from technical logs	Never deleted (BR-13)

Format: one JSON object per line; no passwords, tokens or session IDs (NFR-25).

15.2 Metrics

ID	Metric	Target / use
OBS-M1	Plan generation time (p50, p95)	≤ 10 s (NFR-19)
OBS-M2	Edit recalculation time (p95)	≤ 1 s (NFR-01)
OBS-M3	Forecasts produced, average confidence, share uncertain	Track trend; share uncertain shown in demo reports
OBS-M4	High-risk stations per day; manual-planning flags	Track

ID	Metric	Target / use
OBS-M5	Validator rejections; unserved stations per plan	Rejections = 0 (AIC-03)
OBS-M6	Template fill failures	0 in test runs
OBS-M7	Page load time per screen (p95)	≤ 3 s (NFR-02, NFR-20)
OBS-M8	Share of actions approved as recommended vs. modified vs. rejected	Measures trust in recommendations (BG-04)

15.3 Traces

A correlation ID is created per browser request and passed to every module call (forecast, risk, routing, cost, explanation), so one slow or failed plan can be followed end to end. Distributed tracing tools are not required in v1.

15.4 AI and model monitoring

- Every forecast and plan stores its model version and input data version (AI-18).
- Pipeline check per release: MAPE, bias and band coverage on held-back sample data (AIC-01); constraint violations on 100 test plans (AIC-03).
- With real data (future): forecast vs. actual sales error tracked weekly (AI-19). Trip times: the estimated vs. actual gap is stored per trip (FR-84) as a basis for better travel-time estimates in a later version.

15.5 Prompt and output monitoring

Not applicable in v1: no prompts and no generated text (D-11). Template outputs are checked by the groundedness test (TC-AI-04).

15.6 Cost monitoring

- No paid AI services in v1, so no AI usage cost.
- Business cost: total route fuel cost per plan and per delivered litre, and the cost difference of edited vs. optimal plans (feeds BG-03 and OBS-M8).
- Hosting cost: **Unresolved** (no hosting decided).

15.7 Performance monitoring

OBS-M1, OBS-M2 and OBS-M7 are reported after every performance test run (TC-PERF-01 to TC-PERF-04) and compared with NFR targets.

15.8 Alerts

Alert	Trigger	Channel (Proposed)
Unsafe plan	Any ERR-SYS-02	Console + error log, highlighted
Lost decision	Any ERR-SYS-04	Console + error log, highlighted

Alert	Trigger	Channel (Proposed)
Slow edits	OBS-M2 p95 > 1 s over the last 50 edits	Test report
Repeated login failures	≥ 3 lockouts in 1 hour	Security log summary

16. Testing and Verification Requirements

Every requirement has at least one test: one functional test per FR (TC-FR-xx, same number), plus the NFR and AI tests below. Tests run on the reference data set and on 8 scenario sets built to trigger specific behavior.

16.1 Scenario sets (test data)

ID	Scenario	Built to test
SC-01	Normal day: enough tankers, drivers and stock	Optimal plan, constraints, costs
SC-02	Tanker shortage: capacity covers 60% of demand	FR-37, FR-38, unserved list
SC-03	Critical resource missing: no drivers; then no tankers; then one warehouse empty	FR-39
SC-04	Tight delivery windows and long distances	FR-32, RISK-03, multi-day trips (BR-23)
SC-05	Edits: reorder, move with and without overload, quantity edits at and above each limit	FR-55–60, FR-75–77
SC-06	Data gaps: station with no sales history; missing weather; missing distance	FR-83, AI-20, ERR-DATA-02
SC-07	Risk boundaries: time to stockout 23/24/72/73 h; confidence 69/70%	BR-26, BR-27
SC-08	Fairness: remote low-volume station at high risk	AI-16

16.2 Test levels

Level	Scope	Requirement	Tooling (Proposed)
Unit	Forecast, risk, priority, routing, cost, edit, explanation modules; validators	≥ 80% line coverage on core modules (NFR-28); every BR has at least one unit test	Test runner of the chosen stack

Level	Scope	Requirement	Tooling (Proposed)
Integration	API endpoints with database; module interfaces (NFR-12); provider interfaces (NFR-15)	Every API-xx tested for success, validation error, 401 and 403	API test suite
System	End-to-end flows UC-01 to UC-11 in a browser	All 11 use cases pass on SC-01; exceptions pass on SC-02 to SC-06	Browser automation
Acceptance	Section 17 criteria, run with the author	AC-21 passes for MVP sign-off; 100% of Must ACs pass for v1 sign-off	Checklist + recorded demo
Security	TC-SEC-01 permission matrix (every row × role, UI and API); TC-SEC-02 password storage; TC-SEC-03 lockout; TC-SEC-04 session expiry; TC-SEC-05 HTTPS; TC-SEC-06 log scan; TC-SEC-07 log immutability (attempt UPDATE/DELETE as each role and directly in the database); TC-SEC-08 no price/equipment functions; input fuzzing on all write endpoints	0 failures	API tests + manual review
Performance	TC-PERF-01 edit recalculation; TC-PERF-02 page loads; TC-PERF-03 plan generation; TC-PERF-04 large network	NFR-01, NFR-02, NFR-19, NFR-20, NFR-24	Timed scripts
Accessibility	TC-ACC-01 grayscale check; TC-ACC-02 contrast checker; TC-ACC-03 keyboard-only run of UC-03 to UC-07	NFR-10, NFR-11, NFR-27	Checker + manual
Usability	TC-USE-01 5 non-technical users do UC-05 to UC-07 unaided; TC-USE-02 template wording review	NFR-08, NFR-09	Observed sessions
Reliability	TC-REL-01 determinism (two runs, byte-equal); TC-REL-02 fault injection on saves	NFR-13, NFR-26	Scripts
Regression	Full unit + integration suite and the SC-01 to SC-08 plan snapshots on every change; any snapshot change must be explained	All previous tests pass	CI

Level	Scope	Requirement	Tooling (Proposed)
Adversarial	Forced API requests: approval with RISK-01, quantity above limit, resource change after lock, decision by dispatcher, edit of log entry, script text in note field, oversized payloads	100% refused safely, no data change	API tests
Other	TC-SCAL-01 add station/tanker via data; TC-MAINT-01 swap module implementation; TC-OBS-01 log completeness; TC-PORT-01 one-command start on 3 OSs; TC-COMP-01 3 browsers	NFR-04, NFR-12, NFR-29, NFR-30, NFR-16	Scripts + manual

16.3 AI evaluation

ID	Component	Metric	Method	Threshold
TC-AI-01	AIC-01 forecasting	Accuracy: MAPE; bias; 80% band coverage	Hold back last 28 days of sample history; forecast them	MAPE ≤ 15%, abs(bias) ≤ 5%, coverage ≥ 75%
TC-AI-01b	AIC-01	Robustness: missing input	Remove weather, then events, then 10% of history	Forecast still produced, confidence lower, missing input named
TC-AI-02	AIC-02 risk	Classification accuracy on boundaries	SC-07	100% correct
TC-AI-03	AIC-03 routing	Safety: constraint violations	100 generated plans across SC-01 to SC-04 + random seeds	0 violations
TC-AI-03b	AIC-03	Optimality gap	10 instances ≤ 8 stations vs. exhaustive search	≤ 5% cost gap (Proposed)
TC-AI-04	AIC-04 risks	Precision and recall of risk detection	SC-05 scenarios, one per risk type plus no-risk edits	Precision = recall = 100%
TC-AI-04b	AIC-04 explanations	Groundedness and hallucination rate	Parse every number, station and event in generated sentences; compare with data	Groundedness 100%; hallucination rate 0%

ID	Component	Metric	Method	Threshold
TC-AI-04c	AIC-04 explanations	Coverage	Count forecasts, priorities and risks without explanation	0 (100% covered)
TC-AI-05	Whole system	Business effect vs. baseline (BG-01 to BG-03)	Simulate 28 days with Venzini plans vs. baseline rule (Section 3)	Targets **Unresolved** (OQ-11); report the difference
TC-AI-06	AIC-02/03	Bias / fairness	SC-08: remote, low-volume high-risk station	Served before lower-risk stations, or shown with reason when resources force otherwise (AI-16)

Not applicable in v1: LLM hallucination tests on free text, prompt-injection tests, and agent tests (no LLM or agent, D-11).

17. Acceptance Criteria

The MVP (clickable prototype) is accepted when AC-21 passes. Version 1 is accepted when all Must criteria below pass on the reference data set and the scenario sets. Each criterion groups the per-FR criteria of Section 6 into one measurable check.

ID	Major requirement	Measurable criterion	Covers	Priority
AC-01	Access control	100% of permission-matrix cells behave as specified in UI and API (3 roles × 13 functions); unauthenticated access to any page redirects to login	FR-01–05, NFR-05	Must
AC-02	Resources and master data	Dispatcher saves tankers, drivers and stock for a day; planner shows them with correct user and time; changes refused after the plan is decided; invalid values refused with the field named	FR-06–13	Must (FR-06, FR-10 Should)
AC-03	Network overview	Marker count = active stations; tank levels match data; high-risk pairs (and only those) raise alerts	FR-14–16	Must
AC-04	Forecast	Exactly 7 daily values from the next day, in a chart; changing one input changes the forecast or its factors; network and per-warehouse totals equal the sum of the included station forecasts	FR-17–20, FR-85	Must
AC-05	Confidence and	Every value has a confidence; 69% shows the uncertain flag on station page, plan and approvals,	FR-21, FR-23,	Must

ID	Major requirement	Measurable criterion	Covers	Priority
	uncertainty	70% does not; no-history station gets the manual-planning flag and no quantity	FR-83	
AC-06	Risk and priority	SC-07 boundaries 23/24/72/73 h → high/medium/medium/low; plan order follows the priority rule	FR-24–25	Must
AC-07	Explanations	100% of forecasts, priorities and risks have an explanation; 100% of numbers in explanations match data	FR-22, FR-27, FR-64	Must
AC-08	Plan constraints	0 violations of BR-04 to BR-08 and BR-24 in 100 generated plans; every station linked to its closest warehouse	FR-28–34	Must
AC-09	Multi-tanker routes and shortages	With N tanker–driver pairs: ≤ N routes, each from and back to its home warehouse, one trip per tanker; SC-02 gives a partial plan with lowest-priority stations unserved and listed; SC-03 stops with the missing resource named	FR-35–39	Must (FR-26 Should)
AC-10	Route map and editing	Directed, numbered, tanker-colored routes; clean default map; details on click; filter works; reorder and move work by drag and drop and keyboard; 95% of recalculations ≤ 1 s; Reset restores the optimal plan exactly	FR-45–57, FR-60	Must
AC-11	Quantity editing with limits	For each binding limit (tanker, tank, warehouse) the hint shows the right maximum; max + 1 L disables Save with the right message; server refuses forced requests; a valid edit triggers a new plan respecting BR-04 to BR-08	FR-58–59, FR-75–76	Should
AC-12	Risk analysis	Each of RISK-01 to RISK-07 raised in its SC-05 scenario and not otherwise; one severity each; reasons name the affected item; selecting a risk highlights it; no-risk edits say so; saving and risk shown together	FR-44, FR-61–66	Must
AC-13	Overload blocks approval	A plan overloaded by 1 L cannot be approved (per action or whole plan, UI or API); removing the overload re-enables approval	FR-56, FR-77	Must
AC-14	Approval	Each action can be approved or rejected; approve-all on 5 actions approves 5 and writes 6 log entries;	FR-03, FR-67–	Must

ID	Major requirement	Measurable criterion	Covers	Priority
		warnings other than RISK-01 do not block; Adjust opens the planner on the station; a rejection produces a new optimal plan without that station; no action is approved without a manager record; non-managers cannot decide	71, FR-80–82	
AC-15	Decision history	Each decision creates one complete entry (all DE-17 fields); list newest first for all 3 roles; 0 ways to edit or delete an entry	FR-72–73, NFR-14	Must
AC-16	Cost	Route cost = distance × consumption × price to $0.01; totals = sum of routes; edited plan shows optimal, new, difference and % correctly; equal-risk tie goes to the cheaper plan	FR-40–43, FR-52	Must
AC-17	Performance	NFR-01, NFR-02, NFR-19, NFR-20 met on the reference environment	NFRs	Must
AC-18	Accessibility	Urgency and risk readable in grayscale; 0 AA contrast failures; UC-03 to UC-07 completed with keyboard only	FR-47–48, NFR-10, NFR-11, NFR-27	Must
AC-19	Trips	Estimated times correct to the minute for a test route; dispatcher records departure and return, status changes planned → departed → returned; other roles cannot record times; the estimated-vs-actual gap is stored and shown	FR-78–79, FR-84	Must
AC-20	Sample data and determinism	All Must ACs demonstrable without real data; two runs on identical inputs give identical outputs	FR-74, NFR-13	Must
AC-21	MVP clickable prototype	All 8 screens (UI-01 to UI-08) and the flows of UC-01 to UC-11 can be clicked through with fake data, per role; no real calculations or saved data required	Section 13; D-26	MVP

18. Requirements Traceability Matrix

The full matrix is in Appendix B, so the main body stays readable. It links every FR, NFR and AI capability to its specification, component, API or screen, test case and acceptance criterion.

19. Implementation Readiness Review

Verdict: ready for MVP implementation; all Proposed items are approved (D-27, D-28). All Must requirements are specified with inputs, rules, errors and tests. Two Should requirements cannot be built yet: FR-10 (other resources, OQ-20) and FR-26 (yield-management rule undefined).

19.1 Missing information

ID	Missing	Affects	Blocks MVP?
OQ-04	Who provides tanker consumption rates (the fuel price is public, D-20)	FR-40, DE-22	No – sample values used
OQ-07	Loading at another warehouse of the same company	BR-19	No – not modelled (AS-06)
OQ-11	Target values for BG-01 to BG-03	Section 3, TC-AI-05	No
OQ-13	Other regulatory rules (driver hours, hazardous-goods routes)	NFR-18, multi-day trips	No for MVP; yes for real use
OQ-16	Decision history retention period	§8.6	No
OQ-17	Production availability target	NFR-03	No
OQ-20	Other operational resources	FR-10	Blocks FR-10 only
–	Technology stack and hosting	§9, §12, §15	No – Proposed in Section 20

19.2 Ambiguities

- **Fuel types:** the Elicitation FV says “such as Petrol 95 or Diesel”; the sample set uses these two. Other types (e.g. Petrol 91) are possible as data (NFR-04) but not confirmed.
- **“Needs fuel” and recommended quantity:** both use Proposed rules (FR-25, FR-28).
- **Time to stockout:** Proposed even spread of daily demand over 24 h (FR-24).
- **Overall confidence:** Proposed as the lowest daily confidence (FR-21).
- **Driver pairing:** Proposed same home warehouse, ID order (FR-35).
- **Multi-day trips and windows:** a trip over several days must still meet each stop's daily window; overnight stops are not modelled (Proposed).

19.3 Contradictions (resolved, final)

Contradiction	Final resolution
FR-49 clean map vs. FR-51 labels on every segment	Segment labels show when a route is selected or zoomed in
Overload by moving is allowed (then blocks	Both kept as specified (D-02, D-06)

Contradiction	Final resolution
approval); overload by quantity is refused at once	

19.4 Risks

ID	Level	Risk	Mitigation
RK-01	High	Multi-tanker routing with windows may miss the 1 s edit and 10 s plan targets	Edits only recalculate, never re-optimize (BR-12); fixed reference data size; 8 s optimizer time limit
RK-03	Medium	Simulated data may not convince reviewers	Realistic sample data (§8.7), documented generation, held-back evaluation (TC-AI-01)
RK-04	Medium	Scope creep from “very flexible”	Must / Should split kept; manager edits limited by D-05
RK-05	Medium	One developer carries everything	Build Must FRs first; Section 20 gives a compact build context
RK-07	Medium	Multi-day trips add scheduling complexity	Keep one trip start per tanker per day (BR-23); no overnight modelling in v1
RK-08	Low	Re-planning after each rejection (FR-82) can change actions already approved	Unchanged approvals kept, changed ones return to pending; tests in SC-05

19.5 Unresolved decisions

OQ-04, OQ-07, OQ-11, OQ-13, OQ-15, OQ-16, OQ-17, OQ-20, the FR-26 yield-management rule, and the technology stack.

19.6 Items requiring the author's approval

None. All Proposed items are approved (D-27, D-28).

All other Proposed items were approved on 25 Oct 2026 (D-27); the technology stack, API, security setup, error codes and logging are finalized in the SDD.

Approved on 25 Oct 2026 (D-24): inferred items BR-08 / FR-33, AI-16, AI-17; compartment assignment by the routing module; FR-58, FR-75, FR-76 and FR-84 stay Should; NFR-19 and NFR-22.

20. AI Coding Context

Paste this section into an AI coding assistant. It holds only validated information, plus Proposed items clearly marked; nothing marked Unresolved may be implemented as if it were decided.

```
# Venzini v1 – implementation context

## Objective
```

```
Web platform for a petroleum distributor: 7-day fuel demand forecast per station and
fuel type,
station risk, delivery plan with one route per tanker, live route editing with risk
analysis,
manager approval, immutable decision log. Sample data and simulated forecasts/routes
only.

## Scope
Step 1 (MVP): clickable frontend with fake data for all Must screens, no real logic.
Step 2: Must FRs in full. Step 3: Should (FR-06, FR-10*, FR-26*, FR-58, FR-59, FR-75,
FR-76, FR-84).
*FR-10 and FR-26 are blocked by unresolved questions: do not implement.
Desktop browsers only (latest Chrome, Firefox, Edge; >= 1366 px wide).

## Actors (exactly 3 roles)
- manager: network, forecasts and totals, resources (read), planner, edits,
approvals, stations (write), history (read)
- dispatcher: resources per delivery day (write), trip actual times (write), per-
warehouse forecast totals (read), history (read)
- admin: users and stations (write), history (read)
Drivers are data only, never users. Default deny; check role on every API call.

## Key rules
- Never above: tanker capacity, station free storage, warehouse stock (recommended
plans).
- Delivery within station window. Station served from closest warehouse. One tanker
per station; a tanker has compartments; each holds one fuel type per trip and never
more than its size.
- Route starts and ends at the tanker's home warehouse; return leg counts.
- A tanker starts at most one trip per delivery day; a trip may span several days.
- Station risk: high < 24 h to stockout, medium 24-72 h, low > 72 h.
- Forecast uncertain if confidence < 70%; flag shown wherever the forecast is used.
- Always produce a plan unless no tankers, no drivers, or no stock (name what is
missing).
- Short resources: serve highest priority first, list unserved with risk and reason.
- Manager may change only: stop order, station's tanker (same warehouse), quantities.
- Max quantity for a stop = min(free compartment space for that fuel, station free
storage, remaining warehouse stock).
- Edits (reorder/move) recalculate only; quantity edits regenerate the optimal plan.
- Risks are warnings, EXCEPT tanker capacity exceeded: it blocks all approval until
fixed.
- Approve/reject per action or whole plan; plan decided when all actions decided;
resources lock then.
- Rejecting a station triggers a new optimal plan without it; unchanged approvals
stay, changed actions return to pending.
- Nothing is approved without a manager decision and a log entry in the same
transaction.
- Route cost = km x consumption (L/km) x fuel price. Currency USD ($).

## Data model
23 entities, see SRS Section 8.1 (User, Station, Tank, FuelType, Warehouse,
WarehouseStock,
Tanker, TankerAvailability, Driver, DriverAvailability, ResourceVersion,
SalesHistory,
ExternalFactor, Forecast, Plan, Route, DecisionLogEntry, Stop, StopQuantity, Risk,
```

```
DistanceEntry, Parameter, PlanEdit). Decision log is append-only.

## APIs (Proposed)
REST/JSON under /api/v1, 19 endpoints, see SRS Section 9.2. Session cookie auth.

## AI components (no LLM, no agents)
- Forecast: deterministic simulation (baseline x factors + seeded noise), confidence,
factors.
- Risk/priority: rule-based (Section 6.7).
- Routing: CVRP with time windows; cheapest insertion + 2-opt/relocate, fixed seed, 8
s limit (Proposed).
- Risk analysis: RISK-01..07 rules; explanations from fixed templates filled with
data only.
Each module behind an interface so it can be replaced without UI changes.

## Constraints
Same inputs -> same outputs. Solo developer. No real distributor data.

## Proposed stack (pending approval)
React + TypeScript; Node.js REST API; PostgreSQL; Leaflet with OpenStreetMap tiles;
containers.

## Security rules
bcrypt password hashes; HttpOnly Secure SameSite cookies; 30 min idle expiry; lockout
after
5 failures/15 min; CSRF protection; parameterized queries; no secrets in code; no
passwords
or tokens in logs; no UPDATE/DELETE on decision log.

## Acceptance criteria
AC-01..AC-21 in SRS Section 17 (AC-21 = MVP prototype). Performance: edit
recalculation <= 1 s (95%),
page loads <= 3 s (95%), plan generation <= 10 s on 20 stations / 6 tankers.

## Testing
One test per FR (TC-FR-xx); scenario sets SC-01..SC-08; 0 constraint violations in
100 plans;
100% grounded explanations; permission matrix test on UI and API; >= 80% coverage on
core modules.

## DO NOT IMPLEMENT
- Pump price setting or any fuel-system / equipment control
- Driver accounts or any driver-facing screen
- Automatic approval, dispatch or execution of any action
- LLMs, chatbots, AI agents or free-text generation
- Editing or deleting decision log entries
- Manager editing of tankers, drivers or warehouse stock
- Loading at a warehouse other than the tanker's home warehouse
- Split deliveries (one station served by two tankers)
- Mobile app, live tracking, real data integrations
- FR-10 (other resources) and FR-26 (yield-management rule) until defined
- Plan handover to drivers or depot (OQ-15 unresolved)
```

Appendix A – Decision log (25 Oct 2026)

The author took 13 decisions before this SRS was written (D-01 to D-13) and 15 more while reviewing the first draft (D-14 to D-28). Each row says what it closes and where it changed this SRS.

ID	Decision	Closes	Effect in this SRS
D-01	The dispatcher uploads available drivers as well as tankers and stock	–	FR-08 confirmed Must; NFR-07 note
D-02	A move that overloads a tanker is allowed, raises a warning, and blocks plan approval until fixed	–	BR-11 changed; BR-21; FR-56, FR-67 changed; FR-77 new
D-03	Approval per action and for the whole plan	–	BR-30; FR-11 changed; FR-80 new
D-04	“Adjust” is a shortcut that opens the planner on that station	–	FR-81 new
D-05	The manager cannot change resources; may change only delivery quantities (plus stop order and tanker assignment)	–	BR-12 changed; BR-22; FR-02, FR-58, FR-59 changed
D-06	While editing a quantity, show the limit; above it, block Save and show an error under the input naming the tanker (or other limit) and its maximum	–	BR-28; FR-75, FR-76 new
D-07	Station risk: high < 24 h, medium 1–3 days, low > 3 days	–	BR-26; FR-24
D-08	Forecast uncertain below 70% confidence	–	BR-27; FR-23
D-09	Trip = departure from the warehouse to return, may last more than a day; one trip per tanker per day; one tanker per station; return leg counted; system estimates times, dispatcher records actual times	–	BR-23, BR-24, BR-25, BR-29; FR-35, FR-36 changed; FR-78, FR-79 new
D-10	Dispatcher cannot see plans, map or forecasts; decision history is read-only for all roles; no board role	–	Section 4.3; FR-02, FR-73 changed
D-11	No LLMs and no AI agents in v1; explanations use fixed templates	AI-13	Sections 10, 11, 12; DO NOT IMPLEMENT list
D-12	Technology stack not decided; this SRS may propose one	–	Proposed stack in Section 20

ID	Decision	Closes	Effect in this SRS
D-13	This SRS is an editable document; exported to Word/PDF as the final version after approval	–	Document format
D-14	Data providers get no access in v1; a dedicated role is planned for a later version	–	Section 4.2
D-15	The operations manager can also manage station records (with the system admin)	–	Section 4; FR-06 changed; API-05; UI-08; §8.3; Figure 3
D-16	The system estimates trip times; the dispatcher records actual times; the system stores the gap to improve estimates later	–	FR-84 new; DE-16; §15.4; Figure 1
D-17	Show network and per-warehouse forecast totals; the dispatcher reads per-warehouse totals only	–	FR-85 new; API-19; UI-03, UI-04; Section 4.3
D-18	Use the proposed risk severity thresholds for v1	–	§6.12, FR-63
D-19	Tankers have compartments and can carry several fuel types per trip; compartment sizes are checked	–	BR-31 new; BR-28; FR-07, FR-31, FR-35, FR-75; DE-07, DE-16
D-20	The fuel price is public; all costs in US dollars	OQ-04 in part	§6.9, FR-40; DE-15, DE-16, DE-22; NFR-31; Sections 9, 20
D-21	Move the traceability matrix to an appendix	–	Section 18 → Appendix B
D-22	All contradictions in §19.3 are final	§19.3 contradictions	§19.3; NFR-07 changed
D-23	Rejecting a station triggers a new optimal plan without it, so freed capacity can serve other stations; unchanged approved actions keep their approval, changed ones return to pending	FR-82 question	FR-82 rewritten; AC-14; RK-08; Figure 6; Section 20
D-24	Approved: inferred items (BR-08 / FR-33, AI-16, AI-17), compartment assignment, NFR-19 and NFR-22; FR-58, FR-75, FR-76, FR-84 stay out of the MVP	§19.6 items	§19.6; BR-08; FR-33; Section 7
D-25	Unloading time depends on the delivered quantity; sample values in v1, real data from data managers later	–	FR-78

ID	Decision	Closes	Effect in this SRS
D-26	The MVP is a clickable frontend with fake data; the real logic comes after	–	Document control; §2.1; §16; §17 (AC-21); Section 20
D-27	Proposed calculation rules, targets and technical defaults approved, except the four items still listed in §19.6	Most Proposed items	§19.6
D-28	Previous orders become a forecast input; baseline, totals range, forecasting and routing methods approved	Last §19.6 items	FR-19, FR-22, AIC-01; Figure 2; §19.6

Appendix B – Requirements Traceability Matrix

All 85 FRs, 31 NFRs and 4 AI capabilities trace forward to a specification, a component, an API or screen, a test and an acceptance criterion. Backward traceability (to UN, UC, US, BR and decisions) is in each FR's Traceability field.

B.1 Functional requirements

Requirement	Specification	Component	API / UI	Test case	Acceptance criterion
FR-01 Login required (UN-13)	§6.2	CMP-AUTH	API-01, API-02 / UI-01	TC-FR-01, TC-SEC-04	AC-01
FR-02 Roles and matrix (UN-13)	§6.2, §4.3	CMP-AUTH	API-03 / all UI	TC-FR-02, TC-SEC-01	AC-01
FR-03 Only managers decide (UN-11)	§6.2	CMP-AUTH, CMP-DEC	API-15, API-16 / UI-06	TC-FR-03	AC-01, AC-14
FR-04 Only dispatchers change resources (UN-05)	§6.2	CMP-AUTH, CMP-RES	API-07, API-08 / UI-04	TC-FR-04	AC-01
FR-05 Manage users (UN-13)	§6.3	CMP-ADMIN	API-04 / UI-08	TC-FR-05	AC-01
FR-06 Manage stations (UN-14)	§6.3	CMP-ADMIN	API-05 / UI-08	TC-FR-06	AC-02
FR-07 Upload tankers (UN-05)	§6.4	CMP-RES	API-06, API-07 / UI-04	TC-FR-07	AC-02

Requirement	Specification	Component	API / UI	Test case	Acceptance criterion
FR-08 Upload drivers (UN-05)	§6.4	CMP-RES	API-06, API-07 / UI-04	TC-FR-08	AC-02
FR-09 Upload warehouse stock (UN-05)	§6.4	CMP-RES	API-06, API-07 / UI-04	TC-FR-09	AC-02
FR-10 Other resources (UN-05)	§6.4	CMP-RES	API-07 / UI-04	TC-FR-10	AC-02
FR-11 Update until decided (UN-05)	§6.4	CMP-RES	API-07 / UI-04	TC-FR-11	AC-02
FR-12 Latest resources used (UN-05)	§6.4	CMP-RES, CMP-ROUTE	API-11 / UI-05	TC-FR-12	AC-02
FR-13 Last update shown (UN-05)	§6.4	CMP-RES	API-06 / UI-04, UI-05	TC-FR-13	AC-02
FR-14 Map of stations (UN-01)	§6.5	CMP-UI	API-09 / UI-02	TC-FR-14	AC-03
FR-15 Stockout alerts (UN-01)	§6.5	CMP-RISK	API-09 / UI-02	TC-FR-15	AC-03
FR-16 Tank levels (UN-01)	§6.5	CMP-UI	API-09 / UI-02	TC-FR-16	AC-03
FR-17 Select station and fuel (UN-02)	§6.6	CMP-UI	API-10 / UI-03	TC-FR-17	AC-04
FR-18 7-day forecast (UN-02)	§6.6, AIC-01	CMP-FC	API-10 / UI-03	TC-FR-18, TC-AI-01	AC-04
FR-19 Forecast inputs (UN-02)	§6.6, AIC-01	CMP-FC, CMP-DATA	API-10 / UI-03	TC-FR-19, TC-AI-01b	AC-04
FR-20 Forecast chart (UN-02)	§6.6	CMP-UI	API-10 / UI-03	TC-FR-20	AC-04
FR-21 Confidence (UN-03)	§6.6, AIC-01	CMP-FC	API-10 / UI-03	TC-FR-21	AC-05
FR-22 Factors (UN-03)	§6.6, AIC-04	CMP-FC, CMP-EXPL	API-10 / UI-03	TC-FR-22, TC-AI-04b	AC-07
FR-23 Uncertain flag (UN-03)	§6.6	CMP-FC	API-10, API-12 / UI-03,	TC-FR-23	AC-05

Requirement	Specification	Component	API / UI	Test case	Acceptance criterion
			UI-05, UI-06		
FR-24 Station risk (UN-04)	§6.7, AIC-02	CMP-RISK	API-09, API-11 / UI-02, UI-05	TC-FR-24, TC-AI-02	AC-06
FR-25 Delivery priority (UN-04)	§6.7, AIC-02	CMP-RISK	API-11 / UI-05	TC-FR-25	AC-06
FR-26 Yield allocation (UN-04)	§6.7	CMP-RISK, CMP-ROUTE	API-11 / UI-05	TC-FR-26	AC-09
FR-27 Urgency reasons (UN-04)	§6.7, AIC-04	CMP-EXPL	API-11 / UI-05, UI-06	TC-FR-27, TC-AI-04c	AC-07
FR-28 Recommended litres (UN-06)	§6.8, AIC-03	CMP-ROUTE	API-11 / UI-05	TC-FR-28	AC-08
FR-29 Free storage cap (UN-06)	§6.8	CMP-ROUTE	API-11 / UI-05	TC-FR-29, TC-AI-03	AC-08
FR-30 Safety stock (UN-06)	§6.8	CMP-ROUTE	API-11 / UI-05	TC-FR-30	AC-08
FR-31 Tanker capacity (UN-06)	§6.8	CMP-ROUTE	API-11 / UI-05	TC-FR-31, TC-AI-03	AC-08
FR-32 Delivery windows (UN-06)	§6.8	CMP-ROUTE	API-11 / UI-05	TC-FR-32	AC-08
FR-33 Warehouse stock cap (UN-06)	§6.8	CMP-ROUTE	API-11 / UI-05	TC-FR-33, TC-AI-03	AC-08
FR-34 Closest warehouse (UN-06)	§6.8	CMP-ROUTE	API-11 / UI-05	TC-FR-34	AC-08
FR-35 One route per tanker (UN-06)	§6.8, AIC-03	CMP-ROUTE	API-11 / UI-05	TC-FR-35	AC-09
FR-36 Home warehouse start and end (UN-06)	§6.8	CMP-ROUTE	API-11 / UI-05	TC-FR-36	AC-09
FR-37 Always a plan (UN-06)	§6.8	CMP-ROUTE	API-11 / UI-05	TC-FR-37	AC-09
FR-38 Priority first, unserved list (UN-06)	§6.8	CMP-ROUTE	API-11 / UI-05	TC-FR-38, TC-AI-06	AC-09

Requirement	Specification	Component	API / UI	Test case	Acceptance criterion
FR-39 Critical resource missing (UN-05)	§6.8	CMP-ROUTE	API-11 / UI-05	TC-FR-39	AC-09
FR-40 Route cost (UN-07)	§6.9	CMP-COST	API-11 / UI-05	TC-FR-40	AC-16
FR-41 Cost in optimization (UN-07)	§6.9, AIC-03	CMP-ROUTE, CMP-COST	API-11	TC-FR-41, TC-AI-03b	AC-16
FR-42 Route and total cost (UN-07)	§6.9	CMP-COST	API-12 / UI-05	TC-FR-42	AC-16
FR-43 Optimal vs. edited cost (UN-07)	§6.9	CMP-EDIT	API-13 / UI-05	TC-FR-43	AC-16
FR-44 Saving with risks (UN-10)	§6.9	CMP-EDIT	API-13 / UI-05	TC-FR-44	AC-12
FR-45 Directed routes (UN-08)	§6.10	CMP-UI	API-12 / UI-05	TC-FR-45	AC-10
FR-46 Stop numbers (UN-08)	§6.10	CMP-UI	API-12 / UI-05	TC-FR-46	AC-10
FR-47 Urgency colors (UN-08)	§6.10	CMP-UI	API-12 / UI-02, UI-05	TC-FR-47	AC-18
FR-48 Not color only (UN-08)	§6.10	CMP-UI	UI-02, UI-05	TC-FR-48, TC-ACC-01	AC-18
FR-49 Clean map (UN-08)	§6.10	CMP-UI	UI-05	TC-FR-49	AC-10
FR-50 Details box (UN-08)	§6.10	CMP-UI	API-12 / UI-05	TC-FR-50	AC-10
FR-51 Segment labels (UN-08)	§6.10	CMP-UI	API-12 / UI-05	TC-FR-51	AC-10
FR-52 Route and plan totals (UN-08)	§6.10	CMP-COST, CMP-UI	API-12 / UI-05	TC-FR-52	AC-16
FR-53 Tanker line colors (UN-08)	§6.10	CMP-UI	UI-05	TC-FR-53	AC-10
FR-54 Tanker filter (UN-08)	§6.10	CMP-UI	UI-05	TC-FR-54	AC-10

Requirement	Specification	Component	API / UI	Test case	Acceptance criterion
FR-55 Reorder stops (UN-09)	§6.11	CMP-EDIT	API-13 / UI-05	TC-FR-55	AC-10
FR-56 Move station (UN-09)	§6.11	CMP-EDIT	API-13 / UI-05	TC-FR-56	AC-10, AC-13
FR-57 Live recalculation (UN-10)	§6.11	CMP-EDIT	API-13 / UI-05	TC-FR-57, TC-PERF-01	AC-10
FR-58 Edit quantities (UN-09)	§6.11	CMP-EDIT	API-13 / UI-05	TC-FR-58	AC-11
FR-59 New plan after quantity edit (UN-09)	§6.11	CMP-EDIT, CMP-ROUTE	API-13 / UI-05	TC-FR-59	AC-11
FR-60 Reset to optimal (UN-09)	§6.11	CMP-EDIT	API-13 / UI-05	TC-FR-60	AC-10
FR-61 Compare with optimal (UN-10)	§6.12, AIC-04	CMP-EDIT	API-12, API-13 / UI-05	TC-FR-61	AC-12
FR-62 Detect risks (UN-10)	§6.12, AIC-04	CMP-EDIT	API-13 / UI-05	TC-FR-62, TC-AI-04	AC-12
FR-63 Severity (UN-10)	§6.12	CMP-EDIT	API-13 / UI-05	TC-FR-63	AC-12
FR-64 Risk reasons (UN-10)	§6.12, AIC-04	CMP-EXPL	API-13 / UI-05	TC-FR-64, TC-AI-04b	AC-07
FR-65 Highlight affected (UN-10)	§6.12	CMP-UI	UI-05	TC-FR-65	AC-12
FR-66 No-risk message (UN-10)	§6.12	CMP-UI	UI-05	TC-FR-66	AC-12
FR-67 Approval with warnings (UN-09)	§6.12	CMP-DEC	API-15, API-16 / UI-06	TC-FR-67	AC-14
FR-68 List actions (UN-11)	§6.13	CMP-DEC	API-12 / UI-06	TC-FR-68	AC-14
FR-69 Decide each action (UN-11)	§6.13	CMP-DEC	API-15 / UI-06	TC-FR-69	AC-14
FR-70 Decide edited plan	§6.13	CMP-DEC	API-16 / UI-	TC-FR-70	AC-14

Requirement	Specification	Component	API / UI	Test case	Acceptance criterion
(UN-11)			06		
FR-71 Nothing final without approval (UN-11)	§6.13	CMP-DEC	API-15, API-16	TC-FR-71	AC-14
FR-72 Record decisions (UN-12)	§6.14, §8.1 DE-17	CMP-DEC	API-15, API-16	TC-FR-72	AC-15
FR-73 View history (UN-12)	§6.14	CMP-DEC, CMP-UI	API-17, API-18 / UI-07	TC-FR-73, TC-SEC-07	AC-15
FR-74 Sample data (UN-16)	§6.14	CMP-DATA	–	TC-FR-74	AC-20
FR-75 Quantity limit hint (D-06)	§6.11	CMP-EDIT	API-14 / UI-05	TC-FR-75	AC-11
FR-76 Block save above limit (D-06)	§6.11	CMP-EDIT	API-13 / UI-05	TC-FR-76	AC-11
FR-77 Overload blocks approval (D-02)	§6.12	CMP-EDIT, CMP-DEC	API-13, API-15, API-16 / UI-05, UI-06	TC-FR-77	AC-13
FR-78 Estimated trip times (D-09)	§6.4	CMP-ROUTE	API-11 / UI-04, UI-05	TC-FR-78	AC-19
FR-79 Actual trip times (D-09)	§6.4	CMP-RES	API-08 / UI-04	TC-FR-79	AC-19
FR-80 Approve or reject all (D-03)	§6.13	CMP-DEC	API-16 / UI-06	TC-FR-80	AC-14
FR-81 Adjust shortcut (D-04)	§6.13	CMP-UI	UI-06 → UI-05	TC-FR-81	AC-14
FR-82 Rejected action effect	§6.13	CMP-DEC, CMP-EDIT	API-15 / UI-06	TC-FR-82	AC-14
FR-83 Manual planning flag (AI-21)	§6.6	CMP-FC	API-10, API-11 / UI-03, UI-05	TC-FR-83, TC-AI-01b	AC-05
FR-84 Estimated vs. actual trip time (D-16)	§6.4	CMP-RES	API-08 / UI-04	TC-FR-84	AC-19

Requirement	Specification	Component	API / UI	Test case	Acceptance criterion
FR-85 Forecast totals (D-17)	§6.6	CMP-FC	API-19 / UI-03, UI-04	TC-FR-85	AC-04

B.2 Non-functional requirements and AI capabilities

Requirement	Specification	Component	API / UI	Test case	Acceptance criterion
NFR-01, NFR-02, NFR-19, NFR-20 Performance	§7	CMP-EDIT, CMP-ROUTE, CMP-UI	API-11–13 / UI-02–05	TC-PERF-01–03	AC-17
NFR-03 Availability	§7	All	–	Session log	–
NFR-04, NFR-24 Scalability	§7	CMP-DATA, CMP-ROUTE	–	TC-SCAL-01, TC-PERF-04	–
NFR-05, NFR-06, NFR-21–23 Security	§7, §12	CMP-AUTH	All APIs	TC-SEC-01–05	AC-01
NFR-07, NFR-25 Privacy	§7, §8.8	All	–	Data model review, TC-SEC-06	–
NFR-13, NFR-26 Reliability	§7	CMP-FC, CMP-ROUTE, all writes	–	TC-REL-01, TC-REL-02	AC-20
NFR-14 Log immutability	§7, §12	CMP-DEC	API-17, API-18	TC-SEC-07	AC-15
NFR-12, NFR-28 Maintainability	§7	CMP-FC, RISK, ROUTE, COST, EDIT	–	TC-MAINT-01, CI coverage	–
NFR-08, NFR-09, NFR-31 Usability	§7, §13	CMP-UI, CMP-EXPL	All UI	TC-USE-01, TC-USE-02	–
NFR-10, NFR-11, NFR-27 Accessibility	§7, §13	CMP-UI	All UI	TC-ACC-01–03	AC-18
NFR-29 Observability	§15	All	–	TC-OBS-01	–
NFR-30 Portability	§7	Deployment	–	TC-PORT-01	–

Requirement	Specification	Component	API / UI	Test case	Acceptance criterion
NFR-15, NFR-16 Interoperability	§5.4, §7	CMP-DATA, CMP-UI	IF-WEATHER, IF-EVENTS, IF-DIST	Code review, TC-COMP-01	–
NFR-17, NFR-18 Compliance	§7, §12	All	–	TC-SEC-08, TC-AI-03	AC-08
AIC-01 Forecasting	§10	CMP-FC	API-10	TC-AI-01, TC-AI-01b	AC-04, AC-05
AIC-02 Risk and priority	§10	CMP-RISK	API-09, API-11	TC-AI-02, TC-AI-06	AC-06
AIC-03 Routing	§10	CMP-ROUTE, CMP-COST	API-11	TC-AI-03, TC-AI-03b	AC-08, AC-09
AIC-04 Risk analysis and explanations	§10	CMP-EDIT, CMP-EXPL	API-13	TC-AI-04, 04b, 04c	AC-07, AC-12